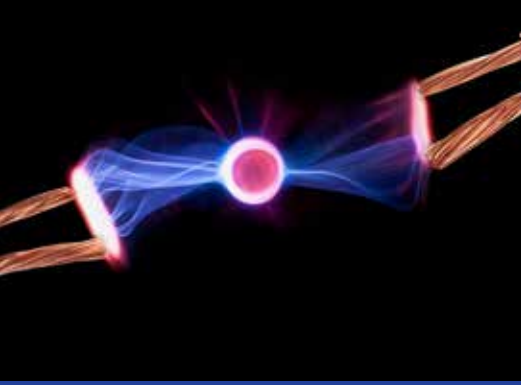




STRAND

Energy and Change



13 Forces



Key questions

- What is a force?
- What effect can a force have on an object?
- Do forces have to be between objects which are touching?

Keywords

- force
- net force
- contact force
- non-contact force
- newton



Did you know?

The Earth's atmosphere has changed overtime. Our oxygen-rich atmosphere was formed by algae millions of years ago.



Did you know?

The Newton is defined as the force needed to accelerate 1 kilogram of mass at 1 metre per second squared (m/s^2)

13.1 Forces

What is a force?

Think of the following situation: You are all helping your teacher to rearrange the classroom and she asks you to move her desk from one side of the classroom to the other. How would you do that? The desk is too heavy for you to lift so how do you get it across the classroom?

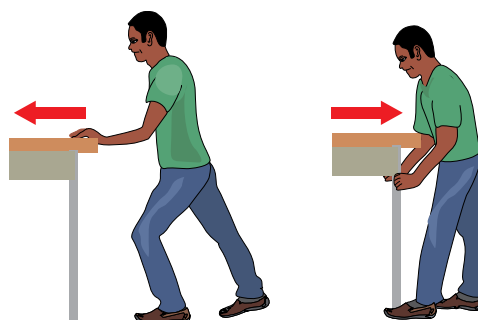


Figure 13.1

That is right, you are going to either push it or pull it across the room. In doing so, you have exerted a force on the desk to get it to move.

A force is a push or a pull on an object. The unit in which we measure force is a **newton (N)**.

The newton is named after Sir Isaac Newton, an English physicist and mathematician. Sir Isaac Newton is recognised as one of the most influential scientists of all times. The unit of force is named after him in recognition of his work in mechanics and his three laws of motion.



Figure 13.2: Sir Isaac Newton (25 December 1642 – 20 March 1727).

We use forces every day of our lives. Our own bodies rely on forces. Our muscles pull on our bones to allow us to move. Our feet push on the ground when we walk. To open doors, to pick up our food – everything we do involves some kind of force.

What can forces do? Let's experiment with forces and see what we can do.

ACTIVITY What can forces do?

This activity is all about experimenting with different objects and seeing what happens to them when we push and pull them.

Materials

- blocks (wooden or metal)
- sponge or piece of foam
- ball
- blown up balloon
- putty or play dough

Instructions

1. Work in groups of 2 or 3 as you follow the instructions and describe the effects of the forces that you are applying.
2. Start with the ball and place it on the ground. Push it towards your partner.
3. What were you able to cause the ball to do by pushing it?
4. When one of you pushes the ball to the other, the third person must give the ball another push at an angle to the direction in which it is already moving. What were you able to do to the direction in which the ball was moving?
5. Exert a force in the opposite direction to its movement while it is already moving. What are you able to cause the ball to do?
6. Exert a force in the same direction to its movement while it is already moving. What are you able to cause the ball to do?
7. Pick up the piece of putty or play dough. Exert pulling or pushing forces on it. Try this out with the blown up balloon too. What are you doing to the shape of the putty or play dough and the blown up balloon?
8. Push and pull the wooden blocks. Are you able to change their motion?
9. Are you able to change their shape?
10. Pick up the piece of sponge and twist it. This is also a type of force which changes the sponge's shape.
11. Press the sponge between both hands. This is called compression.



Did you know?

In 1687, Newton published *Philosophæ Naturalis Principia Mathematica*, which is often thought of as one of the most important books in the history of science. In it he describes universal gravitation and the three laws of motion.

Effects of forces

From the last activity, you should have seen that forces can have the following effects:

- Forces can change the **shape** of an object. This is called deformation.
- Forces can change the **motion** of an object. If an object is stationary, a force can cause the object to start moving. Or, if an object is already moving, a force can cause an object to speed up or slow down.
- Forces can change the **direction** in which an object is moving.

How do we describe the motion of an object? When an object is moving, we say it has a **speed**. Speed is the rate of change of the position of an object. Speed describes only how fast an object is moving.

An object can move at constant speed. This means it travels at the same speed. For example, a car travelling at 100 km/h has a constant speed. However, what happens when the car moves faster or slows down?

Keywords

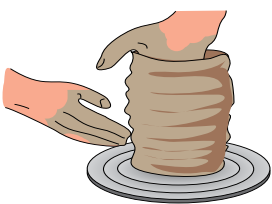
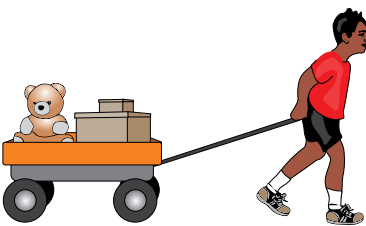
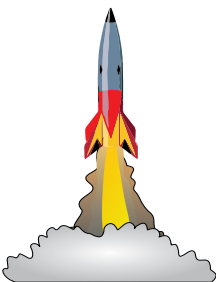
- speed
- acceleration
- deformation

We saw in the last activity that we could change the motion of an object by applying a force to make it speed up or slow down. The speed of the object is changing over time due to a force acting on it. This is called **acceleration**. Acceleration is the rate of change of a body's speed with time. In other words, it is a measure of how an object's speed changes every second.

ACTIVITY Is it a push or a pull?

Instructions

1. Look at the pictures in the table. Copy the table in your exercise books.
2. Describe the action in each image.
3. Decide if the force being exerted is a push or a pull.
4. Describe the effect of the force.

Action	Push or pull?	Effect
 Kicking a ball.		
 Moulding clay.		
 Playing with a toy wagon.		
 A rocket blasting off into space.		

Action	Push or pull?	Effect
 Exercising on a bar.		

Questions

1. In the example of the girl kicking the ball, which is the object experiencing the force and which is the agent of the force (i.e. the body which is applying the force)?
2. When moulding clay, which object is experiencing the force and which object is the agent?

Pairs of forces

We are now going to do another practical activity to investigate another concept about forces.

ACTIVITY Pairs of forces

Instructions

1. Work in groups of three for this activity. First, go up to your classroom wall and push against it. Describe what you feel.
2. When you push on the wall, do you think the wall is pushing back on your hands? How does this force compare to the force you are exerting on the wall?
3. Stand in a triangle with your two partners and hold hands. Pull on each other's hands. Do you feel your partners' hands pulling back as you pull?
4. Still standing in a triangle, place your palms up against each other and push against each other's hands. Do you feel your partners' hands pushing back as you push?
5. Next, stand shoulder-to-shoulder with your two partners. The two learners on the outside must push against the shoulders of the learner in the middle.
6. What happens when you both push with the same force?
7. What happens when one of you pushes with a harder force than the other?



Figure 13.3



Did you know?

What we have described here is actually Newton's Third Law of Motion. The law states that when one body exerts a force on a second body, the second body simultaneously exerts a force equal in strength and opposite in direction to that of the first body.

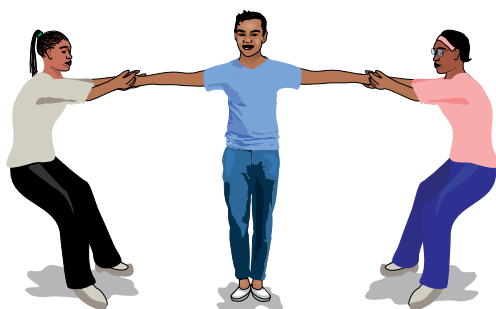


Figure 13.4

8. Next, the learner in the middle must stretch out his or her arms. The learners on the outside must pull on the middle learner's hands in opposite directions.
9. What happens when you both pull with equal force?
10. What happens when one of you pulls with a stronger force than the other?
11. What happens when one of you pulls and the other pushes?

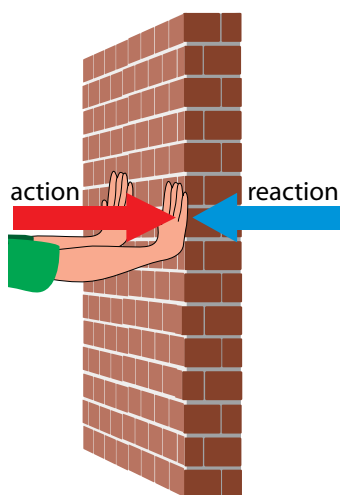


Figure 13.5

What we saw in the last activity is that whenever one object exerts a force on a second object, the second object exerts a force back on the first object. You saw this when you pushed against a wall. We say that forces act in pairs. Newton called the one force the **action**, and the other force the **reaction**, as shown in the following diagram.

We also saw that when you exerted a force on the wall, you experienced the wall exerting a force back on you. Forces act in pairs on different objects. The force exerted by the second object is **equal in strength** and **opposite in direction** to the first force.

Types of forces

So far, we have looked at forces acting on an object when the object causing the force is in contact with the object experiencing the force. Do we always have to be in contact with an object in order to exert a force?

ACTIVITY Can forces act over a distance?

Materials

- bar magnets
- metal paper clips

Instructions

1. Place one of the bar magnets on the table.
2. Bring the north end of another bar magnet close to the south end of the first bar magnet. What happens?
3. Bring the north end of one bar magnet close to the north end of the other bar magnet. What happens?
4. Place the paper clips on the table.
5. Bring a bar magnet over the paper clips. What do you observe?

Questions

1. Did you have to touch the bar magnets together before they would attract each other?
2. Did the paper clips move towards the magnet?
3. What caused the movements?

There were forces exerted by the magnets but they did not have to touch each other. That means that you do not have to be in contact with something in order to exert a force on it.

There are two types of forces:

- **Contact forces:** objects are in contact with each other and exert forces on each other.
- **Non-contact (field) forces:** objects are not in contact with each other and exert forces on each other. We are now going to look at these two broad groups of forces in more detail.

13.2 Contact forces

Contact forces are forces between objects which are touching each other. Most of the forces that we looked at in the previous section were contact forces, for example, when you push a desk, or pull a go-kart. You are touching the object.

Friction

What happens when you kick a ball across the grass? The ball moves quickly at first but then slows down again. Something has caused the ball to slow and stop moving. If the motion of the ball has changed then a force must have been exerted on it. The force which opposes motion is called **friction**. Friction forces always act in the opposite direction to the motion of the object. Friction resists movement when the object and surface are in contact. What does that mean? It means that if the ball is moving forward then friction acts backwards on the ball.

The following image shows a ball which has just been kicked. Copy the diagram into your books and draw an arrow to show in which direction friction would be acting.

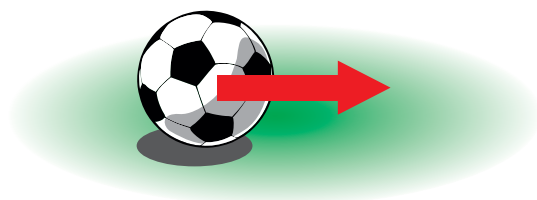


Figure 13.6

There is another force acting on the go-kart. Think of when you stand on the ground: you feel the ground beneath your feet. This contact force is preventing you from penetrating the ground. This is called the **normal force**.

The normal force always acts perpendicularly to the surface that the object is resting on.

In simple situations such as when you are standing on the ground or the go-kart is travelling along a level surface, the normal force is equal to the weight of the object, but in the opposite direction. Think back to what we learnt about forces acting in pairs. On a flat, level surface, the normal force is the reaction force to the weight of the object. This is shown in the diagram for a box resting on the floor.

Do you think there is a relationship between the friction that a body experiences and the normal force? Let's investigate.

Keywords

- friction
- normal force
- tension
- compression



Take note

As we will see later on, there are also other forces acting on this system such as friction and weight.

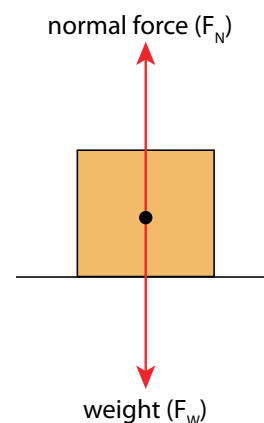


Figure 13.7



Investigation

What is the relationship between the normal force and friction?

Investigative question

What is the relationship between the normal force and friction?

Aim

To determine the relationship between the normal force and the size of the frictional force.

In the situations that we are going to investigate, the object will be pulled along on a flat, level surface. We will increase its mass and measure the resulting frictional force. But how does this relate to the normal force?

Where the object is on a flat level surface, the normal force is equal to the weight. As you learnt in Grade 8 *Earth and Beyond*, and will see in the next section, we can calculate the weight of an object. We can therefore calculate the normal force acting on the object.

For this investigation, calculate the weight using the formula $W = m \times g$, where m is the mass of the object in kg and g is $9,8 \text{ m/s}^2$. We will learn more about this in the next section.

Apparatus

- wooden blocks with different known masses, or mass pieces
- wooden block with a hook
- spring balance
- triple beam balance or electronic scale

Method

1. Measure the mass of the wooden block with a hook with the triple beam balance. Record the mass in the table. Calculate and record the normal force.
2. Put the wooden block with the hook on the table. Attach the spring balance to the hook. Make a small mark on the desk from which to start pulling the block.
3. Pull sideways to the point that the block just starts moving.
4. Copy the table under **Results** (on the next page) into your exercise book, and record the force readings for each mass. Repeat this three times for the wooden block.
5. Put a mass piece on top of the wooden block. Record the total mass. Calculate and record the normal force. Remember that the normal force acting on the object is equal to the weight of the object. Pull it sideways to the point that it starts to move. Record the force. Repeat this three times. In each case, start the block from the same position and pull gently.

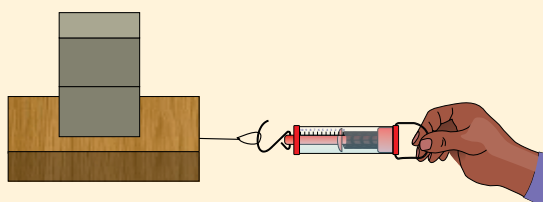


Figure 13.8



Take note

We take three readings and then calculate an average. This increases the reliability of the results.



Take note

Remember that the normal force acting on the object is equal to the weight of the object.

6. Repeat the experiment for larger masses.

Results

Table to record the force required to overcome the frictional force and move the block.

Mass (kg)	Normal force (N)	Reading 1 (N)	Reading 2 (N)	Reading 3 (N)	Average (N)

Plot a graph of the average force applied to the block at rest against the normal force of the block in your exercise books.

1. Which is the dependent variable?
2. Which is the independent variable?

Analysis

1. Draw a labelled free-body diagram of all the forces acting on the block just as it is about to start moving.
2. Why is the weight on the block being changed when the aim of the investigation is to find out how the normal force affects frictional force?
3. Why are three readings taken for each setup and an average calculated?
4. What is the shape of your graph?
5. What does the shape of the graph tell us about the relationship between the normal force of the block and the friction force?
6. What do you think would happen if the block was not placed on the smooth desk, but rather on a rough surface, or a much smoother surface? Would this affect the results?

Conclusion

1. Write a conclusion for this investigation.
2. Repeat the investigation and take some readings from the spring balance once the block is moving. How do the readings for the stationary block and a moving block compare? Is there a difference?

The force of friction depends on the type of surface on which an object is moving and the normal force. In order to get an object to move, a force greater than the frictional force needs to be applied in order to overcome the friction between the object and the surface.



Did you know?

We are measuring the static friction, which is the friction between two objects that are NOT moving. We measure it as the minimum force required to start the block moving.



Did you know?

Friction between two surfaces also causes the objects to heat up. Try this out by rubbing your hands vigorously together and then feel your palms afterwards.



Take note

The force applied to the point at which the block starts to move is equal to the frictional force acting on the block.

Friction is advantageous for a number of reasons. For example, the friction between our feet and the ground enables us to move forward and prevents us from slipping. Friction is also involved in keeping cars from skidding as the tyres experience friction between the tread and the roadway.

Tension and compression

There are other forces which are contact forces. Look at the following drawing of a boy pulling on a block with a rope.

The person is pulling the rope which is pulling the block. The person is not touching the block directly. The person is pulling the rope and the block is pulling back on the rope in the opposite direction. This causes a **tension** force to exist in the rope. The rope is tight and so there is tension in the rope. Tension is a contact force. The tension in the rope pulls the block across the floor.

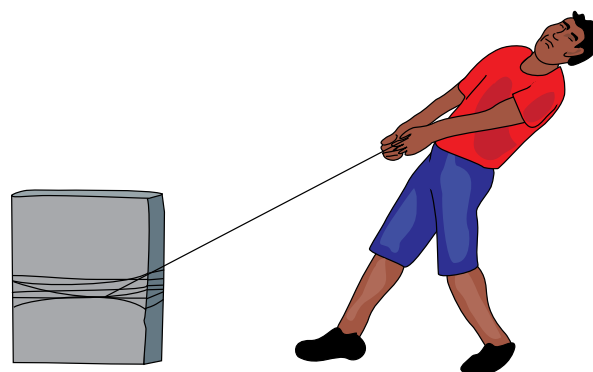


Figure 13.9



Figure 13.10: Crushing dough.

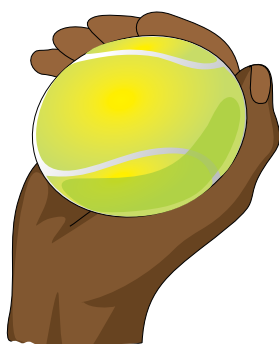


Figure 13.11: Compressing a tennis ball.

Another example of a contact force is **compression**. A compression force is a force which acts to deform or squash an object. Let's think of some examples.

If you take a ball of dough and crush it with your fingers, you are exerting a compression force on the dough. The dough changes shape. It deforms. Another example is crushing a tennis ball or a cooldrink can between the palms of your hands.

Can you think of some other examples?

The following diagram summarises the difference between tension and compression.

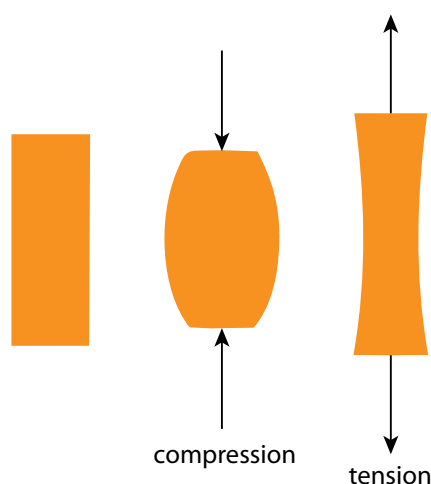


Figure 13.12: Tension forces are two forces acting on one object in opposite directions (away from each other) to stretch the object. Compression forces are two forces acting on one object in opposite directions (towards each other) to compress or deform the object.

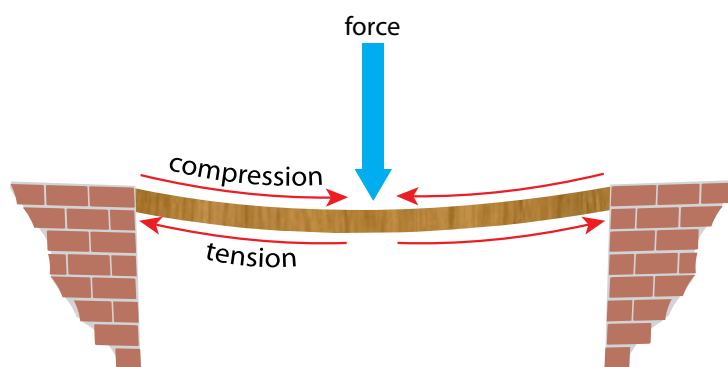


Figure 13.13

There are many other examples of compression forces in everyday life. A bridge experiences both compression and tension due to the weight of the cars and other vehicles which pass over it as shown in the following diagram.

We have been looking at contact forces, which include friction, normal forces, compression and tension. We are now going to look at the forces that act between bodies which do not touch.

13.3 Field (non-contact) forces

A field is a region in space where an object (with certain properties) will experience a force. Field forces are non-contact forces. Non-contact forces are forces which act over a distance. Object's experiencing field forces need not be in contact.

The most common examples of fields are:

- gravitational field
- magnetic field
- electric field

When we discussed contact forces, we spoke about pushes and pulls. However, with field forces, it is better to talk about repulsion and attraction.

Keywords

- gravitational force
- magnetic force
- electrostatic force
- field forces
- attraction
- repulsion



Did you know?

Newton developed his Law of Universal Gravitation, describing the force of attraction between bodies with mass, in 1687. Newton's work on describing a theory of gravitation may have been inspired by watching an apple fall from a tree

Gravitational forces

Have you ever wondered why things fall down and not up?

The force which causes things to fall down towards the Earth and prevents us from falling off the planet is the gravitational force. Gravitational forces exist between any two objects with mass and they are forces of attraction (pull).

The gravitational force is a force that attracts objects with mass towards each other. Any object with mass exerts a gravitational force on any other object with mass. The Earth exerts a gravitational pull on you, the desks in your classroom and the chairs in your classroom, holding you on the surface and stopping you from drifting off into space.

Factors affecting gravitational force

The Earth's gravitational force pulls everything down towards the centre of the Earth which is why when you drop an object like a book or an apple, it falls to the ground. However, do you know that you, your desk, your chair, and the falling apple and book exert an equal but opposite pull on the Earth?

Why do you think these forces on the Earth do not cause the Earth to move noticeably?



Figure 13.14: The arrows show the direction of the gravitational field of the Earth. The arrows all point towards the centre of the Earth because the gravitational force is always attractive



Figure 13.15: These army skydivers have just jumped out of the back of a plane and fall towards the Earth due to gravity.

The Earth attracts us because it has such a large mass and so we are attracted downwards towards the centre of the Earth all the time.

The bigger the mass of the objects, the greater the force between them. This means that two small objects would have a very weak gravitational attraction and so it has no noticeable effect. However, bigger objects such as the Moon and the Earth have a much greater gravitational force.

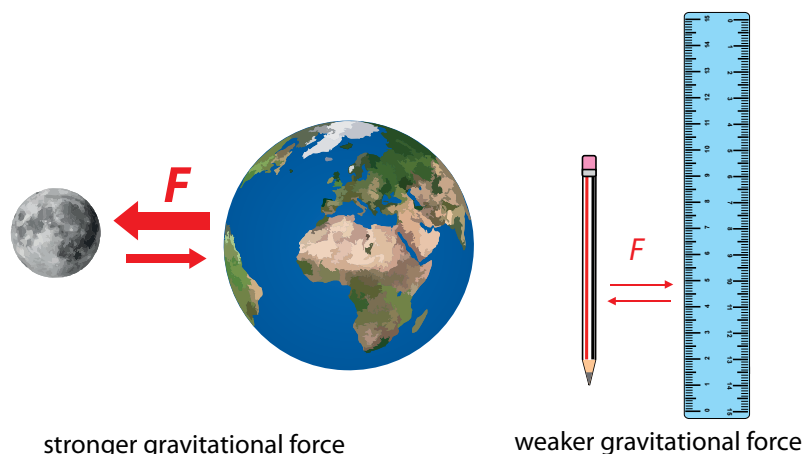


Figure 13.16

As we know from Planet Earth and Beyond, all the planets in our solar system are held in orbit around the Sun by the gravitational force of attraction between the Sun and planets.

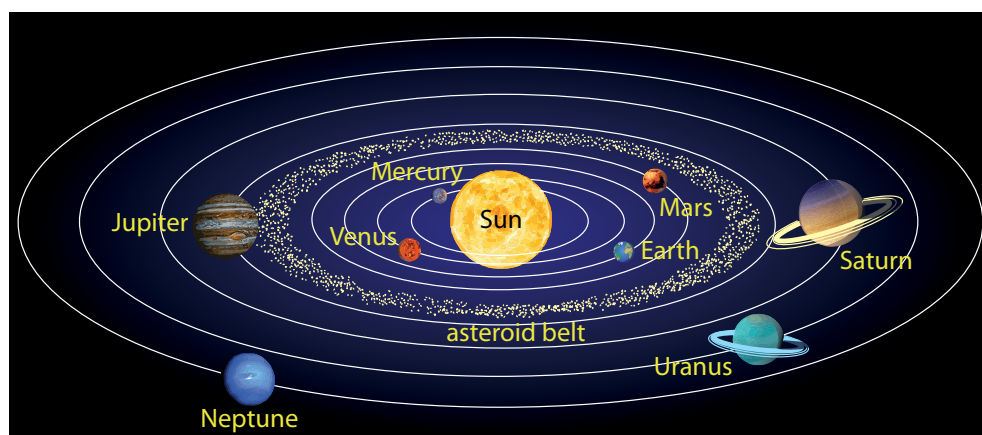


Figure 13.17: The planets move around the Sun in our solar system. There is a gravitational force of attraction between the Sun and planets, and between planets and their moons.

The second factor which affects the gravitational force of attraction between objects is the distance between them. The further objects are away from each other, the smaller the gravitational force.

All the components in our Universe are held together by a gravitational force. There are two factors affecting the gravitational force acting on the object:

- The mass of the object. The bigger the mass, the greater the gravitational force.
- The distance between the objects. The closer the objects, the greater the force.

? Did you know?

There is a gravitational force of attraction between us and the Sun, but we do not notice it as we are so far apart, and we are very small.



Investigation Dropping objects

Investigative question

Do different objects fall at the same rate?

Hypothesis

What do you think will happen?

Materials and apparatus

- hammer
- feather
- two balls of the same mass, different volumes (one set per pair)
- two balls of the same volume, different masses (one set per pair)

Method

1. Work in pairs, taking turns to be the person who drops an object (experimenter) and the person who observes the objects dropping (observer).

2. Copy the following table into your exercise book, then complete the 'prediction' column.
3. Experimenter: stand on top of a chair or desk and take the two balls of the same mass, with one in one hand and the other in the other hand.
4. Experimenter: hold the two balls up at the same height in front of you and drop them at exactly the same time.
5. Observer: note what happens, in particular which lands first.
6. Swap positions and repeat the experiment using two balls which have the same volume but different masses.
7. Your teacher will now do a demonstration for you and drop a hammer and a feather. Before your teacher drops the hammer and feather, record the prediction column for the hammer and feather drop.
8. Record what happened with the hammer and feather and answer the questions that follow.

Results and observations

1. What did you keep constant in this experiment?
2. What did you change in this experiment?
3. Before conducting your experiment, assume that you drop each pair of objects from the same height, at the same time. What do you think will happen?
4. Which object do you think will land first? Why?

Objects	Prediction	Observation
Balls: same mass, different size		
Balls: different mass, same size		
Hammer and feather		

Evaluation

How reliable was your experiment? How could you improve your method?

Conclusion

Write a conclusion for this investigation.

Questions

1. Considering the balls of the same mass, which landed first, the larger one or the smaller one?
2. Considering the balls of the same volume, which landed first, the heavier one or the lighter one?
3. Why do you think the two dropped balls always landed at the same time?
4. Why do you think the hammer landed before the feather?

Mass and weight

You have probably heard the term ‘weight’ used many times before, either in your Natural Sciences classroom, or in conversation with others. Many people use the term weight incorrectly in everyday language. For example, a relative may say to you ‘My weight increased by 2 kg over the holiday period as I ate too much food.’ What is wrong with this statement? Discuss this with your class and teacher.

The **mass** of an object is the amount of matter in the object. It tells you how many particles you have. Do you remember learning about atoms in Matter and Materials? So, for example, the mass of a wooden block tells us how many atoms there are. Mass is measured in kilograms (kg) and is independent of where you measure it. A wooden block with a mass of 10 kg on Earth also has a mass of 10 kg on the Moon.

However, an object’s **weight** can change as it depends on the mass of the object and also the strength of the gravitational force acting on it. Weight is measured in newtons (N) as it is the gravitational force of attraction exerted on an object by the Earth (or Moon or any other planet). Therefore, the weight of an object will change when weighed in different places.

The weight of a 10 kg block on Earth will be different from its weight on the Moon. Why do you think this is? Will the weight be more or less than on the Moon?

Keywords

- weight
- mass
- free-fall
- gravitational acceleration



Did you know?

The word ‘mass’ comes from the Greek word ‘maza’, which means a lump of dough or cake.



Investigation

What is the relationship between the mass of an object and its weight?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- four mass pieces in increments of 500 g (one of 500 g, one of 1 kg, one of 1,5 kg and one of 2 kg)
- spring balance or force meter
- triple beam balance

Method

1. Measure the mass pieces on the triple beam balance.
2. Measure the weight of each mass piece with the spring balance.

3. Copy the table below in your exercise books and record the mass and matching weight.

Results

Mass (kg)	Weight (N)

1. What is the dependent variable?
2. What is the independent variable?
3. Draw a graph of your results.
4. Your graph should be a straight line. Calculate the gradient of your graph.

Conclusion

Write a conclusion for this investigation.

Weight is the force of gravity pulling you towards the centre of the Earth. It is measured in newtons. On Earth the gravitational force causes us all to accelerate towards the centre of the Earth. The acceleration is called gravitational acceleration.

On Earth it is $9,8 \text{ m/s}^2$. The gradient that we calculated in the last investigation should have given you a number close to $9,8 \text{ m/s}^2$ which is gravitational acceleration.

Objects are in free-fall when the only force acting on them is the gravitational force.

Weight (W) is calculated by multiplying an object's mass (m) by the gravitational acceleration (g):

$$W = m \times g$$

But what if you went to the Moon?

The Moon also has its own gravity. The strength of gravity on the surface of the Moon is one-sixth of that on the surface of the Earth, and so you would weigh one-sixth of what you do on Earth on the Moon. On Jupiter you would weigh 2,5 times more than you do on Earth as Jupiter's gravity is 2,5 times that of the Earth's. Even



Figure 13.18: The Moon is 6 times smaller than the Earth.



Did you know?
The mass of the Earth is $5,972 \times 10^{24} \text{ kg}$.

though you would weigh different amounts (and feel lighter on the Moon and heavier on Jupiter) your actual mass would stay the same in both cases.

An astronaut's mass remains the same wherever it is measured. The astronaut's weight, however, depends on where you measure it.

So how much would an astronaut weigh on the Moon? Suppose that an astronaut has a mass of 120 kg. Your weight on Earth would be $120 \times 9,8 = 1\,176\text{ N}$. The gravitational acceleration on the Moon is $1,6\text{ m/s}^2$, so your weight would be $120 \times 1,6 = 192\text{ N}$ on the Moon.

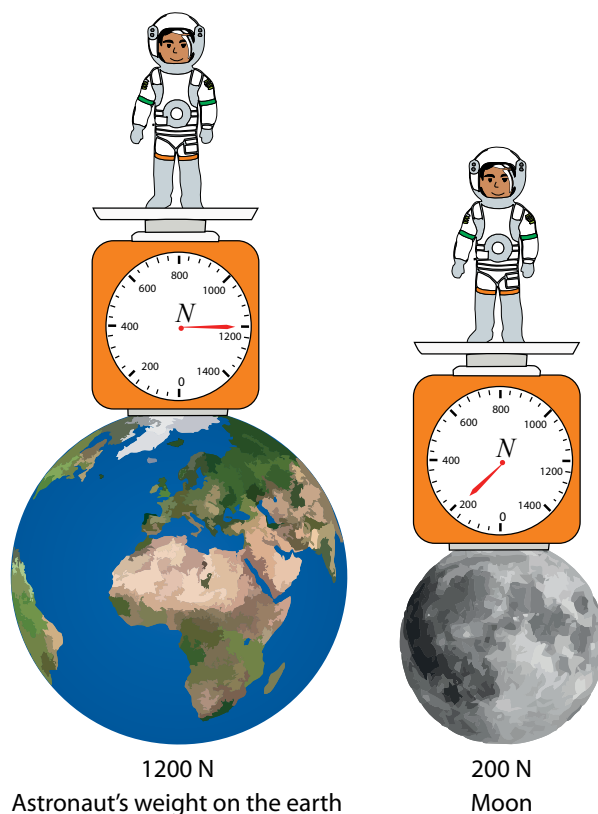


Figure 13.19

Take note

A bathroom scale actually measures weight and converts it to mass.

ACTIVITY Weight and mass calculations

Questions

1. A Ferrari has a mass of 1 485 kg. What is its weight on Earth?
2. Lindiwe has a mass of 50 kg on Earth. What is her mass on the Moon?
3. Ian has a mass of 78 kg. His friend Sam says that he would weigh 24 N on the Moon. Is Sam correct? Explain by using a calculation.
4. You have an apple with a mass of 220 g, what is its weight on Earth and on the Moon?
5. If a cow weighed 1 340 N on the Moon, what is its mass?



Figure 13.20: A Ferrari.



Figure 13.21: A jersey cow.

Have you ever wondered what it would be like to walk around on other planets? Find out how much you would weigh on other planets in the next activity.

ACTIVITY How much would you weigh on other planets?

Materials

- weighing scales
- calculator

Instructions

1. Measure your mass in kilograms. Copy the table below in your exercise books and record the values.
2. Use the values for the acceleration due to gravity on various planets to calculate what you would weigh on that planet.

Planet	Your mass (kg)	Value of g (m/s^2)	Your weight (N)
Earth		9,8	
Mercury		3,6	
Venus		8,8	
Mars		3,8	
Jupiter		26	
Saturn		11,2	
Uranus		10,5	
Neptune		13,3	

Questions

1. On which planets would you feel heavier than you do on Earth?
2. On which planets would you feel lighter than you do on Earth?



Figure 13.22: Astronauts experiencing weightlessness.

Your weight (N)

The weight of a person is the force of gravitational attraction to the Earth that person experiences. Someone in free-fall feels weightless but they have not lost their weight. They are still experiencing the Earth's gravitational attraction.

The only reason the astronauts float is because they are in free-fall and their moving spacecraft is also in free-fall with them, falling at the same rate. Therefore, the astronauts appear to float when compared with the spacecraft because they are both falling at the same rate.

Magnetic forces

Certain materials have strong magnetic fields around them. These are called magnets. All magnets have two poles, a north and a south pole.

Other materials are strongly attracted to magnets. These materials are said to be magnetic. Magnets exert forces on other magnets and magnetic materials.

Which materials are magnetic? Let's investigate.

Keywords

- magnet
- magnetic material
- alloy



Figure 13.22: An example of a bar magnet with a north and South Pole.



Investigation Magnetic or non-magnetic materials

Investigative question

Which materials are magnetic, and which are not?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- bar magnets
- paper
- wood
- plastic
- iron
- aluminium
- steel

Method

1. Hold the different items close to the bar magnet (not touching) to see if they are attracted to the magnet.
2. Complete the table indicating whether or not the items are attracted to the magnet.

Results

Copy and complete the following table in your exercise books.

Material	Magnetic (YES/NO)
paper	
wood	
plastic	
iron	
aluminium	
steel	
copper	

Conclusion

What can you conclude from your results?

Not all metals are attracted to magnets. Those that are attracted to magnets are known as magnetic materials. There are very few magnetic materials. They are iron, nickel and cobalt. Alloys which include any of the magnetic materials can also be attracted to magnets. Steel is an alloy which contains iron so steel can be attracted to a magnet.

So now we know that magnetic forces can act over a distance, but can they still act if there is something in the way? Let's find out.



Take note

An alloy is a mixture of metals.

ACTIVITY Can a magnetic force act through substances?

Materials

- bar magnets
- paper
- thin piece of wood
- thick piece of wood
- foil
- paperclips

Instructions

1. Hold two north poles close together. What do you notice?
2. Hold two south poles close together. What do you notice?
3. Hold a north pole and a south pole close together. What do you notice?
4. Put the paper clips on the desk.
5. Try to pick up the paperclips with the magnet but put one of the other materials between the magnet and the paper clips. Are the paperclips still attracted to the magnet?
6. Try each of the different materials between the magnet and the paper clip.

Questions

1. Which materials prevented the magnet from picking up paper clips?
2. What does this activity tell us about the nature of the magnetic force?

In the last activity we saw that like poles repel each other but opposite poles attract each other. We have also seen that the magnetic force acts over a distance. The magnet does not need to touch something in order to exert a force on it. So a magnetic force is a non-contact or a field force.

What is a field of a force? Can we see it? Let's investigate if it is possible to see a magnetic field.

ACTIVITY Visualising magnetic fields

Materials

- iron filings
- two bar magnets
- paper

Instructions

1. Put one of the bar magnets on the table.
2. Put the paper over the magnet.
3. Shake the iron filings onto the paper.
4. Take note of the pattern and draw it in your exercise books.
5. Lift the paper away from the magnet.
6. Place a second magnet next to the first so that different poles are facing each other.
7. Put the paper back over the magnets.
8. Move the iron filings around the two magnets, especially between the magnets..
9. Draw the pattern in your exercise books.
10. Lift the paper away from the magnet.
11. Move the second magnet so that the same poles are facing each other.
12. Put the paper back over the magnets.
13. Move the iron filings around the two magnets, especially between the magnets.
14. Draw the pattern in your exercise books.

As we have seen, it is possible to visualise the magnetic force field around a magnet. We know from our previous activities that the magnetic force acts over a distance. The field is the space around a magnet in which it can attract or repel another magnet.

How do we draw a force field? The pattern you saw with your magnets can be represented by **field lines**. Field lines are used to show something we can't actually see. The closer the field lines are drawn together, the stronger the field being described. The more field lines that are drawn, the stronger the field. The field lines go from the North Pole to the South Pole. The following diagram shows the field lines around a bar magnet.

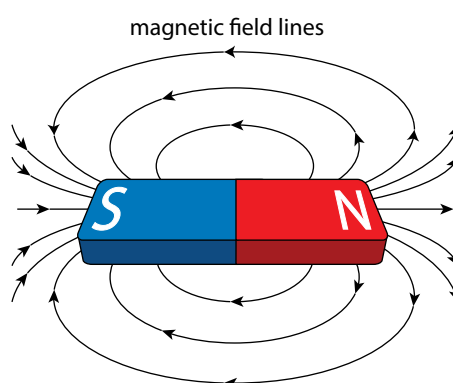


Figure 13.24



Take note

In the last activity, the iron filings showed a two-dimensional view of the field but the field is actually all around the magnet, in three dimensions.

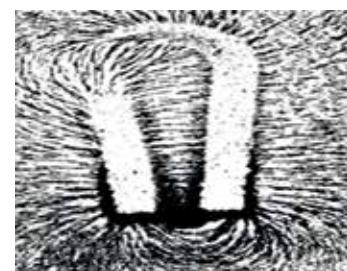


Figure 13.23: The magnetic field around a horseshoe magnet.

The next diagrams show the field lines between bar magnets which are attracting and those which are repelling.

Opposite poles attract.

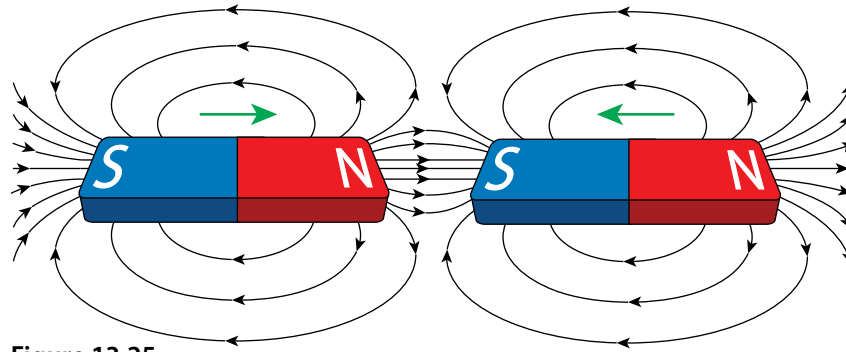


Figure 13.25

Like poles repel.

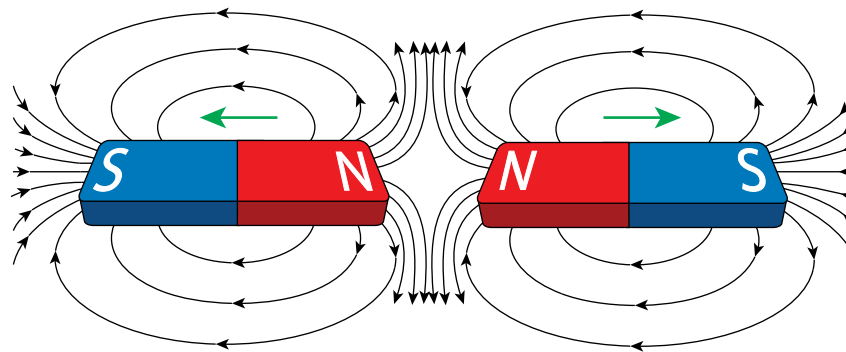


Figure 13.26

A field is strongest next to the magnet and gets weaker further away from the magnet.

Did you know that the Earth is like a bar magnet with a North and a South Pole? The Earth has a magnetic field. You can imagine Earth's magnetic field as though there is a bar magnet running through the core with the magnet's South Pole under Earth's North Pole. No one knows for sure, but the theory is that the superhot liquid iron in the Earth's core moves in a rotational pattern, and these rotational forces lead to the weak magnetic forces around the Earth's rotational axis.

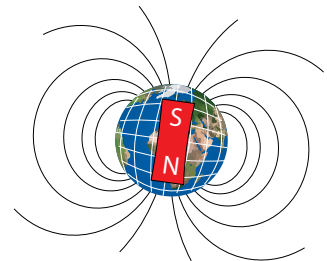


Figure 13.27: Earth has a magnetic field, as though there is a big bar magnet running through the core, with its South Pole below Earth's magnetic North pole.



Take note

The Southern lights are also called the Aurora Australis and the Northern lights are called the Aurora Borealis.

The Earth's magnetic field is the reason why we can use compasses to tell direction.

A plotting compass has a needle with a small magnet. The needle points to magnetic north because the small magnet is attracted to the opposite magnetic field and can be used to determine direction.

Have you heard of the Southern or Northern Lights before? Do you know how these phenomena occur?



Figure 13.28: A compass with the needle pointing North.



Figure 13.29: The Southern Lights, viewed from the International Space Station.



Figure 13.30: An example of a ferrofluid, a liquid that can become magnetised in a magnetic field.

For enrichment

- Charged particles escape from the surface of the Sun and move outwards in all directions. When the charged particles reach Earth, some are trapped by Earth's magnetic field in areas in space around Earth's atmosphere, called belts.
- Sometimes the charged particles escape the belts and spiral along the magnetic field lines towards the magnetic poles where they enter Earth's atmosphere. They then interact with atmospheric gas particles, causing beautiful light shows.
- Some liquids can also become magnetised in the presence of a strong magnetic field. They are called ferrofluids.

Electrostatic forces

Do you remember learning about static electricity in Grade 8? Let's do a quick activity to revise some of the concepts we already know.

ACTIVITY Charging objects

Materials

- balloons (or a plastic comb)
- glass rod
- piece of knitted fabric (wool)
- PVC rod
- plastic ruler
- small pieces of paper
- water tap

Instructions

1. Work in pairs.
2. Blow up a balloon and tie it closed so that the air does not escape.
3. Hold the balloon a short distance away from your hair. What do you notice?
4. Rub your hair with the balloon.
5. Now hold the balloon a short distance away from your hair. What do you see?

6. Next, hold the glass rod over the small pieces of paper. What do you notice?
7. Rub the glass rod with the knitted fabric.
8. Hold the glass rod over the pieces of paper. What do you notice?
9. Rub the glass rod with the knitted fabric again.
10. Open the tap so that a thin stream of water is flowing.
11. Hold the glass rod close to the stream of water. What do you notice?

Questions

1. What did you do to make your hair stick to the balloon?
2. What happens when you rub the glass rod with the knitted fabric?
3. Why did the glass rod attract the stream of water?

Let's look at the example of brushing your hair in more detail to understand what is happening. You have dragged the surface of the plastic comb against the surface of your hair. When two surfaces are rubbed together there is **friction** between them. The friction between two surfaces can cause electrons to be transferred from one surface to the other.

In order to understand how electrons can be transferred, we need to remember what we learnt about the structure of an atom.

- Where are the electrons positioned in the atom?
- What is the type of charge on a proton?
- What is the type charge on an electron?
- What is the charge on a neutron?

The atom is held together by the **electrostatic attraction** between the positively charged nucleus and the negatively charged electrons. Within an atom, the electrons closest to the nucleus are the most strongly held, while those further away experience a weaker attraction.

Normally, atoms contain the same number of protons and electrons. This means that atoms are normally **neutral** because they have the same number of positive and negative charges, so the charges balance each other out. All objects are made up of atoms and since atoms are normally neutral, objects are also usually neutral.

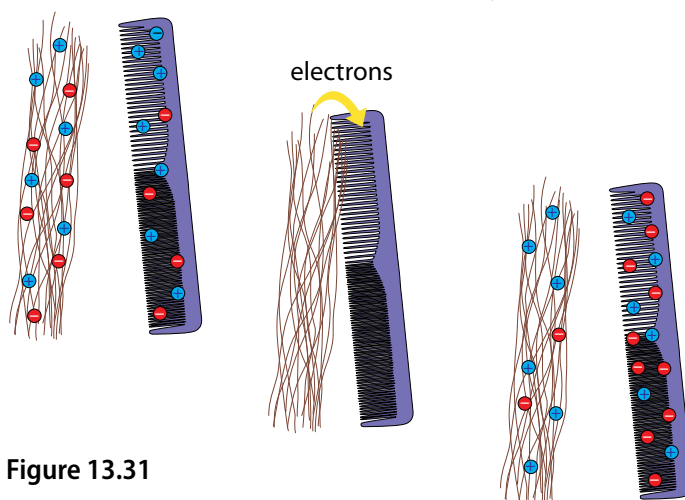


Figure 13.31

However, when we rub two surfaces together, such as when you comb your hair or rub a balloon against your hair, the friction can cause electrons to be transferred from one object to another. Remember, the protons are fixed in place in the nucleus and so they cannot be transferred between atoms. Only electrons can be transferred between atoms. Some objects give up electrons more easily than other objects. Look at the following diagram which explains how this happens.

Which object gave up some of its electrons in the diagram?

Does this object now have more positive or more negative charges?

Which object gained electrons in the diagram?

Does this object now have more positive or more negative charges?

- When an object has more electrons than protons, we say that the object is **negatively charged**.
- When an object has fewer electrons than protons, we say that the object is **positively charged**.

Have a look at the following diagrams which illustrate this.

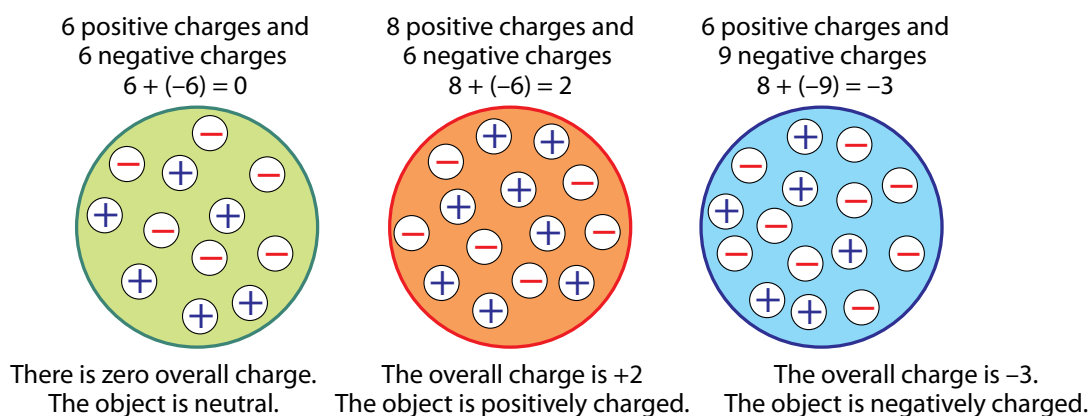


Figure 13.32

We now understand the transfer of electrons that takes place as a result of friction between objects. But how did that result in your hair rising when you brought the charged balloon close to your hair in the last activity? Let's look at what happens when oppositely charged objects are brought together.

ACTIVITY Turning the wheel

Materials

- two curved watch glasses
- two Perspex rods
- cloth: wool or nylon
- plastic rod
- small pieces of torn paper

Instructions

1. Place a watch glass upside down on the table.
2. Balance the second watch glass upright on the first watch glass.
3. Rub one of the Perspex rods vigorously with the cloth.
4. Balance the Perspex rod across the top of the watch glass.
5. Rub the second Perspex rod vigorously with the same cloth.
6. Bring the second Perspex rod close to the side of the first Perspex rod that was charged. What do you see happening?

Take note

Remember it is only the outer electrons which move, and not the protons, which are located in the nucleus of the atom.

Take note

Remember, like charges repel and opposite charges attract.

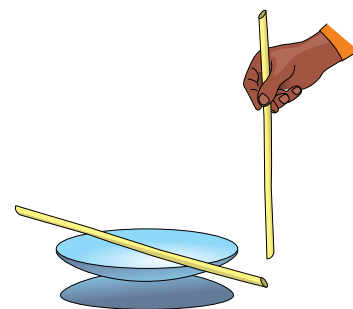


Figure 13.33

7. Repeat the activity but instead of the second Perspex rod, use the plastic rod. What do you see happening?
8. Next, bring a rod that you have rubbed closer to small pieces of torn paper on the table. What do you observe?

Questions

1. What happened when you brought the second Perspex rod close to the first Perspex rod?
2. What happened when you brought the plastic rod close to the first Perspex rod?
3. What happened when you brought the plastic rod close to the pieces of paper?

When we rubbed the Perspex rods with the cloth, electrons were transferred from the Perspex to the cloth. What charge do the Perspex rods now have?

Both the Perspex rods now have the **same** charge. Did you notice that objects with the same charge tend to push each other away? We say that they are **repelling** each other. It is an electrostatic force of repulsion.

When we rubbed the plastic rod with the cloth, electrons were transferred from the cloth to the plastic rod. What charge does the plastic rod now have?

The perspex rod and the plastic rod now have **opposite** charges. Did you notice that objects with different charges tend to pull each other together? We say that they are **attracting** each other. It is an electrostatic force of attraction.

As with gravitational and magnetic force, the distance between charged objects affects the strength of the electrostatic force. The closer the charged objects are, the stronger the force. The more charged the objects are, the stronger the electrostatic force between them.

We have now observed the fundamental behaviour of charges. In summary, we can say:

- If two negatively charged objects are brought close together, then they will repel each other.
- If two positively charged objects are brought close together, then they will repel each other.
- If a positively charged object is brought closer to a negatively charged object, they will attract each other.
- Have you ever wondered where lightning comes from? Let's demonstrate an electrostatic spark.



Did you know?

The fundamental idea of using friction in a machine to generate a charge dates back to the 17th century, but the generator was invented by Robert Van de Graaff only in 1929 at Princeton University.

ACTIVITY Van de Graaff generator

Materials

- Van de Graaff generator

Instructions

1. Turn on the generator.
2. Bring the small metal globe close to the generator. What do you see?

Did you see sparks? The Van de Graaff generator can be used to demonstrate the effects of an electrostatic charge. The big metal dome at the top becomes positively charged when the generator is turned on. When the dome is charged it can be discharged by bringing another insulated metal sphere close to the dome. The electrons will jump to the dome from the metal sphere and cause a spark.

How does this little spark relate to a massive lightning strike?

During a lightning storm, clouds become charged. Friction between the clouds and the moisture in the clouds cause the clouds to become charged. The bottom of the clouds (closest to the ground) become negatively charged and the top of the cloud becomes positively charged. When the build-up of charge becomes too large, the electrons move from the bottom of the cloud to the ground where they are 'earthed'. The energy transfer is massive and results in extremely bright light, heat and sound. A lightning flash is a massive discharge between charged regions within clouds, or between clouds and the Earth. The thunder-clap, which we hear, is the air moving as a result of electrons moving.

Lightning is extremely dangerous. If the electrons move through a person on their way to the ground, then the large amounts of energy cause significant damage. That person can suffer serious injury, even death.

What precautions should we take during a lightning storm? Lightning can strike far from the rain shadow of the storm. This means that even if the storm seems to be far away, it is better to take precautions anyway. The safest place to be in a lightning storm is indoors. Stay away from windows and metal objects. If you cannot get inside, avoid standing next to tall objects or metal objects because if lightning strikes it will usually hit the tallest object in the area. If you are travelling in a car during a storm, stay inside the car until the storm subsides.



Figure 13.34: A Van de Graaff generator.



Figure 13.35: Lightning is a huge electrostatic discharge.



Did you know?

South Africa has one of the highest incidences of lightning strikes in the world.

Summary

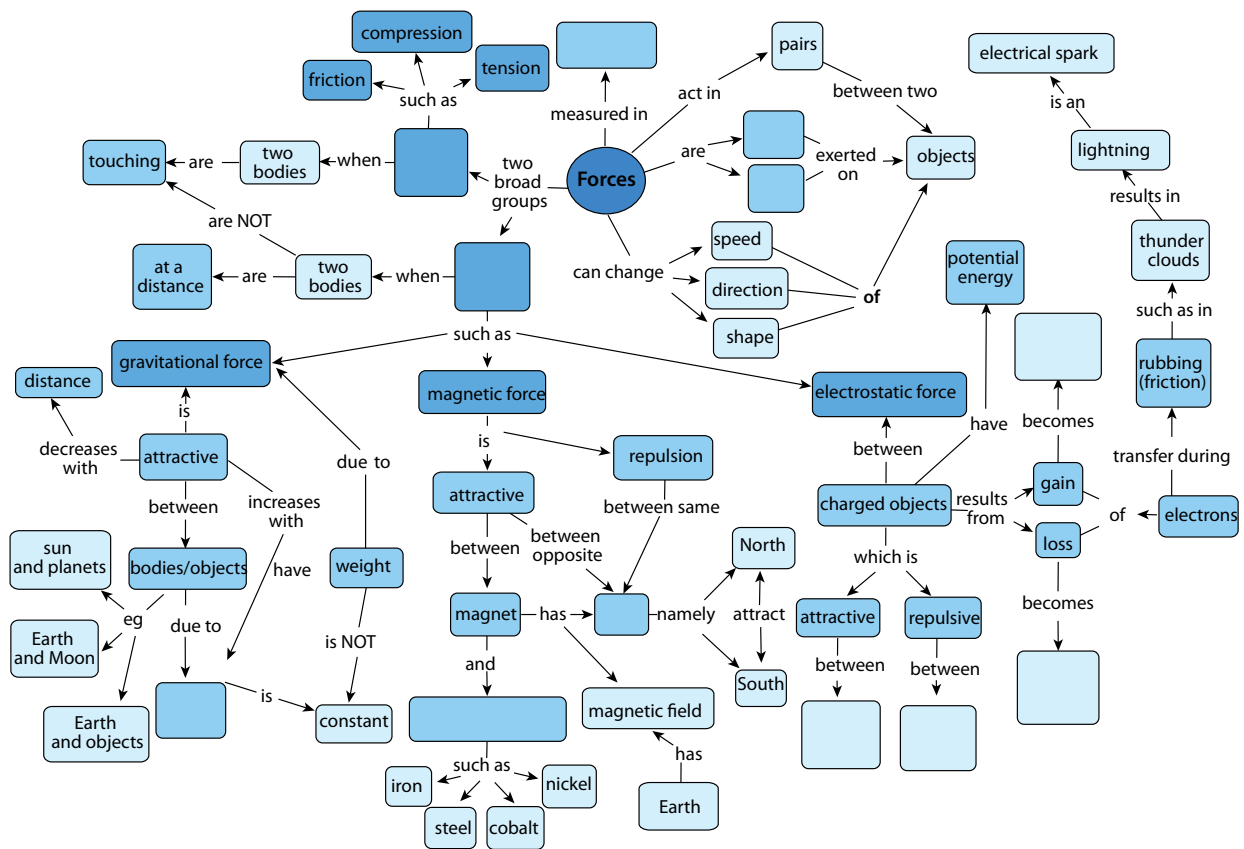
Key concepts

- A force is defined as a push or a pull on an object.
- Forces are measured in newtons (N).
- A force can change the shape, direction and motion of an object.
- Forces act in pairs. The force acting on the object is called the action and the force that the object exerts back in the opposite direction and equal in magnitude is the reaction.
- More than one force can act on an object. The net or resultant force is the sum of all the forces acting on the object.
- The forces acting on a body can be represented as a free-body diagram where the arrows indicate the direction and magnitude of the different forces.
- There are two main groups of forces: contact and non-contact (field) forces.
- Contact forces act when objects are in contact (touching) with each other. Friction, tension and compression are examples of contact forces.
- Friction is the force opposing motion between two surfaces as they rub against each other.
- Compression forces are two forces acting on one object, moving in opposite directions (towards each other) to compress or deform the object.
- Tension forces are two forces acting on one object, moving in opposite directions (away from each other) to stretch the object.
- Non-contact forces can act over a distance and objects do not have to be touching each other. Common examples of field forces are magnetic, electrostatic and gravitational forces.
- Non-contact forces are known as field forces. A field is a region in space where a certain object with certain properties will experience a force.
- Gravitational force is a force of attraction between two bodies due to their mass. The gravitational force increases with mass and decreases with the distance between the bodies.
- The weight of a body is the gravitational force exerted on an object by the Earth (or Moon or other planet). The weight will vary depending on where it is measured.
- The mass of an object is a measure of how much matter it contains. The mass stays constant no matter where it is determined.
- Weight is calculated as $W = m \times g$, where g is the gravitational acceleration. On Earth, $g = 9,8 \text{ m/s}^2$.
- A magnet is a material which has a strong magnetic field around it.
- Magnetic forces of attraction exist between a magnet and a magnetic substance, such as iron, steel, cobalt and nickel.
- A magnet has two poles, a north and south pole. Opposite poles attract each other and like poles repel each other.
- The Earth has a magnetic field around it. We can use compasses to tell direction as the needle is a magnet which points to magnetic North.
- When certain materials are rubbed together, the friction between them causes electrons to move from one material to the other. The objects then have an electrostatic charge, due to either the loss or gain of electrons.
- A charge is a fundamental property of matter. Electrons carry negative charges and protons carry positive charges.
- An object which has gained electrons will be negatively charged. An object which has lost electrons will be positively charged.

- There is an electrostatic force of attraction between objects with opposite charges, and repulsion between objects with like charges.
- Thunder clouds can become charged as the water and air particles rub against each other. A lightning strike occurs when there is a huge discharge between the thunderclouds and the ground.
- Lightning is dangerous and safety precautions should be adhered to during lightning storms.

Concept map

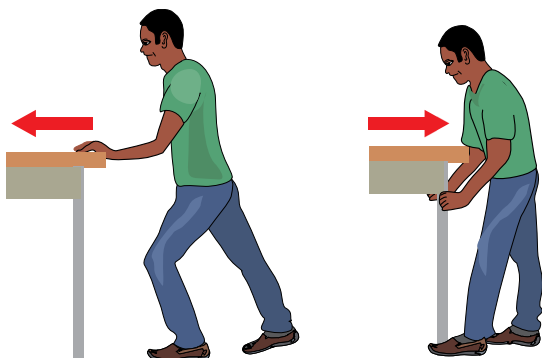
- Complete the concept map to summarise what you have learnt about forces in this unit. You can also use the space around the concept maps to add some of your own notes on these to help form more comprehensive summaries. This will help you prepare for exams when you need to revise everything from the year.



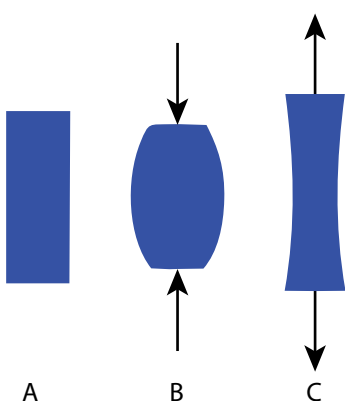
Revision

1. Give one term for each of the following descriptions. [5]
 - a) An influence that can deform a flexible object or change the motion of an object with mass.
 - b) A region in space where an object with certain properties will experience a force.
 - c) When the only force acting on an object is the force due to gravity.
 - d) The two opposite ends of a magnet.
 - e) A fundamental property of matter that comes in two types and is carried by protons and electrons.
2. Four possible answers are given for each of the following questions. There is only one correct answer. Write the letter in your exercise books. [6 × 2 = 12]
 - a) Which ONE of the following statements is FALSE?
 - A. In order for a non-moving object to start moving, a net force must act on that object.
 - B. Contact forces are strongest when the objects experiencing the force are touching.
 - C. Field forces act over distances, but they can also act when objects are touching.
 - D. Forces always act in pairs of equal strength, but these pairs act on different objects.
 - b) Which ONE of the following is NOT a field force?
 - A. Gravitational force
 - B. Frictional force
 - C. Electrostatic force
 - D. Magnetic force
 - c) The correct unit for gravitational force is:
 - A. the newton
 - B. the kilogram
 - C. the newton per kilogram
 - D. the kilogram per newton
 - d) Which ONE of the following substances is magnetic?
 - A. aluminium
 - B. copper
 - C. cobalt
 - D. tin
 - e) The electrostatic force between two charged objects is F. The distance between them is increased. How does the electrostatic force change?
 - A. It increases
 - B. It decreases
 - C. It remains the same
 - D. Not enough information has been provided.
 - f) An astronaut has a mass of 80 kg on Earth. Which ONE of the following regarding mass and weight of the astronaut on the Moon is correct?
 - A. The mass will be the same and the weight will also be the same.
 - B. The mass will be less and the weight will also be less.
 - C. The mass will be the same and the weight will be less.
 - D. The mass will be less and the weight will be the same.

3. Decide which of the following statements are True or False. If they are False, rewrite them to make them true. [5 × 2 = 10]
- A force cannot make a motionless object move.
 - A force can make a moving object change direction.
 - A force can change the shape of an object.
 - A tension force slows down or stops an object because of the surfaces rubbing against each other.
 - Lightning is an application of a magnetic force.
4. Have a look at the following images.



- Which situation shows a push and which situation shows a pull? [1]
 - The boy is pulling the desk with a force of 70 N. There is a frictional force of 20 N. Draw a free-body diagram to show these forces acting on the desk. [4]
 - What is the net force acting on the desk? [1]
5. A force of 50 N and 80 N act on a block. Calculate the net force acting on the block if:
- the forces are acting in the same direction [1]
 - the forces are acting in opposite directions. [1]
6. Write down three different effects that a force can have on an object. [3]
7. What is the following diagram illustrating? Explain your answer. [4]



8. Tabulate the distinction between mass and weight by making use of definitions of mass and weight and comparing the units they are measured in. [5]
9. Write down the formula that relates the weight of an object to its mass. Explain what each symbol represents. [4]
10. What TWO factors affect the gravitational force experienced between two objects? Explain the relationship. [2]

11. An astronaut performs an experiment to determine the relationship between mass and weight on different planets. He takes a scale and sets off in a spaceship and measures his own weight on different planets in the solar system. The following table indicates his results.

Planet	Weight (N)
Mercury	287
Venus	710
Earth	?
Mars	302

Planet	Weight (N)
Jupiter	2 076
Saturn	886
Uranus	854
Neptune	1 126

- a) Calculate the astronaut's weight on Earth if his mass is 80 kg. [3]
- b) Plot an appropriate graph of the astronaut's weight on different planets of the solar system. Your graph will need a suitable heading and labels for the axes. [8]
- c) On which planet is Kevin's weight the smallest? What does this tell you about the size of this planet in relation to the other planets? [2]
12. Draw a diagram to illustrate the magnetic field around a bar magnet. [3]
13. You do an experiment in class to investigate the bar magnets. You place two magnets next to each other on the table and place a sheet of paper over the magnets and sprinkle iron filings over the paper. You then turn one magnet around and do it again. You see the following patterns. What does each photo (A and B) show us? [2]

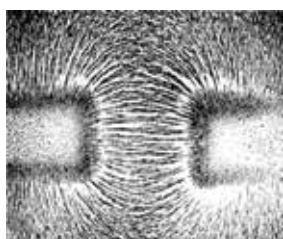


Photo A.

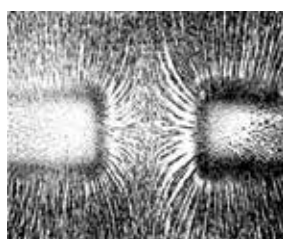


Photo B.

14. Complete the table by determining the overall charge on each object. Show your calculations. State whether the object is positively charged, negatively charged, or neutral, and why. [9]

Object	Overall charge	Why is it positive, negative or neutral?

- 15.** A balloon is rubbed against a jersey and the balloon picks up a negative charge.
- a)** Explain where this negative charge comes from. Make reference to both protons and electrons in your answer. [3]
 - b)** Name the type of force that the balloon and jersey will experience DURING rubbing. [1]
 - c)** Name the type of force that the balloon and jersey will experience AFTER rubbing. [1]
 - d)** Will the force referred to in (c) be attractive or repulsive? [1]
- 16.** What do you think these two girls are touching on the left of the photo? Explain your answer and what is happening to them. [3]



- 17.** Write a short paragraph to explain how lightning forms. [4]
- 18.** What is wrong with the following scene? Explain your answer. [2]



Total [95 marks]

14 Electric cells as energy systems



Key questions

- Where does an electrical circuit get its energy from?
- What is inside a battery?
- How can we build our own electric cells?
- How does an electric cell supply energy?

This term we will be investigating electricity and electrical circuits in more detail. We are going to pay attention to electric cells. You have already come across electric cells in previous grades when looking at electrical circuits. What is the circuit symbol for an electric cell? Draw it in your exercise books. Indicate the positive and negative terminal.

14.1 Electric cells

Keywords

- electric cell
- battery
- electrode
- electrolyte
- half cell
- salt bridge



Take note

Last term in Matter and Materials we looked at many different types of chemical reactions.

What is the source of energy in an electrical circuit?

We use electric cells to supply the energy needed for electrons to move around an electrical circuit. We often talk about batteries in electrical circuits or appliances. A battery is a group of two or more electric cells that are connected together. Where does the energy in a cell come from?

In Grade 8 we spoke about the transfer of energy within electrical systems. We can also call an electric cell a system. Write your own definition of a system in your exercise book.

The electric cell system works to generate electricity. We have spoken before about how electricity is generated using moving parts, such as in a power station, where the moving parts in a generator produce electricity. A cell does not have moving parts to generate electricity. An electric cell generates electricity from **chemical reactions**.

Did you know that we can create our own cell using a fruit? Let's try this out in the next activity.

ACTIVITY Fruit cell

Materials

- lemon (or potato)
- zinc strip or galvanised nail
- copper strip or coin
- LED bulb
- ammeter
- insulated copper conducting wires

Instructions

1. Gently squeeze the lemon to soften the fruit so that the juice is released inside. Be careful not to crush the lemon or break the peel. If you are using a potato, you do not need to squeeze it first.
2. Next, puncture the peel of the lemon with the two different nails (or strips) of **different** metals. If you are using a copper coin, then push it carefully into the lemon so that it breaks the skin.
3. Insert each nail slowly and carefully into either side of the lemon. Push the nails into the lemon so that they almost reach the centre of the lemon, but are not touching.
4. Attach one wire to the zinc (or iron) nail and the other wire to the copper nail or copper coin.
5. Connect the wires to the LED bulb and ammeter if you are using one as shown in the diagram. What do you notice?

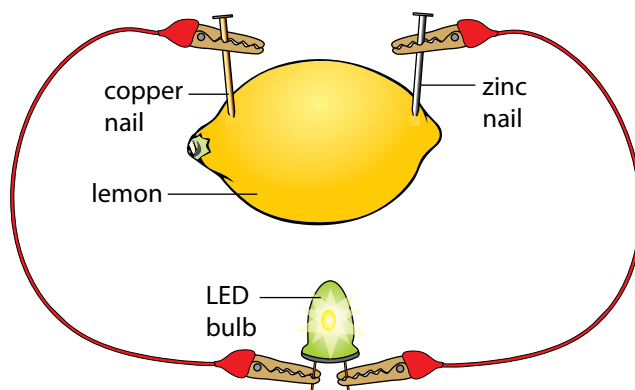


Figure 14.1

Questions

1. If your light bulb does not glow, connect your fruit cell with your partner's cell and reconnect the LED light bulb. Does it glow now? If not, connect several more fruit cells in a series until the LED bulb glows as shown in the diagram. How many cells did you connect in series to cause the LED to emit light?

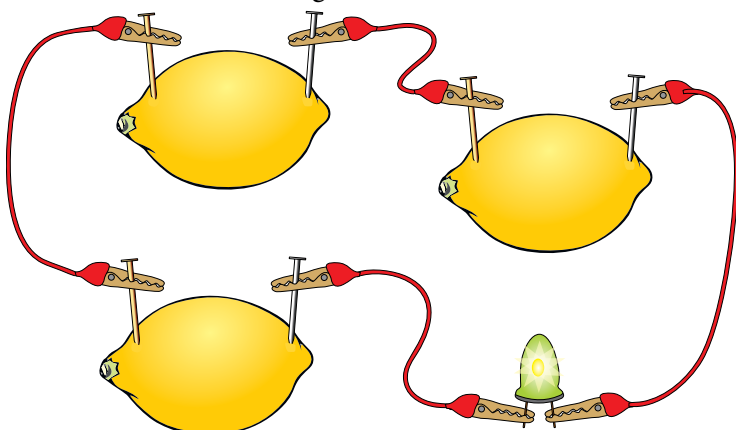


Figure 14.2

2. What do we call the cells connected together?
3. What happens if you replace the copper nail with another zinc nail, so that you have two electrodes of the **same** metal? Are you able to light up the LED light?
4. Experiment further by investigating the effects of pushing the nails deeper into the lemon and placing them at different positions in the lemon (closer together and further apart). Record some of your observations here.



Did you know?

The lemon battery is very similar to the first electrical battery invented by Alessandro Volta in 1800. He used salt water instead of lemon juice.

In the last activity we created a simple electric battery. A chemical reaction takes place inside the lemon which produces electricity. The components in the lemon battery are very similar to those used in a normal battery. The copper and zinc nails (or strips of metal) are called **electrodes**. The lemon juice acts as the **electrolyte**. Citrus fruits, such as lemons, are acidic, which helps their juice to conduct electricity.



Take note

An electrolyte is a special type of solution which is able to conduct electricity.

When the electrodes are connected in a circuit, a chemical reaction takes place within the electrolyte in the lemon, which causes electrons to move in the external circuit. This flow of charge is **electric current**. The chemical reaction causes a potential difference which causes the flow of electrons in the external circuit. This will occur only when the cell is connected in a circuit. Think of a normal battery that you may use in a torch. You can store this battery for a long time, and it will not go flat, as the chemical reactions take place only when it is connected in a circuit.

We are now going to build a more complex cell.

ACTIVITY Zinc-copper cell

Materials

- two 250 ml beakers
- copper sulphate solution
- zinc sulphate solution
- concentrated sodium sulphate or sodium chloride solution
- salt bridge made with a U tube (this can be made from a plastic tube which is bent) or filter paper soaked in the salt bridge solution
- cotton wool
- copper electrode
- zinc electrode
- insulated copper conducting wires with crocodile clips
- LED bulb
- ammeter



Take note

If you do not have a U-tube or plastic tubing, you can use strips of filter paper or a cloth soaked in the sodium sulfate solution with the ends dipped into each beaker.

Instructions

1. Pour about 200 ml of the zinc sulphate solution into a beaker and put the zinc electrode into the solution.
2. Pour about 200 ml of the copper sulphate solution into the second beaker and place the copper electrode into the solution.
3. Fill the U-tube with the sodium sulphate solution and seal the ends of the tubes with the cotton wool. This will stop the solution from flowing out when the U-tube is turned upside down.
4. Connect the zinc and copper electrodes to the ammeter. Does the ammeter record a reading?
5. Place the U-tube so that one end is in the copper sulphate solution and the other end is in the zinc sulphate solution, as shown in the diagram.

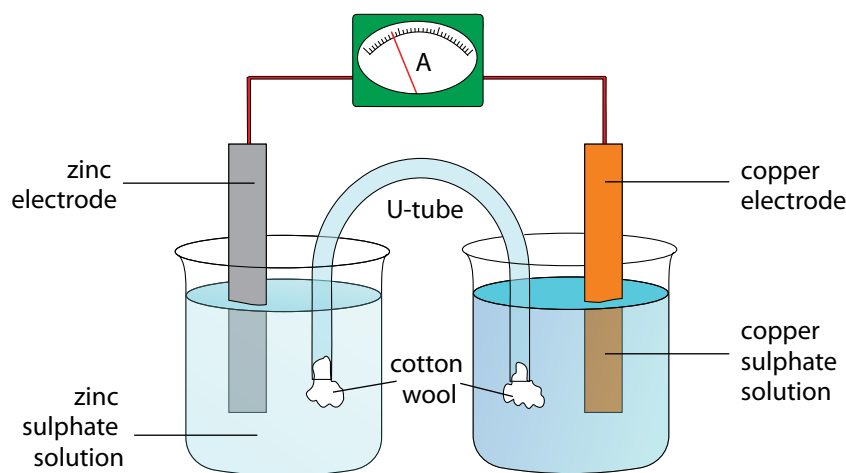


Figure 14.3

6. Remove the ammeter and insert the LED bulb in the circuit. Does it glow? If not, try connecting a few cells in series until the LED lights up.
7. Observe what is happening at the copper electrode and at the zinc electrode. Is there a reading on the ammeter?

Questions

1. What did you notice on the ammeter (or voltmeter) when you connected the circuit with the U-tube?
2. What does the ammeter reading tell us?

In the last activity, we demonstrated a zinc-copper cell. This is made up of a zinc half-cell and a copper half-cell. Together, they make up the whole cell. The purpose of the U-tube is to connect the two half cells. It is called the salt bridge.

How do we explain the chemical reactions taking place in the zinc-copper cell?

When a zinc sulphate solution containing a zinc plate is connected by a U-tube to a copper sulphate solution containing a copper plate, reactions occur in both solutions.

- At the zinc electrode, the zinc metal has gone into the zinc sulphate solution as zinc ions.
- At the copper electrode, copper ions from the solution have deposited onto the electrode as copper metal atoms.

In the zinc-copper cell the important thing to notice is that the chemical reactions that take place at the two electrodes cause an electric current to flow through the outer circuit. In this type of cell, **chemical energy** is converted to **electrical energy**.

As we have said before, an electric battery used in appliances such as a torch consists of two or more electric cells connected together. There are many different battery cell types such as zinc-carbon, nickel-cadmium and nickel-zinc batteries.



Did you know?

An ion is an atom or molecule in which the total number of electrons is not equal to the total number of protons. If there are fewer electrons than protons, this gives an atom a positive charge. If there are more electrons than protons, this gives an atom a negative charge.



Take note

You can also use a voltmeter to measure the potential difference across the cell. The voltmeter will replace the ammeter and LED light.



Did you know?

Rechargeable batteries are recharged by applying an electric current which reverses the chemical reactions that take place during their use in a circuit.

Summary

Key concepts

- An electric cell is a system in which chemical reactions take place to convert chemical energy into electrical energy.
- An acidic fruit can be used to construct a simple cell. The lemon juice acts as the electrolyte.
- An electric cell can be made using two beakers with an electrolyte and electrode in each. The electrolyte solutions in each half-cell are connected by a salt bridge.
- When the electrodes are connected to an external circuit, chemical reactions will take place in each of the beakers, causing a current in the external circuit.
- A battery is a group of cells which are connected together.
- There are many different types of cells, such as zinc-carbon, nickel-cadmium and nickel-zinc batteries.

Concept map

This was a short unit on electrical cells, demonstrating how we can make cells. Draw your own concept map summarising what was covered in this unit. You can refer back to previous units and concept maps from this Learner's Book when thinking about how to construct this concept map.

Revision

1. Write a short paragraph to describe, in your own words, what an electrical cell is and how it causes a current in an external circuit. [3]
2. What is the difference between a cell and a battery? [2]
3. You made an electrical cell from a lemon. How could you generate enough energy from lemons in order to make a light bulb glow? [2]
4. How would you test whether or not a battery or cell is producing energy? [2]
5. Draw a diagram to show how to set up a zinc-copper cell. Include an ammeter in the external circuit. You must use the following labels: zinc electrode, copper electrode, salt bridge/U-tube, zinc-sulphate solution, copper sulphate solution. [8]

Total [17 marks]

15 Resistance



Key questions

- What is resistance?
- What do we use resistors for?
- Does length affect resistance?
- Does temperature affect resistance?
- Does the type of resistor material affect resistance?
- Does the thickness of a resistor affect the resistance?

Keywords

- resistance
- resistor
- electric current
- electric charge
- delocalised
- conductor

15.1 What is resistance?

Think about your school break time. All of the learners are outside on the field, sitting in groups and relaxing. Some of you will be moving around the field from group to group as you greet your friends. The school bell rings, signalling the end of break. You all get up and start moving toward the school building. You are all able to move easily because there is a great deal of space, but what happens as you enter the corridor of the school building?

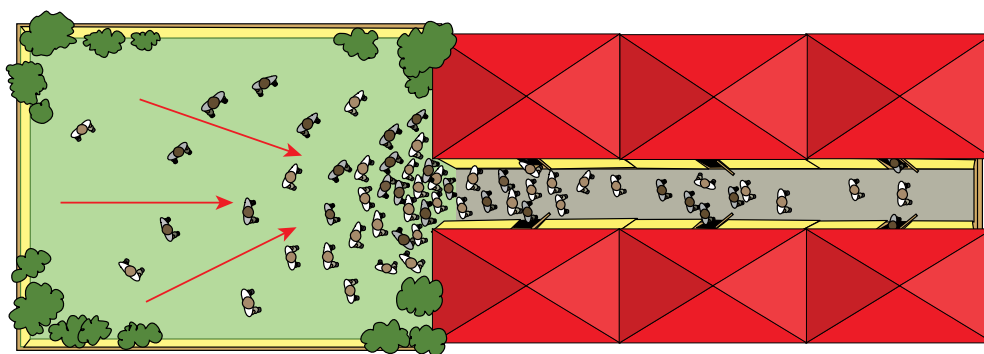


Figure 15.1

Everyone now has to fit through a narrow corridor. Everyone is trying to get to class and so some learners will bump into other learners. As you try to enter your classroom it becomes even more difficult because the doorway is even narrower than the corridor and so only one or two learners can pass through at a time.

The movement of the learners is very similar to the movement of electrons in an electrical conductor. The field offers a very low resistance to the movement of the learners and so the learners are able to move freely. The corridor has a higher resistance to the movement of the learners because fewer learners can now pass through the corridor than through the field. The classroom doorway offers the highest resistance as it only allows a few learners through at a time.

Resistance in an electrical circuit opposes the passage of electrons. The unit of measurement for resistance is the **ohm**, with the symbol Ω .

Do you remember what an electrical conductor is? Write your own definition.

All electrical conductors have some resistance. Some conducting materials have a particular resistance and are used to add electrical resistance to a circuit. An electrical component which adds resistance to a circuit is called a **resistor**.

Resistors are electrical components and have a symbol to represent them in an electric circuit diagram. Do you remember the symbol from Grade 8? Draw it in your exercise books.

On a microscopic level, electrons moving through the conductor collide (or interact) with the particles of which the conductor (metal) is made. When they collide, they transfer kinetic energy. This leads to resistance. The transferred energy causes the resistor to heat up. You can feel this directly if you touch a cellphone charger when you are charging a cell phone – the charger gets warm because its circuits have some resistors in them.



Figure 15.2: Different types of resistors are used to add resistance to an electric circuit.



Take note

Can you see that there are different coloured bands on the resistors? This isn't just to make them look pleasing to the eye. The coloured bands are actually a code that tells us the strength of the resistance of the resistor.

15.2 Uses of resistors

Resistors can be used to control the current in a circuit. Think back to some of the work that you did in Gr 8. If you increase the resistance in a circuit, what happens to the current? Explain your answer.

Another way in which we can use resistors is to provide useful energy transfers. Do you remember looking at energy transfers in a system in Grade 8? The input energy enters the system and then provides an output energy. Some of the output energy is useful to us, and some is wasted energy. For example, a resistor can be used to transfer electrical energy into light (light bulb) or into heat (kettle element). Energy is wasted as it is lost to the surroundings. Resistors are used to provide useful energy transfers.

ACTIVITY Useful resistance

Why do we want to resist the movement of electrons? Resistors can be extremely useful. Think about a kettle. If you look inside you will see a large metal coil.

This metal coil is the heating element. If you plug in and switch on the kettle, the element heats up and heats the water. The element is a large resistor. When the electrons move through the resistor, they release a lot of energy in overcoming the resistance. This energy is transferred to the water in the form of heat. This transfer of energy is useful to us as the thermal energy is used to boil water in the kettle.



Figure 15.3: Looking inside a kettle.

Keywords

- LED
- motor
- variable resistance
- rheostat
- Sankey diagram
- input energy
- output energy



Did you know?

The ohm gets its name from the German physicist Georg Simon Ohm, who noticed that the potential difference across a conductor and the electric current are directly proportional (Ohm's Law).

1. What is the input energy in this system?
2. What is the useful output energy?

Look at the photograph of a light bulb on the left. Can you see there is a small coiled wire in the glass bulb? This is called the filament. The filament is made from tungsten wire. This is an element with high resistance.



Take note

Incandescent means emitting light as a result of being heated.



Figure 15.4: The tungsten filament glowing brightly.



Figure 15.5: An incandescent light bulb.



Did you know?

The inventor Thomas Edison, experimented with thousands of different resistor materials until he eventually found the right material that allowed the bulb to glow for over 1 500 hours.

When the electrons move through the filament they experience high resistance. This means that they transfer a lot of their energy to the filament when they pass through. Describe the energy transfer taking place.

1. What is the useful energy output and what is the wasted energy output in this light bulb?
2. The filament is tightly coiled. Why do you think this is? Discuss this with your class and teacher. Look at the following photo of a toaster.
3. Can you see the glowing filament inside? Why does the element glow?
4. What is the useful output energy in this system?
5. What is the wasted output energy in this system?



Figure 15.6: An electric toaster.

Rheostats are another form of resistor which are commonly used. A **rheostat** is a device which is able to offer a **variable** resistance. Rheostats are used in electric circuits where you want to adjust the current, for example in sound equipment to adjust the volume, in dimmer switches for lights, and in controlling the speed of **motors**. Let's look at how rheostats can be used in a circuit.



Figure 15.7:
An example of a rheostat.

ACTIVITY Make your own rheostat

Materials

- graphite rod or graphite pencil
- torch light bulb
- battery (AA)
- insulated copper conducting wires with crocodile clips
- ammeter

Instructions

Set up a circuit as in the diagram below with the battery, ammeter, light bulb and graphite rod connected in series. Use crocodile clips to attach the wires to each end of the graphite rod.

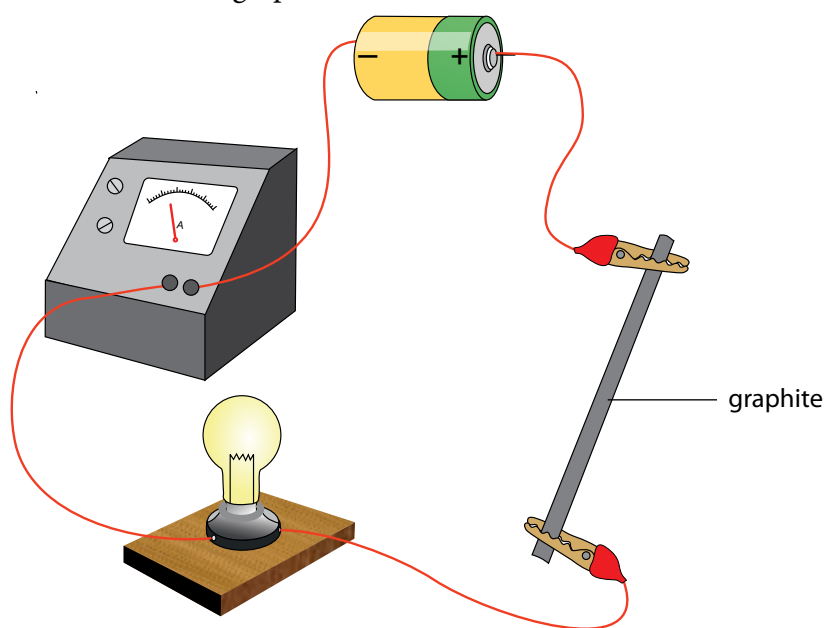


Figure 15.8

1. Does current flow through the circuit? How do you know?
2. The crocodile clips are connected on either end of the graphite rod. Predict what you think would happen if you moved the crocodile clips closer towards the centre of the piece of graphite.
3. Move the crocodile clips closer towards the centre of the graphite rod. What do you observe?
4. How do you think the length of the graphite connected to the circuit has affected the current strength?
5. Draw a circuit diagram to represent this set-up.

The graphite rod was behaving as a rheostat. The resistance of the graphite rod was changed by changing the length connected to the circuit. A dimmer switch often has a dial which can be turned. Turning the dial increases the resistance of the circuit and makes the light dim. Why do you think this happens?



Did you know?

The first electric light was made in 1800 by a man called Humphry Davy. He invented an electric battery, to which he connected wires and a piece of carbon. Being a resistor, the carbon glowed and produced light.

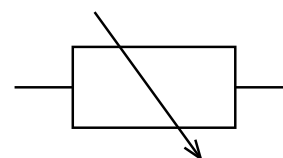


Figure 15.9: The symbol for a variable resistor.



Take note

A diode is an electrical component that has a very low resistance to current flow in one direction, and high resistance in the other direction. Therefore, current can move only in one direction.



Figure 15.10: A small LED light.

Turning the dial in the opposite direction causes the resistance to decrease and so the light burns brighter. Turning the dial changes the resistance of the rheostat in the switch.

Another device which demonstrates the useful application of resistance is in an **LED**. LED stands for light emitting diode.

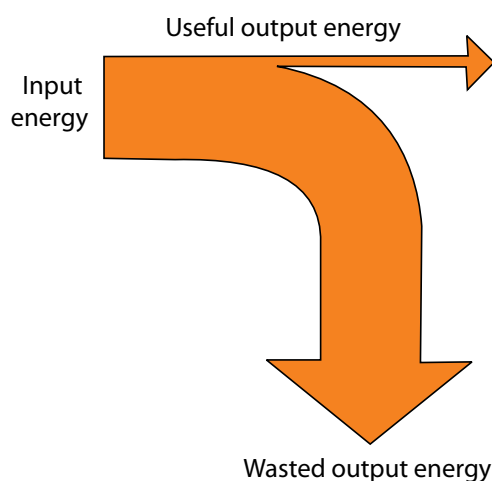
An LED is a diode because it only allows current to pass through it in one direction. This means that it has to be put into a circuit in a very specific way. LEDs are very sensitive to high currents so when they are connected in a circuit, they need to be protected by a large resistor. The resistor is used to control the current which is allowed to travel through the LED. This is another useful application of resistance.

Many households are choosing to replace incandescent light bulbs with LEDs.

Are LEDs a more efficient form of lighting?

ACTIVITY Comparing an LED to a filament light bulb

We can use a **Sankey diagram** to show how the energy is transferred in a system. This gives us a picture of what is happening and shows the input energy and how the output energy is made up of useful energy (arrow at the top) and wasted energy (arrow going to the bottom). Have a look at the following general example.



The width of the arrows tell us something in these diagrams.

The input energy is the width of the original arrow. The width of both the output energy arrows (useful and wasted) add up to the width of the input arrow. Why do you think this is so?

Figure 15.11

Sankey diagrams are drawn to scale so that the width of the arrows gives us a visual idea of how much energy is useful and how much is wasted.

Questions

1. The Sankey diagram for an LED is shown below.

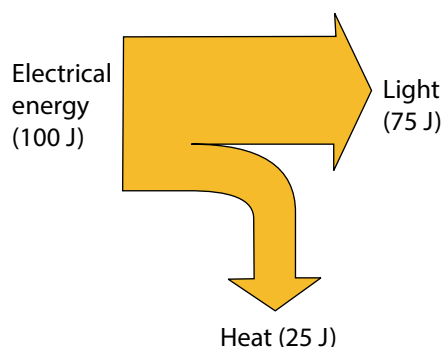


Figure 15.12



Did you know?

Sankey diagrams are named after the Irish Captain Matthew Sankey, who first used this type of diagram in 1898 in a publication on the energy efficiency of a steam engine.

- a) Describe the energy transfer which takes place in an LED, based on the given Sankey diagram.
 - b) Is the LED efficient or inefficient? Explain your answer.
2. The Sankey diagram for an incandescent light bulb is shown below:

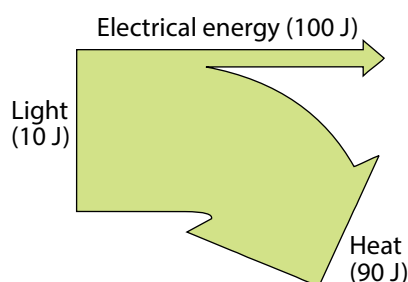


Figure 15.13

- a) Explain the energy transfers in the incandescent light bulb.
 - b) Is the incandescent light bulb efficient? Explain your answer.
3. If you are trying to reduce your electricity consumption in order to save money, which light source would you choose? Why?

When we built our own rheostat, we were able to vary the resistance by changing the length of the graphite rod. This tells us that the length of the rod affected the amount of resistance. Let's look at what other factors which affect the resistance of a conductor.

15.3 Factors that affect the resistance of a conductor

What determines the resistance of a component? Let's investigate some of the factors. There are four different factors which affect resistance:

- The type of material of which the conductor is made
- The length of the conductor
- The thickness of the conductor
- The temperature of the conductor.

Type of material

Conductors can be made of different materials. Do different materials have different resistances?



Take note

We can say that resistance and current in an electric circuit are inversely proportional, because as the one increases, the other decreases, and vice versa.



Investigation

How does the material of the conductor affect the resistance?

How can we measure resistance? Do you remember that in a series circuit, if we increase the resistance, then the strength of the current decreases? This means that we can use the strength of the current in the circuit as an indication of the amount of resistance in the circuit.

Aim

To determine whether different types of conducting materials have different resistances.

Hypothesis

Write a hypothesis for this investigation.

Variables

- Which variables would we need to keep constant in an investigation such as this?
- Which variable is the independent variable?
- Which variable is the dependent variable?

Materials and apparatus

- three 1,5 V cells
- insulated, conducting wires with crocodile clips
- conductors of different materials to test
- ammeter
- light bulb

Method

1. Set up a circuit with the three cells, ammeter and light bulb connected in series.
2. Test each of the conductors by adding each to the circuit individually. Use crocodile clips to connect each conductor to the circuit, as shown below.

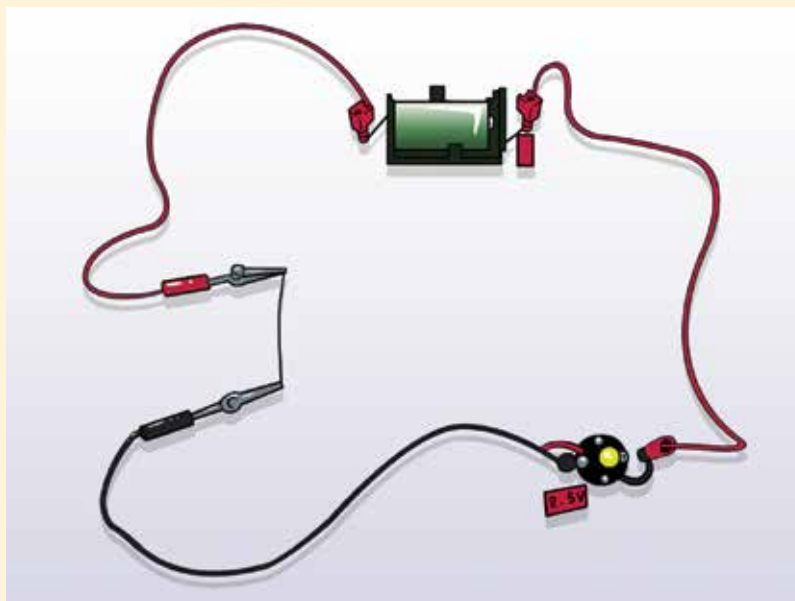


Figure 15.14: A similar setup showing a light bulb, one cell, and a piece of copper wire connected in series.

3. Read the ammeter and record the reading for each test material.
4. Draw a bar graph to show your results.

Results

1. Draw a circuit diagram of the setup.
2. Draw a table showing your results.
3. Draw a bar graph of your results in your exercise books.

Analysis and evaluation

1. Which material offered the most resistance in the electric circuit?
How do you know this?
2. Which material offered the least resistance in the electric circuit?
How do you know this?
3. Are there any potential problems with the way in which this investigation was set up, or are there any ways in which you could have improved the design?

Conclusions

1. What conclusion can you reach from this investigation?
2. Why must the different conductors have the same length and thickness?

Thickness of the conductor

When we investigate the thickness of a conductor, we are looking at the cross-sectional area of the wire, called the gauge. This is shown in the diagram on the right.

Do you think the thickness of a wire will affect the resistance? Let's do an investigation to find out.

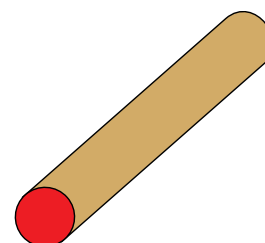


Figure 15.15: The cross-sectional area of a wire is indicated by the red circle.



Investigation

How does the thickness of the conductor affect the resistance?

Aim

To determine whether the thickness of the conductor will affect the resistance.

Hypothesis

Write a hypothesis for this investigation.

Variables

1. Which variables would we need to keep constant in an investigation such as this?
2. Which variable is the independent variable?
3. Which variable is the dependent variable?

Materials and apparatus

- three 1,5 V cells
- insulated, conducting wires with crocodile clips
- conductors of different thickness
- ammeter
- light bulb

Method

1. Assess the lengths of wire that you have and arrange them in order from thickest to thinnest. Label the thickest wire as 1, the next thickest as 2, and so on, so that you can easily record the results.
2. Set up a circuit as in the previous investigation with the three cells, ammeter and light bulb connected in series.
3. Test each of the different wires by adding each to the circuit in turn. Use the conducting wires with crocodile clips attached at the ends to join each conductor to the circuit.
4. Read the ammeter and record the reading for each wire.

Results

Draw a table showing your results.

Analysis and evaluation

1. Which thickness of wire offered the most resistance in the electric circuit?
2. Which thickness of wire offered the least resistance in the electric circuit?
3. Are there any potential problems with the way in which this investigation was set up, or are there any ways in which you could have improved the design?

Conclusions

1. What conclusion can you reach from this investigation?
2. Can you accept or reject your hypothesis?

The thinner the wire, the more resistance it offers. Thicker wires offer less resistance. This is easy to understand if you think back to the example of all the learners filing back into the classrooms after break. If the corridor is narrow (or thin) then it is more difficult for all the learners to move through. A very wide corridor would be easier to move through as it would offer less resistance. This is the same in a conducting wire. A thinner wire is more difficult for electrons to move through than a thicker wire.

Length of the conductor

In each of the previous investigations, we have used the same length for each conductor. It was a controlled variable. Let's now investigate how the length of a conductor affects the resistance.



Investigation

How does the length of a conductor affect the resistance of the conductor?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- piece of resistance wire (110cm) long
- ammeter
- two 1,5 V cells
- metre ruler
- tape
- insulated copper conducting wires

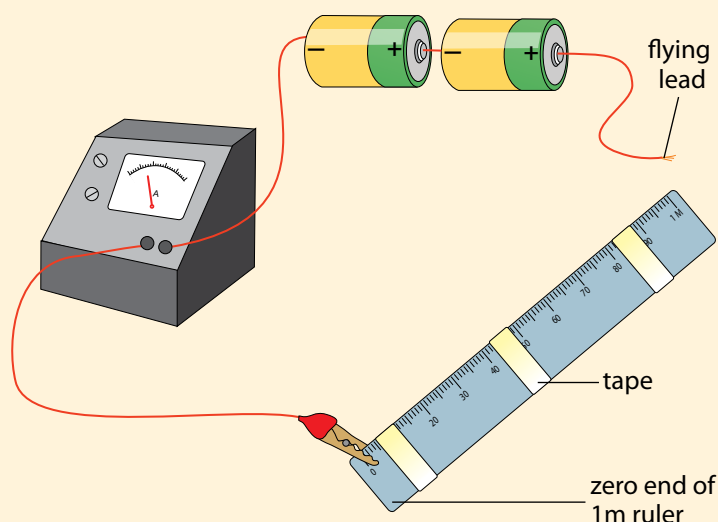


Figure 15.16

Variables

1. Which variables would we need to keep constant in an investigation such as this?
2. Which variable is the independent variable?
3. Which variable is the dependent variable?

Method

1. Tape the resistance wire to the metre ruler. Make sure the wire is stretched flat and that the numbers on the ruler are still visible.
2. Assemble a circuit according to the diagram on the next page.
3. Use the flying lead and touch it to the resistance wire at the 1 m mark. Record the ammeter reading.
4. Use the flying lead and touch it to the resistance wire at the 0,9 m mark. Record the ammeter reading.
5. Move the flying lead in 10 cm intervals until you have 10 readings. Record the ammeter reading each time.

Results

Copy the table and record your results in your exercise books..

Draw a graph to show the relationship between the length of the resistor and the ammeter readings.

Length of wire (m)	Ammeter reading (A)
1,0	
0,9	
0,8	
0,7	
0,6	
0,5	
0,4	
0,3	
0,2	
0,1	

Conclusions

1. Look at your table and graph. What conclusion can you draw?
2. What is causing the decrease in current strength?
3. What can you conclude about the relationship between the length of the resistor and the resistance of the resistor?

The length of the resistor affects how much resistance it offers to the circuit.

The longer the resistor, the more resistance it has. The shorter the resistor, the less resistance it has.



Figure 15.17: A close-up photograph of the tungsten filament in an incandescent light bulb.

Longer wires have more resistance than shorter wires. Let's take a close-up look at the filament of an incandescent light bulb.

You can see that the filament is made up of coils of tungsten wrapped up tightly. We want to fit a very long wire into a small space. The electrons have to travel through this very long, high-resistance wire. How is this more beneficial compared to having a shorter wire? Discuss this with your class.

Temperature of the conductor

The last factor which affects resistance is the temperature of the conductor. The hotter a resistor becomes the more resistance it has. The atoms of the conductor vibrate much more quickly when they are hot because of the increase in kinetic energy. This makes it more difficult for the electric current to move through. Cold resistors offer less resistance to the circuit.



Take note

We are not going to investigate this factor as it is difficult to control the temperature of a wire in an investigation.

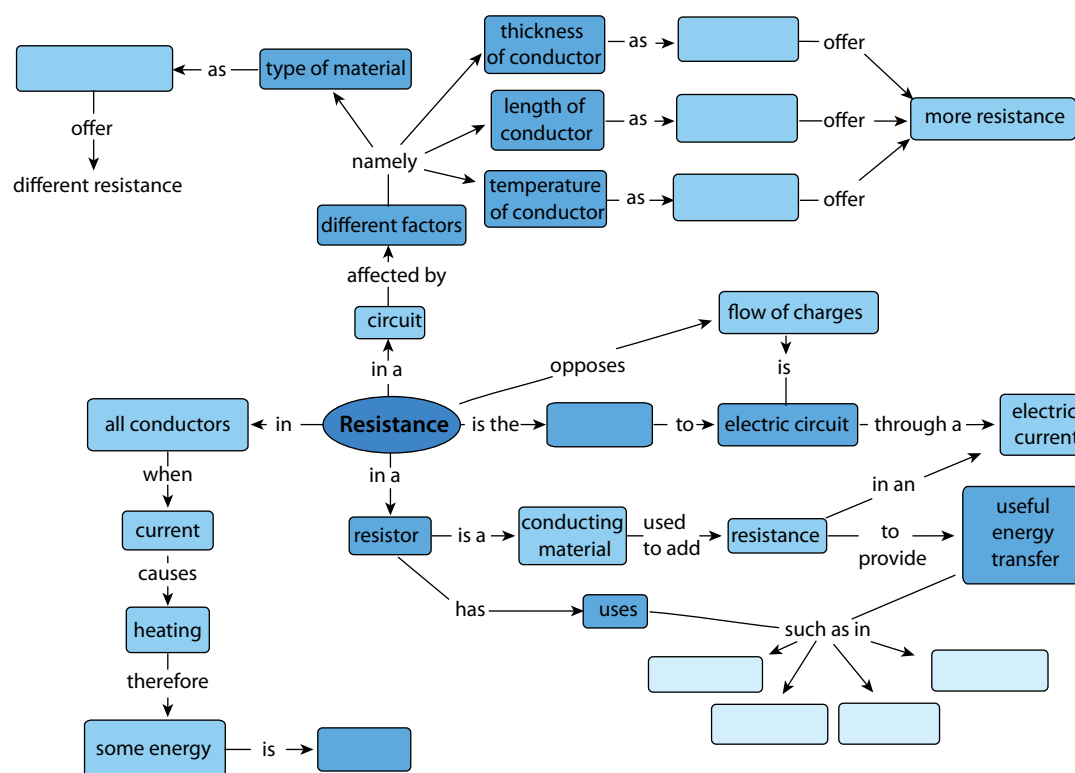
Summary

Key concepts

- Resistance is the opposition to electric current in a circuit.
- A resistor is an electrical component used to add resistance to an electrical circuit.
- Resistance can be useful. For example, the filament in a light bulb and a toaster have a high resistance.
- There are four factors which influence the amount of resistance of a conductor: type of material, length, thickness and temperature.
- Different materials will offer different amounts of resistance.
- Longer resistors will offer more resistance than shorter resistors.
- Thicker resistors offer less resistance than thinner resistors.
- Hot resistors offer more resistance than cold resistors.

Concept map

Complete the following concept map to summarise what you have learnt about resistance in this unit. For example, when looking at the factors that affect resistance, you need to describe the relationship by completing the sentences.



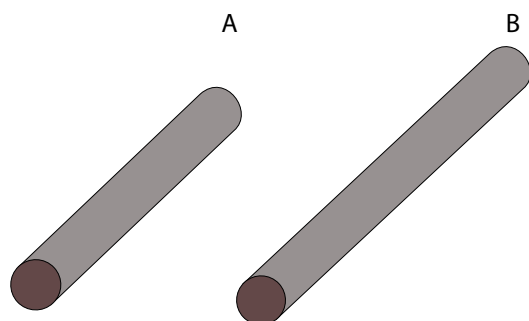
Revision

1. There are many useful applications of resistance. Give two examples of appliances which require large resistances in order to function. [2]
2. Look at the following photograph of an electric toaster.



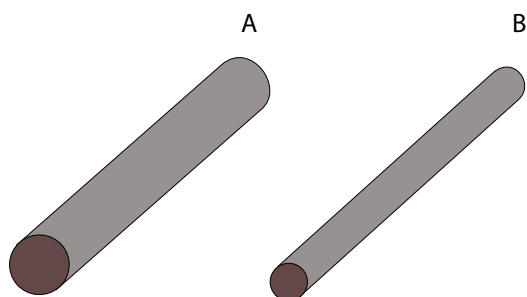
An electric toaster.

- a) Do you think the element in the toaster has a low or high resistance? Explain your answer. [2]
 - b) Explain the energy transfers which take place within the heating element of the toaster. [3]
 - c) Is there wasted energy in this system? If so, what is it, and why can we consider it 'wasted energy'? [2]
3. List the factors which affect the amount of resistance in a resistor. [4]
 4. The pictures below show two pieces of the same type of metal wire with the same diameter.



Which piece has the higher resistance? Explain why. [2]

5. The pictures below show cross-sections of two pieces of the same type of metal wire. The pieces are the same length but have different diameters.



Which piece has the lower resistance? Explain why.

[2]

6. Look at the image of a stove-top heating element. The heating element offers a large resistance to the flow of electric current.



- a) Why is the heating element in the shape of a coil? [2]
- b) What is the input energy in this system? [1]
- c) What is the output energy? [2]
- d) Is all of the energy transferred to the heating element useful? [2]

Total [24 marks]

16 Series and parallel circuits

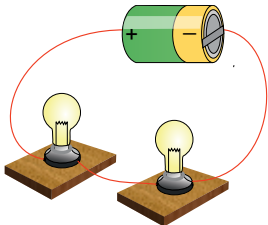


Key questions

- What happens when we add cells in series or parallel?
- What happens when we add resistors in series or parallel?
- What is potential difference?
- How do we connect ammeters and voltmeters in circuits?
- What is electric current?

16.1 Series circuits

A series circuit provides only one path for electric current to move through the circuit.

Series circuit	Draw a circuit diagram for this circuit in your exercise books
	

How many cells and how many resistors are in the above circuit?

What happens when more cells or resistors are added into a series circuit?
We are going to investigate the effects on the current and the potential difference in series circuits.

What is potential difference?

Potential difference is the difference in potential energy per charge between two different points in an electric circuit. Here is a simpler explanation: potential difference tells us how much energy per charge the electrons are losing when they pass through a resistor or how much they are gaining when they pass through the cell or battery. The electrons 'lose' energy because they have transferred it to the resistor in the form of heat, light or sound. Electrons 'gain' energy when they pass through the cell or battery because of the chemical energy from the battery being transferred to the electrons.

The potential difference is measured by a **voltmeter**. The unit of potential difference is the **volt**.

The voltmeter has a very high internal resistance and must be connected in parallel with the component you are measuring. You therefore need to connect it to two different points (which are usually before and after a resistor). The voltmeter calculates the potential difference between those two points. An example illustrating the placement of a voltmeter at two points (A and B) is indicated in the diagram.

Keywords

- potential difference
- voltage
- voltmeter
- electric current strength
- ammeter



Take note

Remember that a battery is a group of cells connected together.



Take note

You may have learnt about potential difference in Technology already, but this is the first time we are investigating it in Natural Sciences.



Take note

People often use the term 'voltage' to describe the potential difference. This is based on the fact that we measure potential difference in volts. Voltage is not the scientific term, and potential difference is the more correct term of use.



Figure 16.1: A voltmeter.

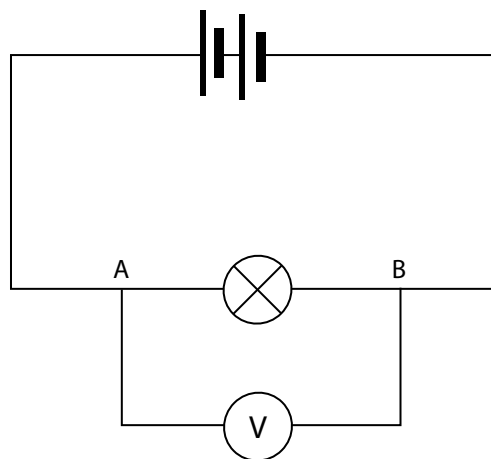


Figure 16.2: This voltmeter is connected in parallel with the bulb and measures the potential difference between points A and B.

Alternatively, a multimeter can be used to measure the current strength across a resistor, and the potential difference across the resistor. A multimeter is a device that can be used to measure the potential difference across a resistor, and the current strength through the resistor.



Did you know?

The volt is named in honour of the Italian physicist Alessandro Volta (1745-1827), who invented the voltaic pile, the first electric battery which could deliver a continuous electric current.

Activity

1. If you connect the voltmeter in series, there will not be two different points as you connect it to the same wire with the same potential. What do you think would happen if you connected the voltmeter in series in the circuit? Explain your answer.
2. What instrument do we use to measure the current strength in a circuit?
3. How do we connect the device named in number 2 in a circuit? Explain why this is so.
4. Draw a circuit diagram in your exercise books to illustrate an ammeter, a light bulb and a cell will should be connected in a circuit.

Electric Current

Electric current is a flow of charges in a circuit. An atom is made up of electrons arranged in the outer space around a central nucleus which comprises of protons and neutrons. The nucleus of the atom is held together by strong electrostatic forces, such that it's almost impossible to move the protons. Therefore, it is the flow of electrons that gives rise to the electric current.

An **ammeter** (shown in Figure 16.3) is an instrument used to measure the strength of



Figure 16.3: an ammeter

the electric current at any point in the electric circuit. Because of its relatively lower resistance, an ammeter allows the charges to flow through with ease. An ammeter should, therefore, always be connected in series with the components of the circuit.

The measuring SI unit of electric current strength is an **ampere** (denoted by A).

Conditions for the electric current to flow in a circuit

- For any circuit to function, that for the electric current to flow, there has to be a **source of energy** in a form of a **cell** or a **battery**.
- There has to be a **complete, cyclic pathway** that connects the components of the circuit to the energy source.

The illustration below summarizes these two conditions through a virtually performed experiment.



Figure16.4: Source, switch open (no complete pathway), no current flows

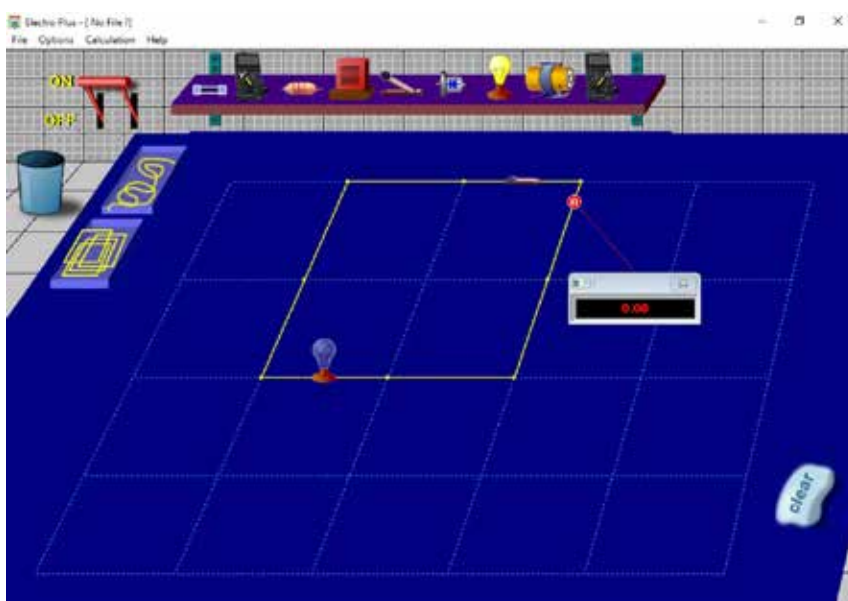


Figure 16.5: No source, switch closed, complete pathway, no current

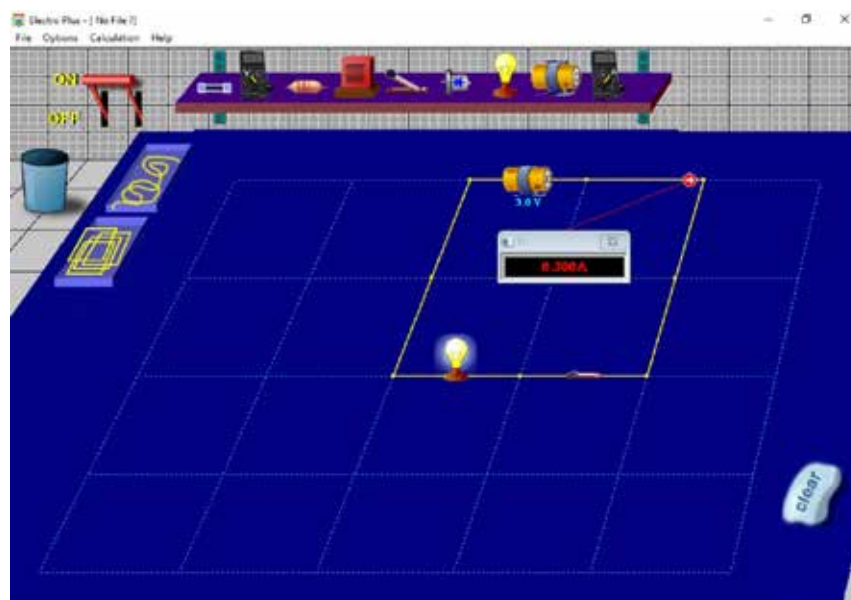


Figure 16.6: Source, switch closed, complete pathway, current flows

Measuring electric current strength in a circuit

We have already discussed that an ammeter is used to measure the electric current strength in the circuit. We have further discussed that an ammeter should be connected in series with the components of the circuit. We even looked at the conditions for the electric current to flow in the electric circuit.

In the next practical activity, we are going to measure the strength current flowing in the circuit.



Practical Activity

Measuring the electric current strength

Aim

Measuring the electric current strength

Apparatus

- 2 x 1,5 V cells
- 1 x Bulb
- Conducting wires
- Switch
- Ammeter

Precautions

- Always keep the switch open if you're not making any observations or taking any measurements. This saves the battery life of the cells.
- Ask for the teacher's assistance to connect the ammeter.

Procedure

1. Construct a circuit as shown in the picture below, but keeping the switch open.

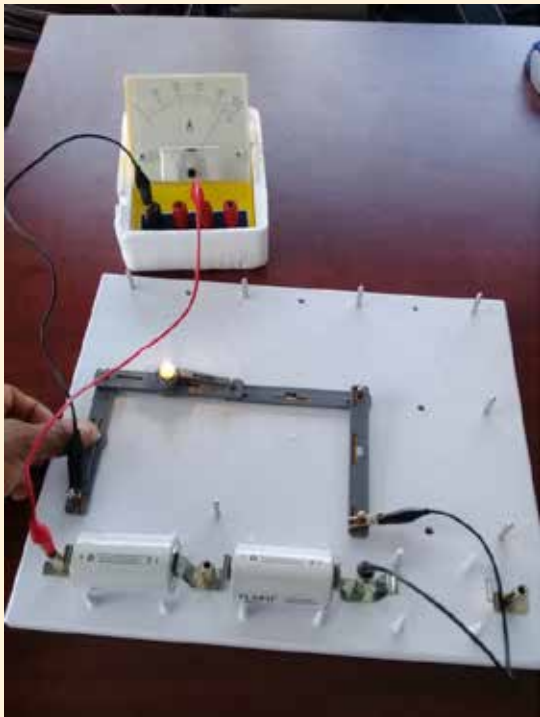


Figure 16.7

2. Observe the bulb and the ammeter reading with the switch open.
3. Closed the switch and observe the bulb lights.
4. Record the ammeter reading and the bulb observation in the table of results below.

Results

	Switch open	Switch closed
Does the bulb light up?		
Ammeter Reading		

Discussion and Conclusion

From the observations you have made when the switch is open and when it is closed, what can you conclude about flow of current in a circuit?

Questions

1. Draw a circuit diagram representing the circuit you have built.
2. What is electric current?
3. What is the name of measuring SI unit of electric current?
4. What instrument is used to measure electric current?
5. How should this instrument be connected in a circuit?
6. List two conditions necessary for the electric current to flow in an electric circuit.

The following photo shows a voltmeter connected in parallel and an ammeter connected in series.

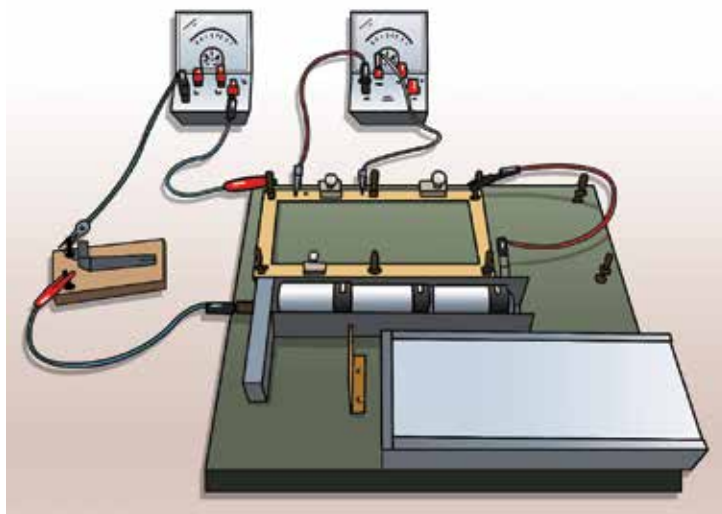


Figure 16.8

Cells in series

When cells are connected together they are called a battery. What happens when we put more than one cell in a circuit? Let's investigate what happens when we add cells, in series, to a circuit.



Take note

If you are using a light bulb and not an ammeter to see the effect, then take note of the brightness of the bulb as you add more cells in series.



Investigation

What is the effect of the number of cells connected in series on current and potential difference?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- three 1,5 V cells
- insulated copper conducting wires with crocodile clips
- ammeter
- voltmeter
- resistor or light bulb

Precautions

- If you using bulbs instead of a resistor, make sure that you use two bulbs. Bulbs have a lower resistance and may blow up as you add more cells in series.
- Always use a switch to control the flow of electric current when you're not taking any readings.

Method

1. Construct a series circuit with 1 cell, a resistor and the ammeter in series.

2. Connect the voltmeter in parallel with the cell as shown in the following circuit diagram.

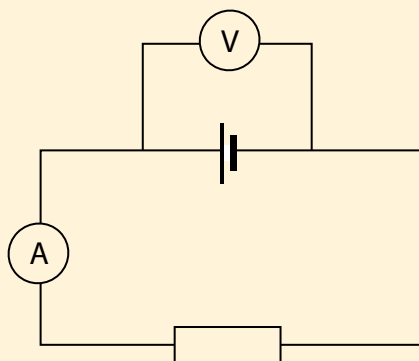


Figure 16.9

3. Record the readings on the ammeter and voltmeter in your exercise books.
4. Add a second cell in series with the first cell.

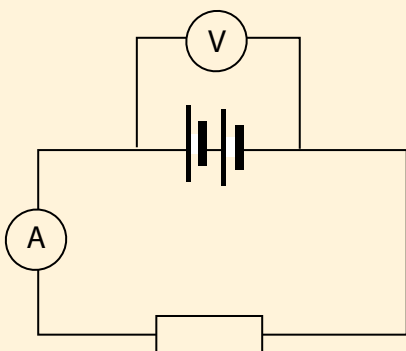


Figure 16.10

5. Record the new readings on the ammeter and voltmeter in your exercise books.
6. Add the third cell in series with the other two cells.

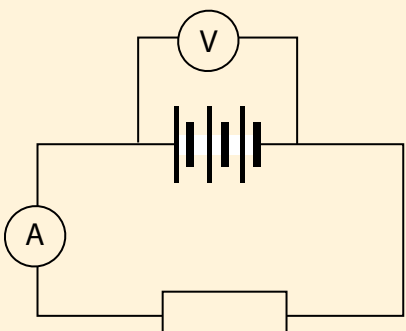


Figure 16.11

7. Record the new readings on the ammeter and voltmeter in your exercise books.
8. Draw a graph of your results.



Take note

Remember that the units of measurement are noted in the heading for a column in a table and not written in each cell in the table.



Take note

A line of best fit (or 'trend' line) is a straight line that best represents the data on a scatter plot. To draw a line of best fit, balance the number of points above the line with the number of points below the line. This line may pass through some of the points, none of the points, or all of the points.

Results

Copy and complete the following table in your exercise books:

Number of cells	Ammeter reading (A)	Voltmeter reading (V)
1		
2		
3		

Use your table to draw two line graphs on the same set of axes. One graph should be the number of cells against the current (ammeter reading) and the other graph should be the number of cells against the potential difference (voltmeter reading). Decide which are your independent and dependent variables in this investigation. Draw a line of best fit through the data points.

If you used a light bulb instead of a resistor, what happened to the brightness of the bulb as you added more cells in series? If you did not do this, predict what would happen.

Conclusion

What can we conclude happens to the current strength and potential difference as more cells are added in series?

What have we learnt? Increasing the number of cells connected together in series increases the strength of the current in the circuit and the potential difference across the cells.



Investigation

The effect of the number of cells connected in series on current strength and potential difference

This is an optional investigation using PhET (Physics Education Technology) online simulations. You might do this in class with your teacher or else you can visit the website and interact with the simulation in your own time.

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- PhET circuit construction kit (DC only) bit.ly/19eKTHf

Method

1. Construct a series circuit with one cell, a resistor and the ammeter in series in the PhET simulation. Drag and drop each component to create the circuit.
2. Connect the voltmeter in parallel with the cell.
3. Record the readings in your exercise books.

4. Add a second cell in series with the first cell.
5. Record the new readings in your exercise books.
6. Add the third cell in series with the other two cells.
7. Record the new readings in your exercise books.
8. Draw a graph of your results.

Results

Copy and complete the following table in your exercise books:

Number of cells	Ammeter reading (A)	Voltmeter reading (V)
1		
2		
3		

Use your table to draw two line graphs in your exercise books. One graph should be the number of cells against the current (ammeter reading) and the other graph should be the number of cells against the potential difference (voltmeter reading). Decide which are your independent and dependent variables in this investigation.

Conclusion

What can we conclude regarding the effect on the current strength and potential difference of adding cells in series into a circuit?

Resistors in series

Let's revise some of the work we covered in Grade 8 about series circuits.

ACTIVITY Increasing the resistance in a series circuit

Materials

- 1,5 V cell
- 3 torch bulbs
- insulated copper conducting wires
- switch
- ammeter

Instructions

1. Construct the circuit with the cell, the ammeter, one bulb and the switch in series.
2. Close the switch.
3. Note how brightly the bulb is shining and record the ammeter reading. Draw a circuit diagram.
4. Open the switch.
5. Add another light bulb into the circuit.
6. Close the switch.

7. Note how brightly the bulbs are shining and record the ammeter reading. Draw a circuit diagram.
8. Open the switch.
9. Add the third light bulb into the circuit.
10. Close the switch.
11. Note how brightly the bulbs are shining and record the ammeter reading. Draw a circuit diagram. Copy the table below in your exercise books and record the ammeter readings:

Number of bulbs (resistors)	Ammeter reading (A)
1	
2	
3	

Questions:

1. What happened to the brightness of the bulbs and the ammeter reading as more light bulbs were added to the circuit?
2. Explain the observations you made in question 1.

If we increase the resistance in a series circuit, by adding more resistors, then the total current decreases. We say the current is inversely proportional to the resistance.

We are now going to look at the potential difference of each resistor.



Investigation

Measuring the potential difference across components in a series circuit

Investigative question

What is the relationship between the potential difference across the battery and the potential difference across the resistors in a series circuit?

Materials and apparatus

- three 1,5 V cells
- insulated copper conducting wires with crocodile clips
- two resistors of different resistances
- three voltmeters
- a switch

Method

1. Construct a circuit with three 1,5 V cells, two resistors and the switch in series with each other.
2. Connect a voltmeter, in parallel, across the three cells. This is voltmeter V_1 .

3. Connect a second voltmeter, in parallel, across one resistor. This is voltmeter V_2 . Take note of whether this is the resistor with the higher or lower resistance.
4. Connect the third voltmeter, in parallel, across the other resistor. This is voltmeter V_3 . Take note of whether this is the resistor with the higher or lower resistance.
5. Record the readings on the 3 voltmeters.

Results and observations

Draw a circuit diagram to illustrate your circuit. Take note of which resistor has the highest resistance.

Reading on V_1 :

Reading on V_2 :

Reading on V_3 :

Record these readings on your circuit diagram as well.

1. What do you notice about the readings on V_2 and V_3 when compared to V_1 ?
2. Add the readings together. What do you notice?
3. Which resistor has the highest potential difference, the one with the higher or lower resistance?

Conclusions

Write a conclusion for this investigation.

What have we learnt? The **sum** of the potential differences across the resistors in a series circuit is equal to the potential difference across the battery. If a resistor has a high resistance, then it will have a large potential difference.

If a resistor has a low resistance, then it will have a small potential difference. We can explain this because the battery provides the electrons with potential energy. The electrons travel through the resistors and lose some of that energy to each resistor in the form of heat, light or sound. There is only one path for the electrons to travel, so they transfer energy to each resistor through which they pass. The higher the resistance of the resistor, the more energy is transferred within the resistor. Therefore, there will be a greater difference in potential energy per charge from before to after the resistor in the series circuit.

Let's look at the example on the right:

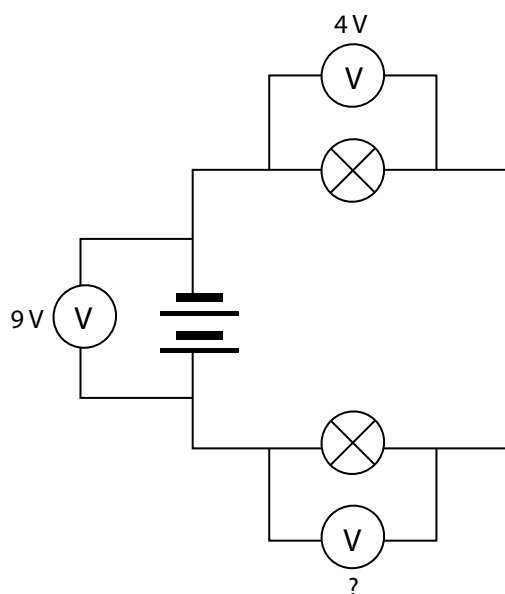


Figure 16.12

If the potential difference across the cells or battery is 9 V and the potential difference across one of the bulbs is 4 V, what would the reading on the third voltmeter be?

We say that resistors in series are **potential dividers**.

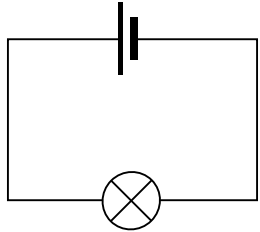
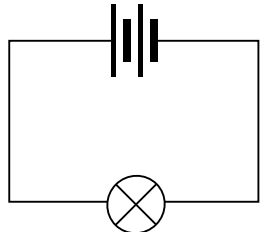
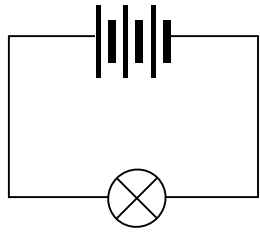
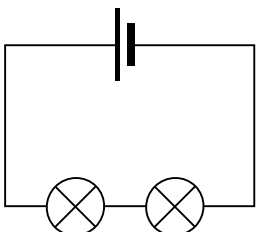
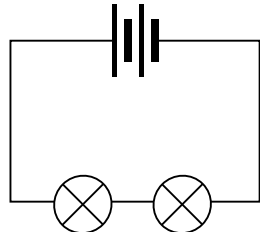
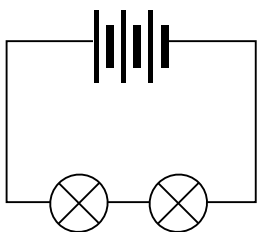
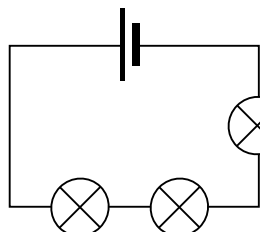
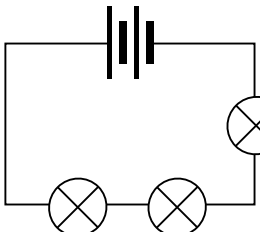
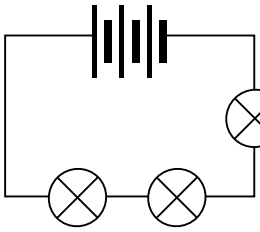
ACTIVITY Check your knowledge of series circuits

Materials

- batteries (or cells)
- torch bulbs
- insulated copper conducting wires
- ammeters

Instructions

Look at the circuit diagrams in the table. Assume that each bulb is the same.

I. 	II. 	III. 
IV. 	V. 	VI. 
VII. 	VIII. 	IX. 

1. Predict in which circuit each bulb will glow the brightest. On what did you base your prediction?
2. Predict in which circuit each bulb will glow the dimmest. On what did you base your prediction?
3. Why will the brightness of bulbs I, V and IX will be the same?
4. Now test each of your predictions by building the different circuits. Include an ammeter in the circuits in order to measure the current.

We have now seen that the current is affected by adding more cells and resistors in series, but so far we have only measured the current at one point in the circuit. Let's see how the current compares at different points in the circuit.

ACTIVITY Current in a series circuit

Materials

- insulated copper conducting wires
- two 1,5 V cells
- two torch light bulbs
- ammeter

Precautions

The longer the cells are used for, the more energy they transfer to the bulbs and that might affect the reading. Always keep the switch closed when you are not taking any readings.

Instructions

1. Set up a series circuit with two cells and two torch light bulbs in series with each other.
2. Insert an ammeter in series between the positive terminal of the cells and the first torch bulb.
3. Measure the current strength using the ammeter. Draw a circuit diagram of this set-up.
4. Insert the ammeter, in series, between the two torch bulbs.
5. Measure the current strength using the ammeter. Draw a circuit diagram of this set-up.
6. Insert the ammeter, in series, between the last torch bulb and the negative terminal of the battery.
7. Measure the current strength using the ammeter. Draw a circuit diagram of this set up.

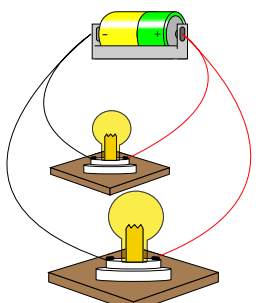
Copy and complete the following table in your exercise books:

Position of ammeter in circuit	Ammeter reading (A)
Between positive terminal of battery and first bulb	
Between two bulbs	
Between negative terminal of battery and last bulb	

What can you conclude from this about the current in a series circuit?

16.2 Parallel circuits

A parallel circuit provides **more than one** path for the electric current to move through the circuit.

Parallel circuit	Draw a circuit diagram for this circuit in your exercise books
	

Cells in parallel circuit

We saw that connecting cells in series increases the amount of energy supplied to the electrons. The potential difference increases. Let's investigate what happens when we add cells in parallel in a circuit.



Investigation

What happens to the current and potential difference in a circuit when adding cells in parallel?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- three 1,5 V cells
- insulated copper conducting wires with crocodile clips
- ammeter
- voltmeter
- resistor

Method

1. Construct a series circuit with 1 cell and the ammeter in series.
2. Connect the voltmeter in parallel with the cell as shown in the circuit diagram.

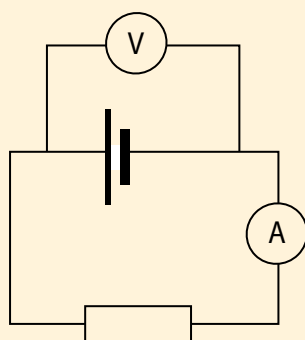


Figure 16.13

3. Copy the table on the next page, then record your results.
4. Add a second cell in parallel with the first cell as shown in the diagram.

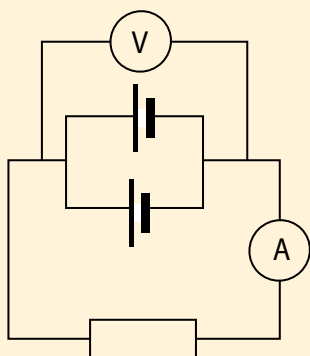


Figure 16.14

5. Record the new readings in your table.
6. Add the third cell in parallel with the other two cells. Draw a circuit diagram for this in the space below.
7. Record the new readings in your table.

Results

Copy and complete the following table in your exercise books:

Number of cells in parallel	Ammeter reading (A)	Voltmeter reading (V)
1		
2		
3		

Conclusion

What can we conclude regarding the effect of adding cells in parallel into a circuit?

What have we learnt?

When we connect two cells in parallel with each other, the overall potential difference is the same as if we had only one cell. Therefore, if both cells are 1,5 V, the overall potential difference for the circuit is still 1,5 V. The current is the same as if there was only one cell because the electrons travel through only one of the cells.

What advantage would we get from connecting cells in this way? Discuss this with your class.

Resistors in parallel

Parallel circuits have more than one pathway for the current. Let's look at how adding resistors in parallel affects the current strength.

ACTIVITY Adding resistors in parallel

Materials

- 1,5 V cell
- 3 torch bulbs
- insulated copper conducting wires
- switch
- ammeter

Instructions

1. Construct the circuit with the cell, ammeter, one bulb, and the switch in series.
2. Close the switch.
3. Note how brightly the bulb is shining, and write down the ammeter reading. Draw a circuit diagram of your circuit.
4. Open the switch.
5. Add another light bulb, in parallel to the first, into the circuit.
6. Close the switch.
7. Note how brightly the bulbs are shining, and write down the ammeter reading. Draw a circuit diagram of your circuit.
8. Open the switch.
9. Add the third light bulb, in a parallel to the first two, into the circuit.
10. Close the switch.
11. Note how brightly the bulbs are shining, and write down the ammeter reading. Draw a circuit diagram of your circuit.

Questions

1. What happened to the brightness of the bulbs and the ammeter reading as more light bulbs were added in parallel?
2. Explain your observations from question 1.

In the last activity, we measured only the current in the main branch of the circuit. What happens to the current in a parallel circuit?

ACTIVITY Current in a parallel circuit

Materials

- insulated copper conducting wires
- two 1,5 V cells
- three identical torch light bulbs
- ammeter

Method

1. Set up a parallel circuit with two cells in series with each other and three torch light bulbs in parallel with each other.
2. Insert an ammeter in series between the cells and the first pathway as shown in the diagram.

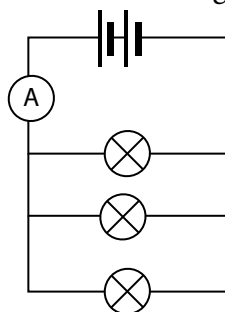


Figure 16.15

3. Measure the current strength using the ammeter.
4. Remove the ammeter and close the circuit again. Insert the ammeter, in series, in the first pathway.

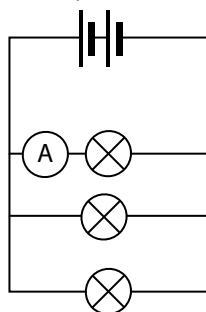


Figure 16.16

5. Measure the current strength using the ammeter.
6. Insert the ammeter, in series, in the second pathway.

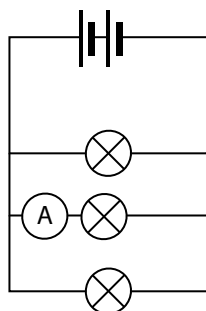


Figure 16.17

7. Measure the current strength using the ammeter.
8. Insert the ammeter, in series, in the third pathway.

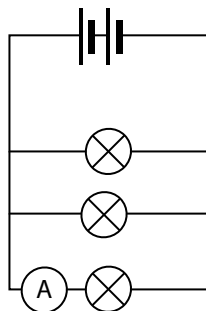


Figure 16.18

9. Measure the current strength using the ammeter.
10. Insert the ammeter, in series, between the first pathway and the batteries on the opposite side to the first reading.

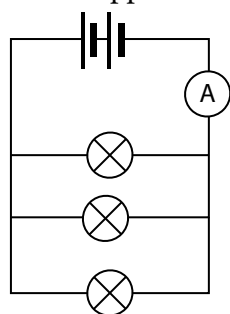


Figure 16.19

11. Measure the current strength using the ammeter.

Draw a table in your exercise books to record your readings.

Questions

1. What do you notice about the current in the main branch and the current in the pathways?
2. Add up the currents in each pathway through a bulb. What do you notice?
3. Use the diagram on the right and write an equation to illustrate the relationship between:
 - a) A_1 and A_4 .
 - b) A_1 , A_2 and A_3 .
 - c) A_4 , A_2 and A_3 .

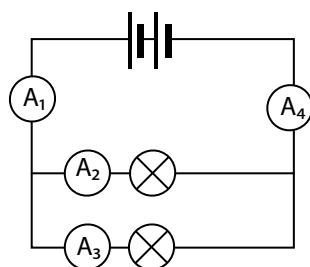


Figure 16.20

When we add resistors in parallel to each other the total resistance decreases and the current increases. Why does this happen? Adding resistors in parallel provides more alternative pathways for the current. Therefore, it is easier for current to move through the circuit than if all the current had to move through one resistor. In conclusion; a parallel circuit is a current divider and that voltage is the same in all the resistors.

As the electrons approach the branch in the parallel circuit, some electrons will take the first path and others will take the other path. The current is divided between the two pathways. We say that resistors in parallel are current dividers. Although both pathways provide resistance, the total resistance is less than if there were just one pathway.

We are now going to look at the potential difference across each resistor in a parallel circuit.



Investigation

Measuring the potential difference across components in a parallel circuit

Investigative question

What is the relationship between the potential difference across the battery and the potential difference across the resistors in a parallel circuit?

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- three 1,5 V cells
- insulated copper conducting wires with crocodile clips
- two torch light bulbs or resistors
- three voltmeters
- a switch
- three ammeters

Method

Construct the circuit.

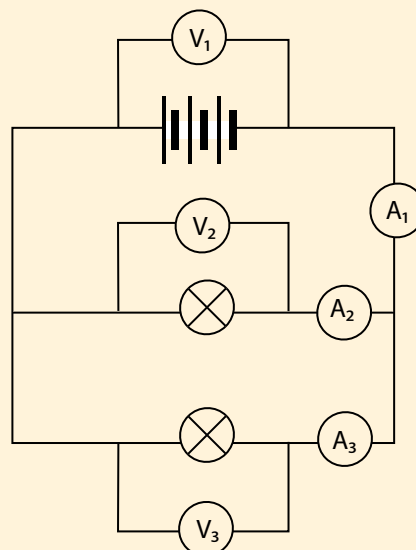


Figure 16.21

Note the readings on the three voltmeters and ammeters.

Results

Copy the circuit diagram above and the table into your exercise books. Record the readings on both.

Voltmeter	Reading (V)	Ammeter	Reading (A)
V_1		A_1	
V_2		A_2	
V_3		A_3	

1. What do you notice about the readings on V_2 and V_3 when compared to V_1 ?
2. Add the readings on A_2 and A_3 together. What do you notice?
3. Explain the behaviour of the electrons in the circuit based on the ammeter readings.

Conclusion

Write a conclusion for this investigation based on the investigative question.

Extension

Do you know that we can calculate the resistance of each light bulb in the circuit used in this investigation? We have seen that the current (I) through a resistor is inversely proportional to the resistance (R) and the potential difference across a resistor (V) is directly proportional to the resistance. This relationship is summarised in the following equation:

$$R = \frac{V}{I}$$

The unit of resistance is the ohm (Ω), which is defined as a volt per amp of current. This can be written as:

$$1 \text{ ohm} = \frac{\text{volt}}{\text{amp}}$$

We can therefore calculate the resistance. An example is shown here using the values in this circuit diagram:

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{3 \text{ V}}{2 \text{ A}} \\ &= 1,5 \Omega \end{aligned}$$

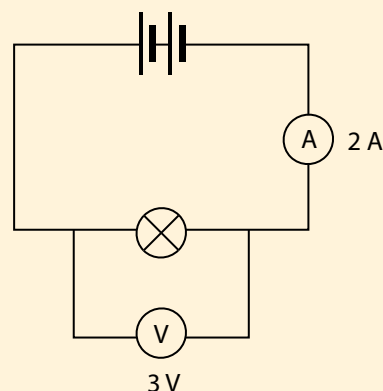


Figure 16.22

Did you know?

The relationship described here between current through a resistor and the potential difference across the resistor, is called Ohm's Law.

In this investigation, you measured the potential difference (in volts) and the current (in amps) for each bulb. Use these measurements to calculate the resistance for each bulb in your circuit.

What have we learnt from this investigation?

- The current in a parallel circuit divides when it enters the separate branches. The total current is the sum of the current in the branches.
- The potential difference across the branches of the circuit is the same as the potential difference across the battery.



Investigation Testing current strength

Investigative question

Is the current strength the same at all points in a series circuit?

Hypothesis

Write a hypothesis for this investigation. What do you think will happen in this investigation?

Materials and apparatus

- insulated copper connecting wires.
- two 1,5V cells
- two torch light bulbs
- ammeter

Method

1. Set up a series circuit with two cells and two torch light bulbs in series with each other.
2. Insert an ammeter in series between the positive terminal of the batteries and the first torch bulb.
3. Measure the current strength using the ammeter. Draw a circuit diagram of this setup.
4. Remove the ammeter and close the circuit again.
5. Insert the ammeter in series between the two torch bulbs.
6. Measure the current strength using the ammeter. Draw a circuit diagram of this setup.
7. Remove the ammeter and close the circuit again.
8. Insert the ammeter in series between the last torch bulb and the negative terminal of the batteries.
9. Measure the current strength using the ammeter. Draw a circuit diagram of this setup.

Results

Copy and complete the following table in your exercise books:

Position of ammeter in circuit	Ammeter reading (A)
Between positive terminal of cell and first bulb	
Between two bulbs	
Between negative terminal of cell and last bulb	

Conclusion

1. Write a conclusion based on your results.
2. Is your hypothesis true or false?

ACTIVITY Series and parallel circuits

Materials

- two 1,5 V cells
- insulated copper conducting wires
- two torch light bulbs

Instructions

1. Set up a series circuit with the two cells and the two torch light bulbs. Are both torch lights shining?
2. Disconnect one of the torch light bulbs. What happens?
3. Set up a parallel circuit with two cells and the two torch light bulbs in parallel with each other. Are both torch lights shining?
4. Disconnect one of the torch light bulbs. What do you notice?

Questions

1. Why did the series circuit stop working when one of the light bulbs was removed?
2. Why did the light bulb in the parallel circuit keep shining after you removed the other bulb?
3. Which type of circuit, series or parallel, would be more useful in a household circuit? Why?

Parallel circuits are useful in household circuits because if one pathway stops working then the other pathways can still work. So if your bathroom light bulb breaks, the rest of the lights or appliances in the house can still be used. If your house used a series circuit then all the lights and appliances in the house would stop working if one item broke. You can also turn lights on in different rooms at different times without having to turn all the lights on in the whole house at once.

An example of a series circuit is a set of tree lights. Each light bulb is connected in series with the others. This means that if even one breaks, all will stop working. To find the broken one and fix it, you would have to test every bulb.



Figure 16.23: Tree lights are sometimes connected in series.

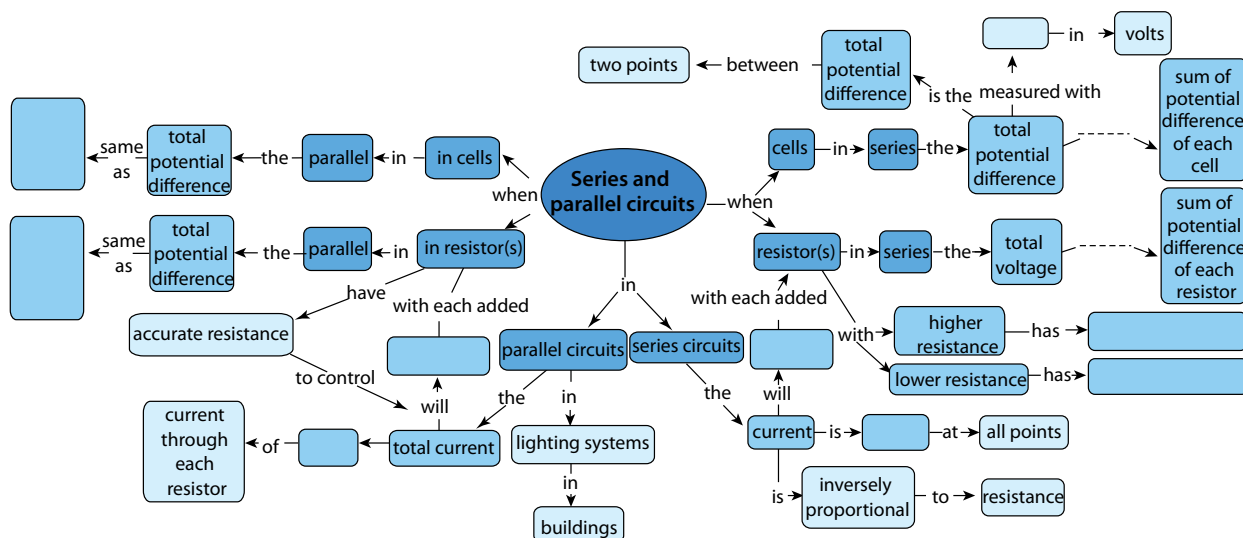
Summary

Key concepts

- A series circuit provides only one pathway for the electrons to move through the circuit.
- Increasing the number of cells connected in series increases both the current strength through the circuit and the potential difference across the cells.
- Increasing the number of resistors in a series circuit increases the overall resistance of the circuit and decreases the current strength.
- Resistors connected in series are potential dividers. The sum of the potential differences of the resistors is equal to the potential difference of the battery.
- The current strength in a series circuit is the same throughout the entire circuit.
- A parallel circuit provides more than one pathway for the electrons to move through the circuit.
- Increasing the number of cells connected in parallel with each other has no effect on the current strength and the potential difference of the circuit.
- Increasing the number of resistors connected in parallel decreases the overall resistance of the circuit.
- Resistors connected in parallel are current dividers. The current has more than one pathway to move along and so the current divides between the paths. The sum of the current strengths in the pathways is equal to the current strength before and after the branch in the pathway.
- The potential difference across each pathway is equal to the potential difference across the battery.
- Parallel circuits are used in the lighting systems in buildings.

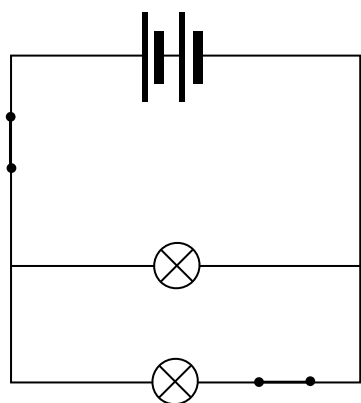
Concept map

Complete the concept map on the next page in your exercise books. Remember that you can also add your own notes on this page to make your summary more comprehensive and easier for you to study from for tests and exams.



Revision

1. Draw the following circuit diagrams.
 - a) A closed circuit with one cell, two light bulbs and a switch in series. [2]
 - b) An open circuit with two cells, two light bulbs and a switch in series. [2]
 - c) A closed circuit with 1 cell and a resistor in series, with an ammeter to measure the current and a voltmeter to measure the potential difference of the cell. [2]
 - d) A closed circuit with two cells in series and two light bulbs in parallel. [2]
 - e) A closed circuit with an ammeter and resistor in series and three cells in parallel, with a voltmeter connected to measure the potential difference across the three cells. [2]
2. Look at the following circuit diagram. Identify the number of bulbs, switches and cells in this circuit. Identify whether they are in series or parallel. [3]



3. Ian has bought a string of Christmas tree lights and has hung them in the tree and plugs them in. One of the light bulbs breaks.
 - a) What happens to the rest of the light bulbs? [1]
 - b) Explain your answer to question a. [2]
4. Household circuits are parallel circuits. Explain why it is better to use a parallel circuit in a house than a series circuit. [2]
5. Answer the following questions on the circuit below.

Circuit	Values for each reading
	$A_1 = 3 \text{ A}$ $A_2 =$

- a) Calculate the missing values. [2]
- b) Explain your answer above. [1]

6. Answer the following questions on the circuit below.

Circuit	Values for each reading
	$A_1 =$ $A_2 = 3 \text{ A}$ $A_3 = 3 \text{ A}$ $A_4 = 1 \text{ A}$

a) Calculate the missing values.

[2]

b) Explain your answer above.

[2]

7. Answer the following questions about the circuit below.

Circuit	Values for each reading
	$V_1 = 12 \text{ V}$ $V_2 = 8 \text{ V}$ $V_3 =$

a) Calculate the missing values.

[2]

b) Explain your answer above.

[1]

8. Answer the following questions about the circuit below.

Circuit	Values for each reading
	$V_1 = 10 \text{ V}$ $V_2 =$ $V_3 =$

a) Calculate the missing values.

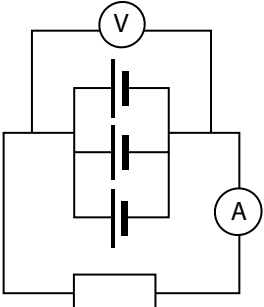
[2]

b) Explain your answer above.

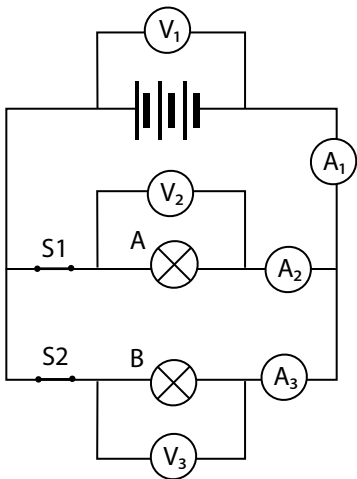
[1]

- c) How many bulbs will still glow if switch S_1 is opened? [1]
 d) How many bulbs will still glow if switch S_2 is opened? [1]

9. Calculate the missing values below. [2]

Circuit	Values for each reading
	Each cell in this circuit is 1,5 V. What is the reading on V?

10. Answer the following questions about the circuit below. [4]

Circuit	Values for each reading
	$V_1 = 9\text{ V}$ $V_2 =$ $V_3 =$ $A_1 = 6\text{ A}$ $A_2 = 2\text{ A}$ $A_3 =$

- a) Work out the missing values. [4]
 b) What would the reading on A_2 show if switch S_2 is opened? [2]
 c) Explain your answer to the previous question. [2]
 d) Which bulb, A or B, has the higher resistance? [2]
 e) Explain your answer to the previous question. [2]
 f) **Extension question:** Calculate the resistance of Bulb A and Bulb B. Show your working. [4]

Total without extension [47 marks]

Total with extension [51 marks]



Key questions

- How safe is my electricity connection?
- What is a short circuit?
- Why do plugs have three wires?

17.1 Safety practices

Imagine you are at home, it is dark and you have switched on one of the overhead lights. You then switch on a second light. Does the first light become dimmer? No, it does not. This is because the electrical circuits in houses are parallel circuits.

Despite the advantages of using parallel circuits in the electrical wiring in buildings, there is a disadvantage. Parallel circuits can become overloaded with too many branches and become a safety hazard. The overloading can cause too much heat which could lead to a fire starting. The fire could spread throughout the house and cause a lot of damage.

Keywords

- armature
- reset
- excess
- fuse
- circuit breaker
- earthing
- earth leakage
- short circuit

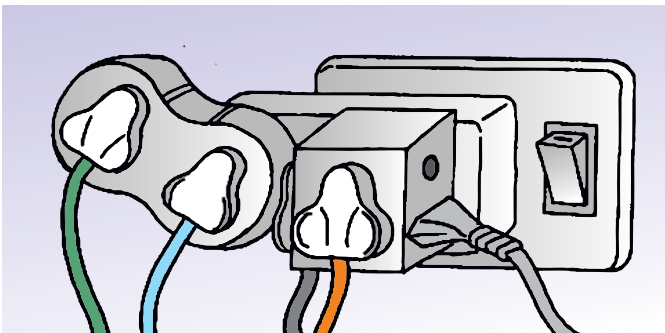


Figure 17.1: This plug is overloaded and is a safety hazard.

Let's now look at some of the safety practices which are followed and employed.

Earthing

What does it mean to **earth** an electrical appliance? Let's consider the example of a washing machine.

The electric circuit inside the washing machine has three different wires:

- a brown **live wire**
- a blue **neutral wire**
- a green-and-yellow striped **earth wire**.

The live and neutral wires provide the potential difference needed for the motor inside the washing machine to turn. The

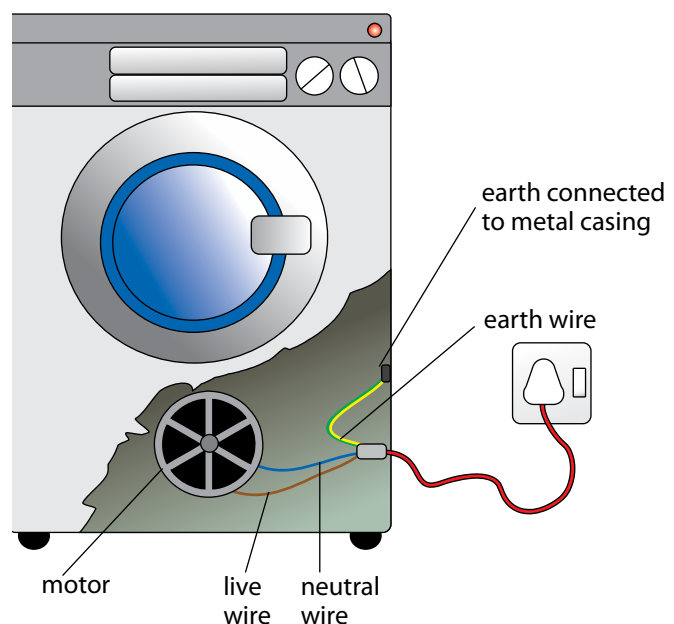


Figure 17.2: The electric circuit inside a washing machine.

earth wire is connected to the metal case of the washing machine. The three wires are encased in a plastic insulation to form one cord, which is plugged into the mains electricity supply at the wall.

The earth wires from all the electric sockets end up in one thick earth wire which is connected to a big metal spike driven into the ground. The earth wire usually does nothing. The only time it is used is when something goes wrong inside the machine.

If the live wire is exposed and touches the metal casing of the washing machine, you could get an electric shock if you touch the metal casing. However, the earth wire is connected to the metal casing so that the current goes through the earth wire and into the ground instead of shocking you. The earth wire has a very low resistance and so a strong current will easily go through it.

The earth wire completes the circuit and connects the live wire to the ground.

This is a **short circuit**. The washing machine will stop working because none of the electricity will flow through the motor.

If there were no earth wire, the metal casing of the washing machine would become part of the electrical circuit and anyone who touched it would get an electrical shock. That is why an earth wire is an important safety feature on any electrical appliance.

What are short circuits? A short circuit usually happens by mistake. An extra electrical pathway is made. This pathway has very low resistance and so the current increases. This increased current can damage appliances and cause overheating. Overheating can lead to fires. There are several safety devices which are used to stop the flow of current if a short circuit occurs. Let's look at some of the safety devices which are commonly used.



Figure 17.3: The green-and-yellow earth wire connected to the metal casing inside an electric motor.

? Did you know?

The colours for the different wires are used universally, so no matter where you are in the world, you will be able to identify the different wires by their colours.

Take note

The earth wire is sometimes just green or just yellow instead of striped.

Take note

How to deal with an electrical shock from an appliance.
bit.ly/1c9pGDg

ACTIVITY Making your own fuse

Materials

- three 1,5 V cells (large voltage battery)
- copper conducting wires with crocodile clips
- steel wool
- heat resistant mat or piece of wood
- torch light bulb
- variable resistor
- ammeter

Instructions

1. Set up a circuit according to the picture on the next page.
2. Twist a few strands of steel wool into a wire. This must not be very thick. Just a few strands will do.
3. Use the steel wool to complete the circuit.
4. Set the variable resistor to its highest resistance.

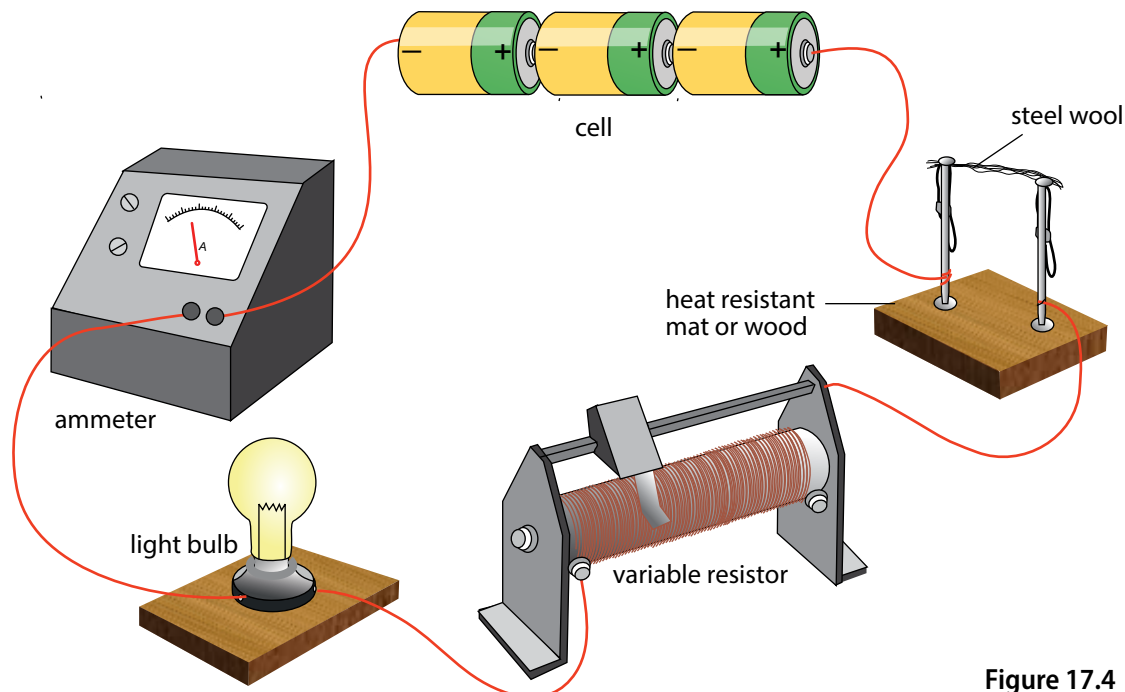


Figure 17.4

5. Close the switch. What do you observe?
6. Take note of the reading on the ammeter which measures the current in the circuit.
7. Open the switch.
8. Set the variable resistance to its lowest resistance.
9. Close the switch. What do you observe?

Questions

1. Draw a circuit diagram for your circuit.
2. Why is the light bulb included in the circuit?
3. When you decreased the resistance, what happened to the current? In other words, what happened to the reading on the ammeter?
4. What do you think happens to the electric current when the steel wool has burnt? Explain your answer.



Figure 17.5: Motor cars also have fuses. Can you see the fuse in this photo showing the battery in a motor car?

A fuse is a wire which will melt if the current travelling through becomes too large due to a fault, such as a short circuit or overload. When the fuse wire melts it breaks the circuit and current stops flowing. This disconnects the appliance to prevent any further damage. Fuses are stamped with the maximum current that they can handle. The photo is of a 5-ampere fuse. It will melt if a current of more than 5 amperes passes through it.



Figure 17.6: A 5-ampere fuse.

ACTIVITY Drawing circuit diagrams with fuses

Instructions

Draw the following circuit diagrams to show various places to insert fuses in circuits.

1. A circuit diagram with two cells and two light bulbs in series with each other. Insert a fuse into the circuit so that if the fuse breaks all the light bulbs will switch off.
2. A circuit diagram with a cell and two light bulbs in parallel with each other. Insert a fuse into the circuit so that if the fuse breaks, only one of the light bulbs will switch off.
3. A circuit diagram with a cell and two light bulbs in parallel with each other. Insert a fuse into the circuit so that if the fuse breaks, both bulbs will switch off.



Figure 17.7: This fuse has blown and has to be replaced.

Circuit breakers

When a fuse melts, it has to be replaced each time. There are other devices which are now more commonly used in households rather than fuses, such as circuit breakers.

Circuit breakers are one of the most important safety devices in our homes today. Without circuit breakers, electricity in our houses and buildings could be dangerous because of the risk of fires and other safety hazards resulting from electrical wiring faults and equipment failures.

A circuit breaker is similar to a fuse except that it can be reset. Once a fuse has melted it is thrown away and a new fuse is put into the circuit. A circuit breaker acts in the same way that a switch would and breaks the circuit if the current surges. You may have seen these switches before on a circuit or distribution board in your home or school.



Figure 17.8: An example of circuit breakers showing the switches.

New word

- electromagnet

Let's take a look at how a circuit breaker functions. Do you remember learning about **electromagnets** in Grade 8 when we looked at the effects of an electric current? An electromagnet is a type of magnet which forms due to an electric current around a bar. The strength of the magnet depends on the electric current. The more current, the stronger the magnet.

A basic circuit breaker consists of a switch connected to an electromagnet. Have a look at the diagram on the right.

When the switch is on, the current flows through the device, from the left through the moving contact and across to the stationary contact. It then goes around the electromagnet and out the other side. The iron catch is holding the moving contact in place so that the circuit is complete. If the current

passing through the circuit breaker increases, the electromagnet becomes stronger. If the current gets to unsafe levels, the electromagnet becomes strong enough to pull the iron catch lever. This releases the moving contact so that the circuit breaks and the electricity is shut off, as shown in the diagram on the right.

There is a reset button which can be pushed in order to push the contacts back together when the fault has been fixed and it is safe to reconnect the electricity.

Earth leakage

We have mentioned the dangers of electric charge in previous units. An electric charge will move from where there is a lot of potential energy to where this is less potential energy. Do you remember learning about lightning? The excess electrons from the clouds move to the ground and transfer a large amount of energy in the process.

The earth leakage circuit breaker is used in the electrical circuits of households and businesses. The circuit breakers for the different parts of the circuit are put into the electrical distribution board. The earth leakage circuit breaker is also on the distribution board.



Figure 17.9: An example of the earth leakage and main switch on a distribution board in a house.



Figure 17.10: A lightning rod on the roof of a house.

The earth leakage circuit breaker is a safety device which can switch off the electricity supply to the house. The earth leakage is able to detect if any current is moving through the earth wire. If current is moving through the earth wire, there is a short circuit somewhere, as explained with the washing machine. The earth leakage circuit breaker then shuts down all the current as a safety measure.

Lightning is always a danger to an electric circuit. In areas where lightning strikes are common, a lightning spike is often used. This is a metal pole which is connected to the house with one end buried in the ground.

If lightning strikes the house then the surge in current will flow through the metal spike and go safely into the ground. This helps to prevent electric fires in households due to a lightning strike.

Wiring a 3-pin plug

In the first section of this unit, we learned about the three wires that are attached to most electrical appliances. In your exercise books, copy and complete the following table to identify the colours of these three wires.



Figure 17.11: The three wires that form the cable leading to a plug.

Wire	Colour
Neutral wire	
Earth wire	
Live wire	

Why are there three wires? For a complete circuit we have wires coming into the house and wires leaving the house. The wire which enters our homes is the live wire. The neutral wire leaves our homes and is earthed as it leaves the house. The earth wire has a very low resistance and is connected through the plug socket to the earth cable of the house. The earth cable leads into the ground. If an electrical appliance becomes charged because of an electrical fault, it can discharge through the earth wire and earth cable and into the ground. This prevents someone from getting an electric shock from a fault in an appliance.



Take note

The neutral wire is earthed as it leaves the house, as shown in the diagram, to protect the national grid from lightning strikes. If lightning strikes the overhead cables or transmission lines, then the current flows into the ground rather than burning up the entire network.

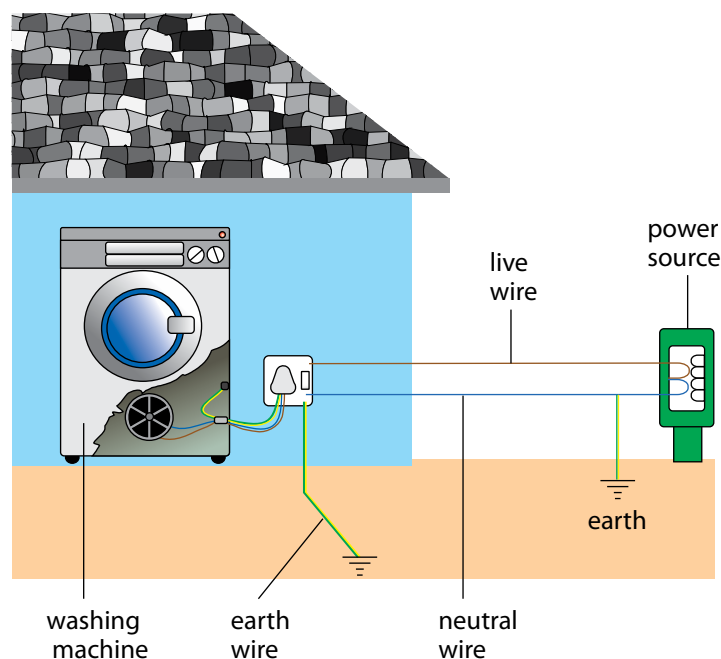


Figure 17.12: Houses and other buildings are connected to the national grid by the live wire entering the house and the neutral wire leaving the house.

The electrical plug has three metal pins. Each pin has a hole in it with a small metal screw. Loosening the screw opens the hole, tightening the screw closes the hole. Let's take a look inside a plug to see how to wire it.

ACTIVITY Wiring a 3-pin plug

Materials

- piece of insulated electrical cord
- wire strippers or craft knife
- 3-pin plug
- small screwdriver

Have a look at the photo of a 3-pin plug.



Figure 17.13: A 3-pin plug.

1. Which pin is the green-and-yellow cable connected to?
2. Which pin is the blue neutral wire attached to?
3. Which pin is the brown live wire always attached to?

We are now going to wire our own plugs.

Instructions

1. Cut about 2 cm of the white insulation off the electrical cord to expose the three wires within it. Do not slice directly into the wire as though you are cutting a loaf of bread. Move the blade carefully around the cord in a circle until you have cut through the insulation.
2. Once you have exposed the three different-coloured wires, cut away about half a centimetre of insulation from each of these three smaller insulated wires to expose the copper wire inside.
3. Twist the copper wires gently with your fingers so that the strand is tight.
4. Open the plug cover.
5. Unscrew the little screws on the 3 metal pins.
6. Insert the copper wire into the metal pins. The green and yellow wire goes into the top pin (often labelled 'E' for earth, or with the symbol for earthing). The blue wire must go into the pin on the right when viewed from the bottom (often labelled 'N' for neutral). The brown wire must go into the pin on the left when viewed from the underneath of the plug (often labelled 'L' for live).
7. Tighten each of the little screws to trap the wires in place.
8. Replace the plug cover.
9. You have now correctly wired a 3-pin plug and attached it to the electrical cable.
10. When you wire a 3-pin plug of an actual appliance, what safety precautions do you think you need to follow? Discuss this with your partner or class and write down your answer here.



Take note

When viewed from the underneath with the pins facing towards you, the green-and-yellow earth wire is always connected to the uppermost pin, the blue neutral wire to the pin on the right and the brown live wire to the pin on the left.

Now that we know more about the safety practices in electrical wiring in buildings, let's practise by designing the wiring for a house.

PROJECT Wiring a house

Instructions

You have made a doll's house for a neighbour's little girl. The doll's house has 2 bedrooms, a bathroom, a lounge and a kitchen. You want to make a simple electrical circuit for the doll's house.

Start off by drawing the floor plan of the house. Once you have this, draw in the wiring system to show how you would put a light bulb in every room. Each light must be able to switch on and off without affecting the other lights in the house. There must be a mains switch located in the kitchen and a fuse to prevent overload.

You should practice this on rough paper before drawing the final design in your exercise books on the next page. Include the labels for each room.

Keywords

- illegal
- provision

17.2 Illegal connections

An illegal electricity connection is made when a person attaches their home's electrical circuit to the national grid without a meter. This is done without the consent or knowledge of Eskom. Eskom cannot monitor the electricity consumption, so the electricity is stolen because these consumers do not pay for the electricity.

Some people make money supplying **illegal** connections and others have no legal way to access electricity and so they resort to illegal connections. Others have access to legal electricity, but prefer not to have to pay for it. Not only are these types of electrical connections illegal and considered energy theft, they are also very dangerous, as you will see in the next activity.



Figure 17.14: Electricity theft is illegal and also very dangerous because of the insecure connections

ACTIVITY Case study on illegal electricity connections

Instructions

1. Read the following newspaper article.
2. Answer the questions that follow.

Doornbach informal settlement celebrates electricity provision

WestCapeNews, July 2012 and fire risks.

There was much celebration in Doornbach, an informal settlement just outside of Cape Town, when the City switched on about 200 new electrical connections over the period of a few months in 2012. The illegal electricity connections had previously been the only supply of electricity to the area. Authorities often encounter fierce opposition when trying to cut down illegal electricity connections in informal settlements. But the Doornbach residents immediately took it upon themselves to cut down the massive web of illegal wires in response to finally receiving formal, legal electricity provision.

Besides the mass of wires running through the informal settlement, many of the wires had been strung across Potsdam Road, the main road running through the settlement. This was very dangerous as the wires hung very low and would often catch on trucks passing through and snap. Fire threats and electrical shocks to passers-by and vehicles were also a safety concern. The use of legal electricity would also help to prevent shack fires as residents would rely less on candles and paraffin stoves.

A fifty-two-year-old Doornbach resident, celebrating the end of illegal connections in the settlement, said that she had lived there for 18 years and had never received any municipal services from the City because they had originally settled on privately owned land, which meant that the City could not, in terms of national legislation, install services on privately owned land. However, the City had bought the land in May 2011, and Eskom could therefore begin the process of providing electricity to households in Doornbach. The fifty-two-year old was very excited about being able to use an electric iron and installing a refrigerator.

As a symbolic gesture, the residents took it upon themselves to remove the illegal wires. Many of the young people climbed up the dangerous makeshift poles in order to collect the wire which they would then sell to scrap yards. Not everyone was celebrating the switching on legal electricity connections in Doornbach. Many residents in neighbouring settlements, living in formal housing, were making money by selling and supplying electricity illegally to Doornbach. Street lighting had also been installed in Doornbach, and it was hoped this would help to reduce the crime rate.

Lastly, the City of Cape Town extended their sincere thanks to the community of Doornbach, as without their support, involvement and cooperation, such a project would not have been possible.

Questions

1. What is an informal settlement?
2. After reading this article, what do you think is the main reason that the people of Doornbach originally set up illegal connections?
3. Why is it dangerous for young people to climb the makeshift electricity poles?
4. What were some of the physical dangers of the illegal connections in Doornbach?
5. Aside from the physical dangers associated with illegal electrical connections, why else are they illegal?

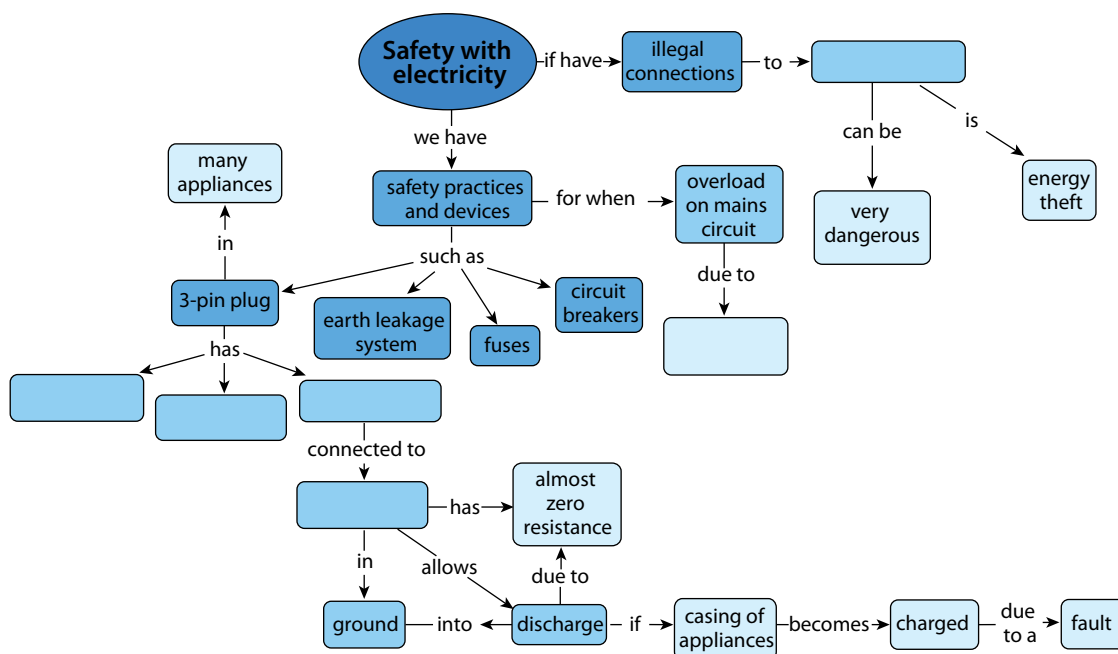
Summary

Key concepts

- Electricity can be dangerous, so we need safety devices such as fuses, circuit breakers and earth leakages to reduce risk.
- A fuse is a safety device with a very low resistance wire, designed to melt if it experiences a large enough current. This breaks the circuit and protects the appliance, as well as preventing a possible fire hazard.
- A circuit breaker is like a fuse, but acts as a switch which breaks the circuit in response to an electrical fault or overload. It can be reset.
- Many electrical appliances with a metal casing have an earth wire attached to prevent electric shocks if there is a short circuit.
- A three-pin plug has three wires: a brown live wire, a blue neutral wire and a green and yellow striped earth wire.
- The earth wire has a very low resistance and is connected through the plug socket to the earth leakage system of the house, and into the ground.
- A plug must be connected properly in order to make sure that it is safe to use.
- Illegal electricity connections are both dangerous and illegal. It is a crime to steal electricity.

Concept map

Complete the concept map in your books on the following page to summarise what you know about safety with electricity.



Revision

1. Explain how a fuse functions to protect an electric circuit. [4]
2. What would happen if you used a 3-amp fuse in an electrical fan heater that needs a current of 8-amps to function? [1]
3. What type of fuse should you use in the 8-amp fan heater? [1]
4. Why are circuit breakers more convenient to use than fuses? [2]
5. When a fuse 'blows', why do you think it is important to fix the problem before replacing the fuse? [2]
6. What is a short circuit? [3]
7. Why is a short circuit dangerous? [2]
8. What is the colour of the live wire in an electrical cable? [1]
9. Write down one safety precaution that should be followed when wiring a 3-pin plug. [1]
10. What is the purpose of the green-and-yellow wire in an electrical cable? [2]
11. Draw an outline of a 3-pin plug and label where each wire is connected and what colour each wire is. [6]
12. Draw a circuit diagram for the following circuits:
 - a) A series circuit with 2 cells and three light bulbs. Insert a fuse that will break the circuit and where all the bulbs won't work if there is a short circuit. [3]
 - b) A circuit with a cell and two light bulbs in parallel with each other. Insert a fuse into the circuit so that if the fuse breaks, only one of the light bulbs will switch off. [3]

Total [31 marks]



Key questions

- How is electricity generated in a power station?
- What energy sources are used in South Africa to generate electricity?
- Is nuclear energy the best way to solve the energy crisis?
- What are the advantages and disadvantages of nuclear power?
- How is electricity distributed from power stations to our homes?

18.1 Electricity generation

Electricity is generated in a power station. In previous grades, we have looked at how electricity is generated within coal-powered power stations and distributed to the country in the national electricity grid. We are going to revise some of these concepts.

The general method for producing electricity is to turn a turbine which will turn a generator. In South Africa most of the power stations use coal for fuel. The coal is mined out of the earth. The coal is transported to the power station in large trucks or trains.

Let's take a closer look at what happens inside a coal-powered power station.

Keywords

- turbine
- generator
- alternative energy
- power station



Figure 18.1: A coal-powered power station.

Have a look at the following diagram.

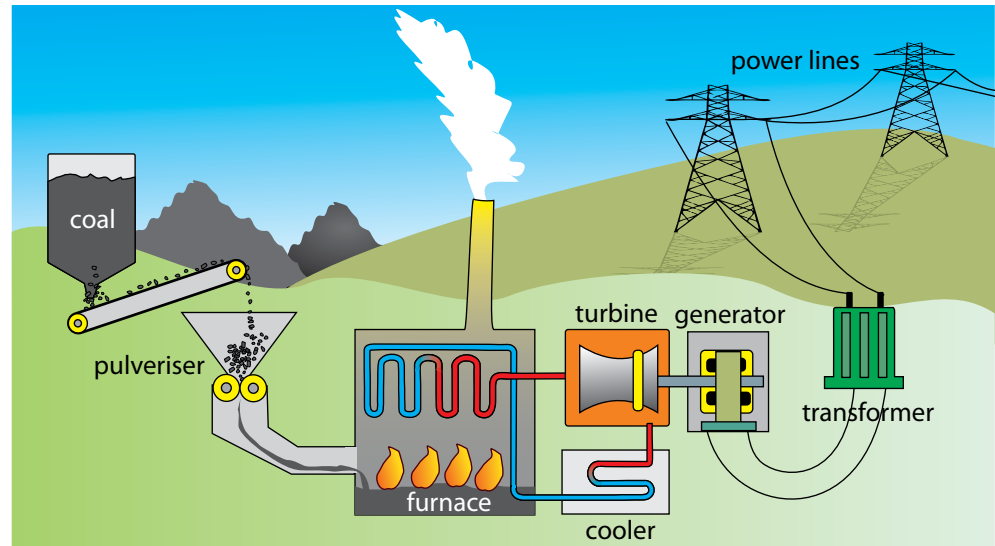


Figure 18.2: Diagram of a coal-powered power station.

An overview of the steps in a coal-powered station:

1. The large chunks of coal are first crushed into a fine powder. This is called pulverisation.
2. The coal is then transported to a **furnace** where it is burnt.
3. The thermal energy from the burning coal is used to boil water and generate **steam**.
4. The steam pushes the blades of the **turbine** and so the turbine spins.
5. The turbine is connected to the shaft of the **generator** which then rotates large magnets within wire coils, which generates electricity.
6. The **electric current** is sent through the **power lines** to businesses and homes.

? Did you know?

The cooling tower on the left in the photo is covered in the largest mural painting in South Africa.



Figure 18.3: This is the Orlando Power Station in Soweto, which served Johannesburg for 50 years from 1951. It is not used any more. The painted cooling towers are seen the most prominently, but the building to the right is also part of the power station.

Is coal a renewable or a non-renewable energy source? Explain your answer.

What are the disadvantages of South Africa's reliance on coal as the main source of energy in power stations?



Figure 18.4: A modern-day steam turbine connected to a generator.

Alternative sources of electrical energy

There are many different ways to get enough energy to turn the turbine. What are some of the alternative energy sources which can be used instead of coal?

List them in your exercise books.

The Gariep Dam on the border of the Free-State and Eastern Cape provinces is a hydroelectric power station which uses the water falling out of the dam to turn the turbine. How does the falling water make the turbine turn? Let's investigate that a little more.

ACTIVITY Hydroelectric power

Materials

- aluminium foil plate or cooking tray
- scissors
- pencil
- adhesive tape or packaging tape
- piece of string about 45 cm long
- eraser
- nut, bolt, or other small mass piece
- source of running water, such as a tap

Instructions

1. Cut out the circular bottom of an aluminium foil plate. If it isn't circular, use a compass to draw a circle in the base and cut it out.
2. Make eight equally spaced cuts toward the centre of the foil circle, as shown with the solid lines in the diagram. End each cut about 2 cm from the centre. You now have eight triangular sections.



Take note

In Grade 8 you might have made miniature turbines and seen how steam is able to make the blades of the turbine turn. You can refer back to this by looking online at www.curious.org.za



Take note

Do you remember how coal is formed? Coal is a fossil fuel, formed over millions of years as prehistoric swamps and vegetation were covered in layers of sediment and compressed.

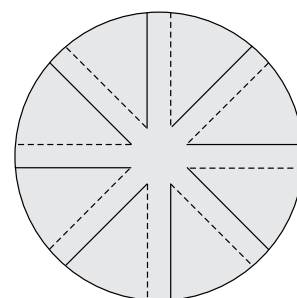


Figure 18.5

3. Fold each of the sections upwards. Use a ruler to help you get a straight edge. Use the dotted lines in the diagram as a guide.
4. Make a small hole in the centre of the plate. Push a pencil through the hole. The pencil should fit tightly into the hole. Use adhesive tape to stick the wheel to the pencil so that if you rotate the pencil, the wheel rotates.
5. Tie a piece of string around one end of the pencil. Tie the small nut or bolt to the other end of the string.
6. Hold each end of the pencil lightly between your thumbs and index fingers.
7. Hold the wheel under a slow stream of water from a running tap. Make sure to hold the wheel so that the blades are in the water, as shown in the a diagram.

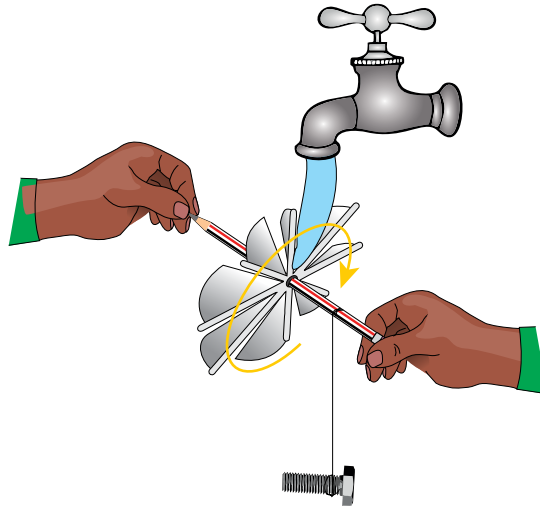


Figure 18.6

Questions

1. What happens to the aluminium wheel when it is placed in the stream of water?
2. What happens to the mass piece when the aluminium wheel is in the water stream?
3. Explain the energy transfers between the falling water and the mass piece lifting.
4. The following diagram shows an example of a hydroelectric power station. Answer the questions that follow.

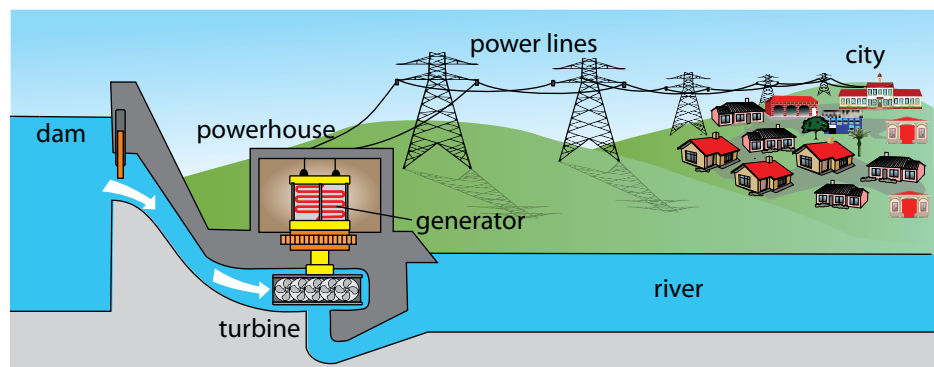


Figure 18.7: A diagram of a hydroelectric power station.

- a) The water in the dam on the left is high up. It has the ability to fall down. What kind of energy does the water have?
- b) As the water flows down the outlet from the dam, describe the transfer of energy.

- c) The flowing water then turns the turbine. This is a mechanical system. What kind of energy does the turbine have?
- d) The generator then transfers the energy between two systems. The kinetic energy in the mechanical system is transferred to electrical energy in the electrical system as it generates electricity. What parts make up the electrical system in the diagram?
- e) What is the output from this whole system? In other words, what does the city get?

A large dam wall is often built to collect water. The water then flows through the hydroelectric power plant.



Figure 18.8: A large dam wall with a hydroelectric power plant.



Figure 18.9: The generators inside the hydroelectric power plant.



Figure 18.10: The water flows out the bottom of the power station and continues down the river.

A turbine can be used to transfer kinetic energy from the falling water to the generator. A generator is a device which converts mechanical energy to electrical energy. A generator consists of large metal coils which move within a magnetic field. In some generators the coils are stationary and the magnet is rotated and in other generators the magnets are stationary and the coil is rotated.

Turning a set of conducting metal coils inside a magnetic field generates an electric current. The modern-day generator works on the principle of

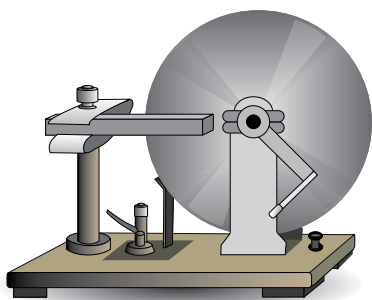


Figure 18.11: A drawing of the Faraday Disk, the first electromagnetic generator. It consisted of a copper disk rotated between the poles of a horseshoe-shaped magnet to generate electricity.



Did you know?

The Faraday Disk was the first electromagnetic generator invented by British scientist Michael Faraday in 1831.



Take note

Portable generators produce exhaust fumes which contain poisonous gases that can kill if they are present in high enough levels. When using a portable generator, make sure it is in a well-ventilated area, and away from water as far as possible.

electromagnetic induction discovered by Michael Faraday in 1831–1832. Faraday discovered that you could cause an electric current to flow by moving an electrical conductor, such as a wire that contains electric charges, in a magnetic field. The movement creates a potential difference between the two ends of the wire or electrical conductor. This then causes the electric charges to flow through the conductor as current.

There are many different types of generators used in different situations, not only in power stations. Some yachts and boats use water or wind-powered generators

to charge their batteries through the use of small propellers in the water or a wind turbine. Portable generators are often used in homes and businesses when there are power outages to keep certain appliances running, such as the lights and refrigerator.

Portable generators run on fuel, such as petrol, and generate electricity.

Small generators, called dynamos, can be rotated by a person rotating a crank, and are used in devices such as portable radios and torches, and especially mining helmets. Dynamos are also used in bicycle lights. The dynamo in a bicycle consists of a permanent magnet and some surrounding coils of wire and is attached to the wheel, which rotates as the bicycle moves. As the dynamo rotates it generates a changing magnetic field that generates electricity in the surrounding coils of wire.

As we have seen, most of the electricity generated in South Africa uses the burning of coal to provide steam to turn the turbines. Some of the electricity is generated using alternative energy sources. Why are they called alternative energy sources? This is because they are not the main source of energy. Most alternative energy sources are renewable forms of energy.



Figure 18.12: A portable generator used in a home.



Figure 18.13: A dynamo on the wheel of a bicycle.



Figure 18.14: Wind turbines out at sea are an alternative energy source.

RESEARCH PROJECT Alternative energy power stations

Instructions

1. Research the different types of power stations in South Africa.
2. Choose one of the alternative energy sources used in South Africa.
3. Alternatively, your teacher may ask you to do this as a research project and present a poster.
4. Write a paragraph here where you discuss the alternative energy source power station you have researched. In your paragraph, include the following information:
 - a) Compare the alternative energy power station to a coal-powered station in terms of sustainability and environmental impact.
 - b) Discuss the advantages and disadvantages of using the alternative energy source rather than coal for generating electricity.
 - c) Include your references.

18.2 Nuclear power in South Africa

For enrichment

South Africa has only one commercial nuclear power station, the Koeberg Power Station in Cape Town. We are going to take a closer look at nuclear power.

Keywords

- nuclear power
- nuclear fission
- nuclear fusion
- radioactive



Figure 18.15: The Koeberg Nuclear Power Station outside Cape Town.

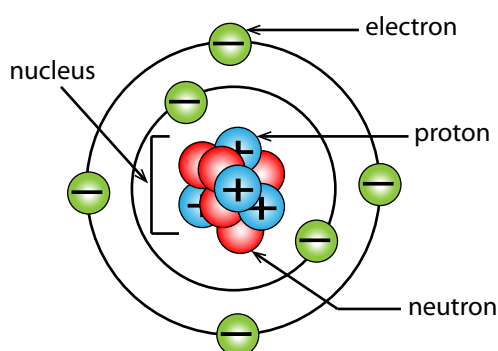


Figure 18.16: The structure of an atom

Before we look at nuclear fission, let's revise the model of an atom which we have already learnt about. Copy and label the diagram of the model of an atom.

Most atomic nuclei are stable. But there are some elements which are not stable. The nuclei in these unstable elements



Did you know?

Atomic bombs use the processes of nuclear fission or nuclear fusion. The devastating effects of nuclear bombs are still seen today in Hiroshima and Nagasaki in Japan where two nuclear bombs were dropped during World War 2.

spontaneously emit particles, referred to as radiation. A nucleus that emits radiation is said to be radioactive. Radioactive decay is the process when an unstable nucleus of an atom emits particles. It then 'decays' into another type of atom with a different mass.

Nuclear power was first pursued for electricity generation in the beginning of the 20th century when researchers discovered that radioactive materials, such as radium and uranium, release large amounts of energy when they decay. For a long time, however, the use of nuclear power was not seen as practical or possible to generate electricity.

This changed in the 1930s with the discovery of nuclear fission. During nuclear fission, scientists split the nucleus of an atom into two smaller atoms. This releases a huge amount of energy. There is also another way in which the energy in an atom can be released during nuclear fusion. Nuclear fusion occurs when two atoms are brought together to make a new, bigger atom. During both of these nuclear reactions, huge amounts of heat and radiation are released.



Figure 18.17: Atomic bombing of Nagasaki on 9 August 1945.

Nuclear power uses nuclear fission, nuclear fusion and nuclear (radioactive) decay. Uranium is an element which is unstable and undergoes radioactive decay at a very slow rate. This makes uranium a good choice to use as a fuel in nuclear power stations. Nuclear power stations therefore use uranium and induce nuclear fission to release the heat and radiation. Let's take a look inside a nuclear power plant, such as Koeberg Power Station.

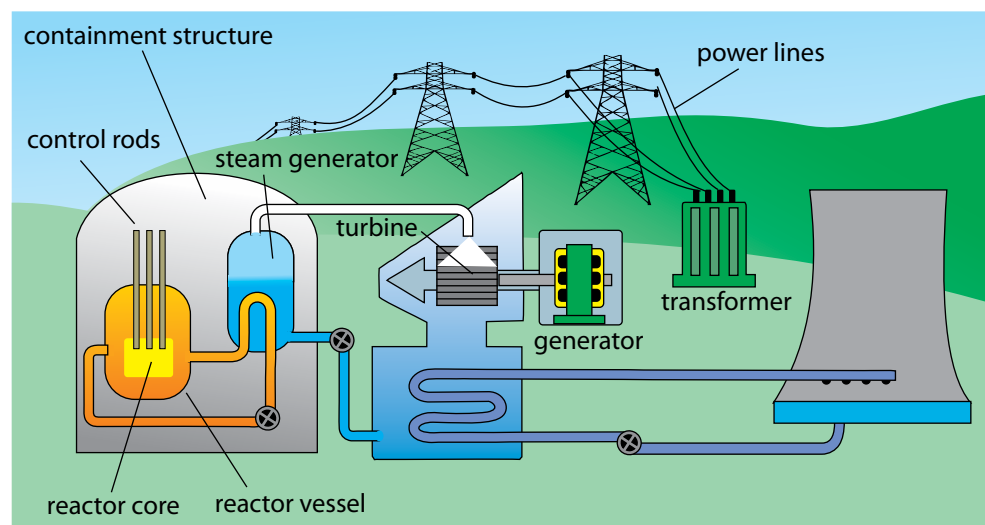


Figure 18.18: Diagram of a nuclear power plant.

The main difference between a nuclear power station and other power stations, such as coal power stations, is the way in which the water is heated to produce the steam.

A nuclear power station has a nuclear reactor vessel. The nuclear power station needs to control the huge amount of energy given off during nuclear fission of uranium in order to generate electrical energy. The nuclear reactor is the device in which the nuclear reactions take place and are controlled. The uranium is formed into pellets which are arranged into long rods, called the reactor core. The rods together make a bundle which is inserted into water to prevent overheating and melting. The bundle of uranium rods also contains control rods which help to control the process.

The large amounts of energy produced by the nuclear fission reactions in the uranium fuel rods heat the water to produce steam. The steam is used to turn large turbines which transfer the kinetic energy to generators, which produce electricity in the same way as in other power stations.

Most of South Africa's power stations use the burning of coal to produce enough heat to boil the water. The only difference in a nuclear power station is how the energy is produced to heat the water and produce steam.

As was mentioned before, the nuclear fuel is radioactive. The radiation that the fuel emits is dangerous and can be very harmful as it can enter our bodies and damage our cells. The workers in nuclear power plants therefore need to take extra precautions. The nuclear reactor is also contained within a special container that acts as a barrier to radiation.

You may have heard some of the debate surrounding the use of nuclear fuel in power stations. There are many supporters but also many critics. Let's look at some of the advantages and disadvantages of using nuclear fuel.

ACTIVITY Advantages and disadvantages of nuclear power

Instructions

1. Discuss and answer the questions that follow.
2. You will then be divided into groups to do some extra reading and research and host a debate for and against the advancement and development of nuclear power in South Africa.

Questions

What are some of the advantages of nuclear fuel? Discuss this with your partner and write down your answers in your exercise books.

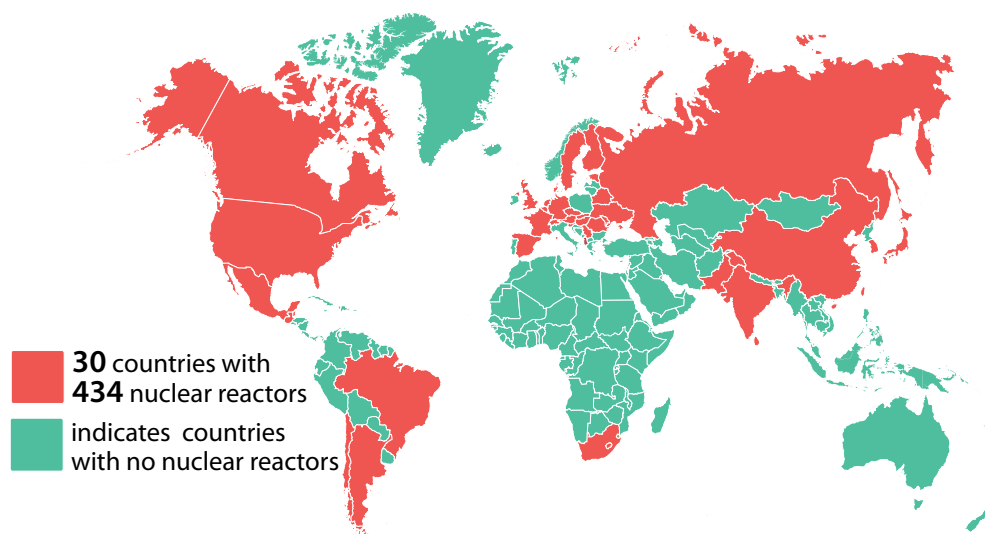
One of the major disadvantages is that once the nuclear fuel has been used, it cannot just be thrown away in a city dump. The spent nuclear fuel is high-level radioactive waste. The radiation can also damage animals and plants. This nuclear waste needs to be disposed of carefully and correctly so that the radiation is not able to cause a lot of damage. Over time, the nuclear waste will decay to safe levels of radioactivity, but this takes thousands of years. In the meantime, it has to be stored so that it does not cause any damage or fall into the hands of nuclear arms manufacturers. This adds to the cost of using



Did you know?

In March 2011, thousands of Japanese citizens evacuated the area surrounding the Fukushima-Daiichi nuclear power plant after a powerful earthquake and resulting tsunami seriously damaged the power station. The water leaked out of the reactor, resulting in overheating and a partial nuclear meltdown.

nuclear fuels. Some people also have reservations about nuclear power plants as the potential for a nuclear reactor meltdown is catastrophic.



Did you know?

Nuclear fission produces roughly a million times more energy per unit mass than fossil fuel alternatives.

Figure 18.19: Operating nuclear power stations worldwide.

Research and debate

You need to conduct further research into the advantages and disadvantages of nuclear power. Your teacher will put you into a group which is either in favour or not in favour of using nuclear power, and developing it further in South Africa. Debate, with your classmates, whether or not nuclear power is the solution to our developing energy crisis. You need to substantiate your point of view and justify your statements either in support of or against nuclear power.

We are now going to look at what happens to the electricity that is generated in a power station, whether it is coal, nuclear, or a hydroelectric power plant.

Keywords

- national electricity grid
- transformer
- transmission lines
- pylon
- power surge

18.3 National electricity grid



Figure 18.20: The national power lines transmit electricity across the country from the power stations.

The national energy grid is a network of interacting parts which form one big system to provide electricity to all sectors of the economy. It starts at the power stations where the electricity is generated. The power stations then feed the electric current into large power lines called transmission lines. There is a massive network of transmission lines that extends throughout the entire country. The transmission lines are supported by pylons.

Very high currents are generated at the power stations. The transmission lines have some resistance. If the power stations transferred electricity at high currents, what do you think would happen in the transmission lines? Hint: Remember what we learnt about the effects of resistance.

The following diagram illustrates a coal-powered power station connected to the national electricity grid.

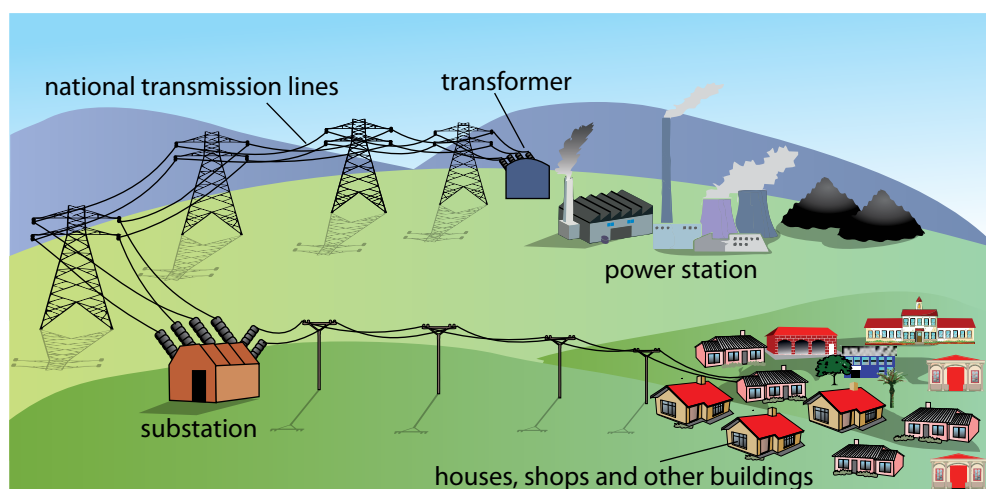


Figure 18.21: A coal-powered power station connected to the national electricity grid.

To prevent the waste of energy, the electric current is rather sent through the transmission lines at very high voltages and low current.

However, these voltages are too high for use in both private homes and commercial buildings. In our homes and buildings we need low voltage and high current.

Transformers are used to change the voltages at different points in the grid. As you can see in the previous diagram, the electricity first goes through a transformer before entering the national transmission lines. This is a step-up transformer as it increases the voltage and lowers the current. When the electricity reaches the substation to be distributed locally, there is a step-down transformer which decreases the voltage again and increases the current.

All the systems in the national electricity grid are connected and this means that a power surge or a grid overload can cause blackouts and disruptions throughout the network.



Did you know?

In 1986, the Ukrainian nuclear reactor in Chernobyl exploded and released 50 tonnes of radioactive material into the area, destroying forests and causing 30 000 people to evacuate the area. Thousands of people subsequently died from cancer and other illnesses.



Take note

In a debate, each person must be given a chance to voice his or her opinion and a chance to respond to other opinions.

Power surge

What is a power surge? A power surge is a sudden increase in the voltage somewhere in an electric circuit. The power surge causes an increase in the current strength. This sudden increase in the current strength can damage sensitive circuits.

Lightning strikes near a transmission line can cause a power surge. Other causes of power surges or a grid overload include faulty wiring, or appliances or equipment that require a lot of energy when they are switched on and off. These sudden demands or excesses of energy cause brief changes in voltage that can cause a surge. There are several points in the national grid which can detect a power surge or grid overload. If these are detected, the power supply to that area is cut off.

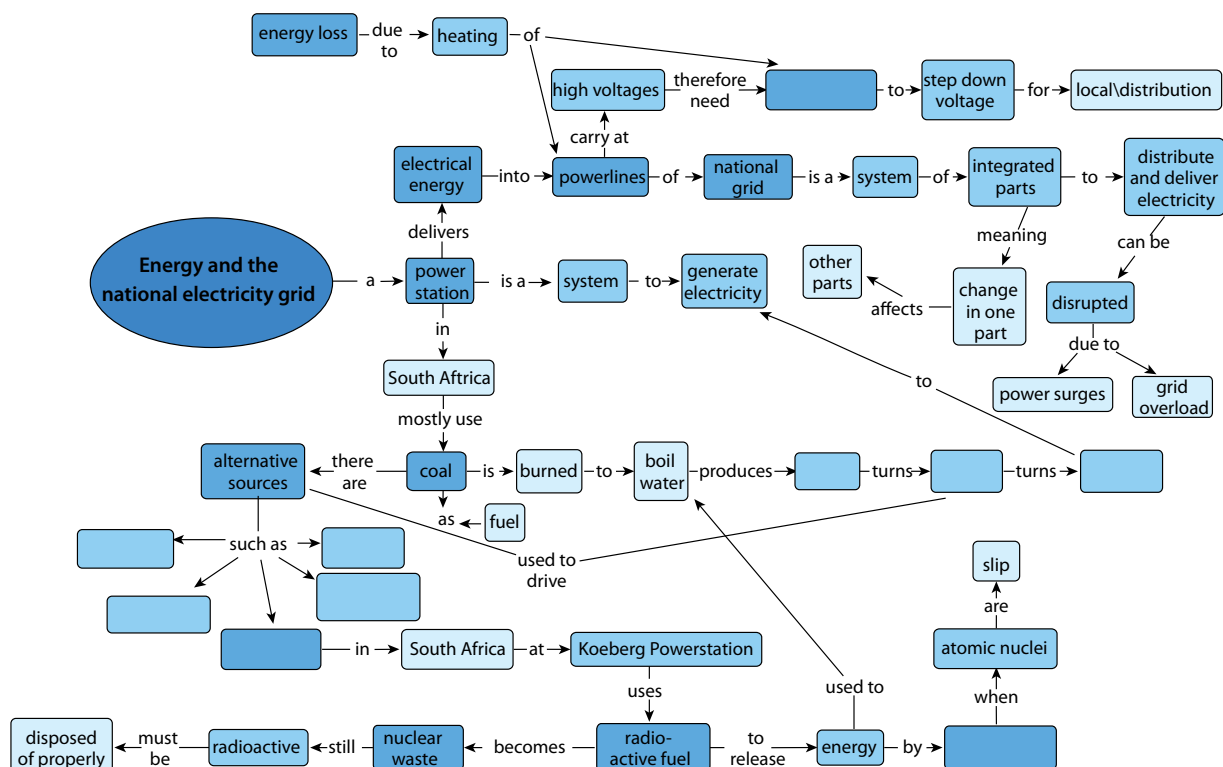
Summary

Key concepts

- The national grid is a network of interacting parts. If one part of the system is damaged, it will affect the entire network.
- Electricity in South Africa is mainly produced using coal-fired power stations.
- In a coal power station, coal is burned to heat water to produce steam. The steam turns a turbine, which turns a generator to produce electricity.
- There are alternative sources of energy besides coal to drive turbines, such as wind, hydropower, sun-heated steam, nuclear power, and tidal energy.
- Koeberg is the only nuclear power station in South Africa.
- Nuclear power stations use nuclear fuels, such as uranium, to generate heat and radiation by nuclear fission. This heats the water to produce steam to turn the turbine.
- Nuclear fuels are very energy-efficient as a large amount of energy is obtained from a very small mass of nuclear fuel. There is no emission of greenhouse gases in the use of nuclear fuels.
- Nuclear power stations generate radioactive waste materials which need to be properly disposed of. There is much debate around the use of nuclear fuels.
- The national electricity grid is a system to deliver electricity around the country.
- Electricity is carried at high voltages and low current through the national transmission lines to reduce the heating effect of the wires and minimise the energy wasted.
- Transformers are needed to step-up the voltage as the electricity leaves the power station and enters the national grid, and to step-down the voltage for local distributors and consumers.

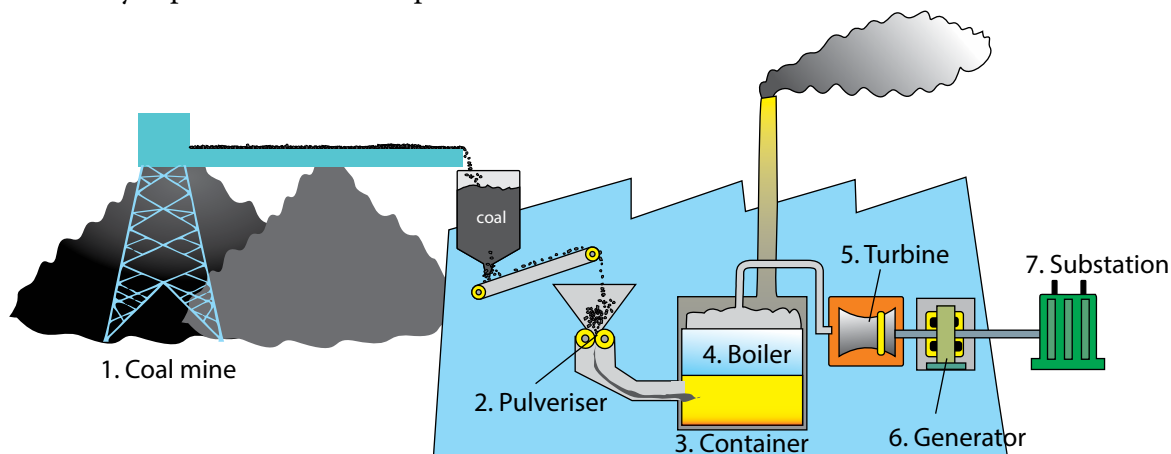
Concept map

Complete the concept map on the following page to summarise this unit on electricity production and delivery.



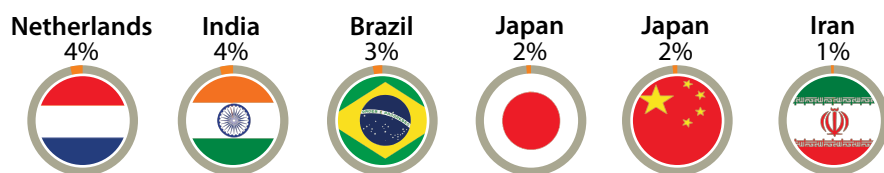
Revision

1. Why do you think we can refer to the national electricity supply as a grid? [2]
2. What is the main source of energy for power stations in South Africa? [1]
3. Why are renewable sources of energy referred to as alternative forms of energy? [2]
4. Look at the diagram of a power station. Write a paragraph to describe the process by which electricity is produced in a coal power station. [6]



5. With the aid of a flow diagram, explain the energy transfer that occur in a coal fired power station. [4]
6. What is nuclear power? [2]
7. Where is South Africa's nuclear power station? [1]
8. What is the difference between nuclear fission and nuclear fusion? [2]
9. Write a paragraph to explain the differences and similarities between a coal-fired power station and a nuclear power station. [4]
10. Write a paragraph to compare how the use of coal impacts on the environment when compared to how the use of nuclear fuels impacts the environment. [4]
11. Study the following diagram below and on the opposite page and answer the questions that follow.





The proportion of electricity generated using nuclear fuels in each country in 2013.

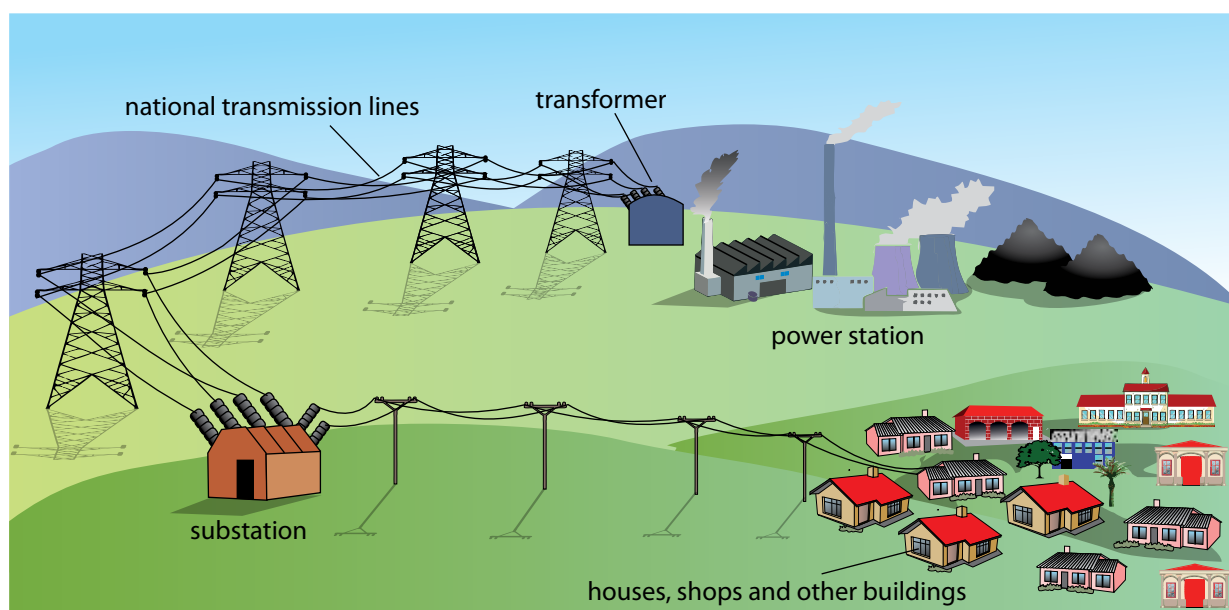
- Which country has the highest percentage of its electricity being produced using nuclear fuels? [1]
- What is the percentage of South Africa's energy generated using nuclear fuels? [1]
- Why are all the countries in the world not shown here in this diagram? [1]
- Draw a bar graph to compare the percentage of electricity generated using nuclear fuels to compare South Africa, France, United States of America, United Kingdom, India and China. [6]

12. Why can we consider the national electricity grid a system? [2]

13. We can divide the national electricity grid into four main stages. These are:

- Generation** (this is where electricity is generated)
- Transmission** (the electricity enters the power lines of the national grids and is transmitted)
- Distribution** (the electricity is distributed from substations to various towns and areas)
- Consumers** (this is where the electricity is transferred to useful energy outputs)

Use this information to write the letters A, B, C and D on the diagram of the national electricity grid to label these stages. [4]



- Why is electricity transmitted through the power lines at high voltages and low current strength? [3]
- Most household appliances need a voltage of 220–240 V. If the electricity in the transmission lines is very high, how are we able to use it? [2]
- Why should you protect your computer from power surges? [2]

Total [50 marks]

19 Cost of electrical energy



Key questions

- What is electrical power?
- How do we measure electrical power?
- Do different appliances use different amounts of energy?
- How do we know how much power we are using?
- How do we measure our use of electrical energy?
- How can we work out how much our electricity costs?
- How can we reduce our energy consumption?

Keywords

- electrical power
- watt
- rate

19.1 What is electrical power?

Electrical power is the rate of electrical energy supply. It is the amount of energy supplied per unit of time. In simpler terms, it is how fast the electrical energy is supplied.

Power is measured in **watts (W)**. We can calculate the power using the formula:

$$\text{power} = \frac{\text{energy}}{\text{time}}$$

Energy is measured in joules and so this means that power is the amount of joules supplied in a certain period of time. When doing calculations of power, you need to have the energy measured in joules and time measured in seconds.

1 watt is the same as 1 joule of energy transferred in a second.
(1 watt = 1 joule per second)

There are 1000 watts in 1 kilowatt (kW).

Different appliances use different amounts of power, depending on their function. All electrical appliances have a stamp or a sticker which indicates the power rating. If you look at your hairdryer or kettle you should find it easily.



Take note

A rate is a ratio where one quantity is compared to time, for example km/h, which compares distance (in kilometres) to time (in hours).



Figure 19.1: The label on an electric pan indicating a power rating of 1400 W.



Figure 19.2: The label on a fan indicating a power rating of 120 W.

Which of the above two appliances uses more power to operate?

ACTIVITY Power rating of different appliances

Instructions

1. Depending on your class and teacher, you may be able to walk around the school or your classroom and look at different appliances.
2. You may also have advertisements from newspapers or magazines to study.
3. Write down the power rating of each appliance that you find. Complete the table to record your findings.
4. Some photographs of the labels on various appliances have been given below. Include these power ratings in your table.
5. Answer the questions that follow.

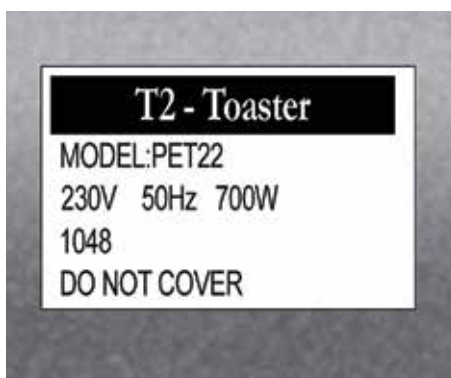


Figure 19.3: The label on the underside of a toaster.



Figure 19.4: The box for an electric beater.



Figure 19.5: The label on the back of a television set.



Figure 19.6: The label on an urn for heating water.

Copy the following table in your exercise books and fill in the power ratings of the various appliances.

Appliance	Power (W)
Toaster	
Electric beater	
Television	
Urn	

Questions

- Copy the following table in your exercise books and complete the conversions between joules and kilojoules:

joules (J)	kilojoules (kJ)
120	
	34
1 230	
	24,6



Take note

If you are given the amount of energy transferred in kilojoules, you need to convert it to joules to calculate the power in watts.

- Copy the following table in your exercise books and complete the conversions between watts and kilowatts:

watts (W)	kilowatts (kW)
1 760	
	4,56
25	
	0,56

- Sequence the appliances listed in your table above from those that use the most power to those that use the least power.
- Did you record the power ratings of any other appliances which involve heating? What do you notice about the power for these appliances?
- The following questions involve calculations based on the equation:

$$\text{power} = \frac{\text{energy}}{\text{time}}$$
 - If 180 kJ of energy is transferred to your bedroom lamp in half an hour, what is the power rating of your lamp? Show your calculations.
 - 100 000 kJ of energy passes through a power station every minute. What is the power rating of the power station? Show your calculations.

Keywords

- Eskom
- power consumption
- tariff
- tiered tariff
- electrical appliance
- survey
- estimate
- VAT
- lumens
- kilowatt-hour

We pay for the electricity that we use in our homes. How do we calculate how much we pay based on our energy consumption?

19.2 The cost of energy consumption

Eskom charges us for the electrical energy we use in our homes. Eskom charges us based on our energy consumption. The more electrical energy we use to run our household electrical appliances, the more Eskom charges us.

How do we work out how much energy we use? Think for example of using a 1 000 W microwave to warm your food for 1 minute. How much energy is transferred?

We can rearrange the following equation:

$$\text{power} = \frac{\text{energy}}{\text{time}}$$

So that we have:

$$\text{energy} = \text{power} \times \text{time}$$

In this formula, energy is measured in joules, time is measured in seconds and power in watts.

Therefore, to calculate the energy consumption when using a 1 000 W microwave, we can calculate it as follows:

$$\begin{aligned}\text{energy} &= \text{power} \times \text{time} \\ &= 1\,000\text{ W} \times 60\text{ s} \\ &= 60\,000\text{ J}\end{aligned}$$

Eskom now wants to work out our energy consumption for the whole month for all the appliances in your home. If 60 000 J of energy were used to warm food for 1 minute, then you can see that we would calculate an extremely large number for our energy consumption for the whole month in joules. This is not practical for electricity bills. We therefore have an alternative unit for energy consumption.

The quantity used for energy consumption is the **kilowatt-hour** (kWh). 1 kWh is the energy used if a 1 000 W appliance is used for 1 hour.

We can calculate the energy consumption of different appliances by multiplying the power rating by the amount of time it was used in hours.

ACTIVITY Calculating energy consumption

Instructions

1. Complete the following table to convert between seconds, minutes and hours.
2. Answer the questions on power consumption showing your calculations.

Seconds (s)	Minutes (min)	Hours (h)
620		
120		
	127	
	940	
		4,5
		12,25

Questions

1. An oven with a power rating of 3 600 W is used to bake a cake for 1 hour. What is the energy consumption?
2. A kettle with a power rating of 2 200 W is used to boil water for 6 minutes. What is the energy consumption?

Take note

The kilowatt-hour is a measure of energy consumption as it is calculated by multiplying power in kilowatts by time in hours.

Take note

1 kWh is equal to 3 600 000 joules.

Take note

A kilowatt-hour requires the unit of time to be in hours. You must therefore convert any times that you are given into hours for your calculations.

3. You use a 3 600 W oven to bake a cake for 1,5 hours. What is the energy consumption?
4. A 120 W light bulb is left on for 2 hours. A 60 W light bulb is left on for 3,5 hours. Which light bulb has a higher energy consumption? Show your calculations.

The cost of energy consumption

We are charged for the number of kilowatt-hours that we use. The cost of energy consumption is charged in cents per kilowatt-hour (c/kWh). The following table gives us the rates at which homeowners are charged for purchasing their power directly from Eskom. As you can see, there are different 'blocks'. The more energy you use per month, the more you pay per kilowatt-hour. This is called a **tiered tariff** system.

Eskom Home Power Tariffs 2013

Different energy consumptions per month	Energy charge (c/kWh)	Environmental levy charge (c/kWh)	Total (c/kWh)
Block 1 [≤ 50 kWh]	67,07	2,28	69,35
Block 2 [51 – 350 kWh]	83,32	2,28	85,60
Block 3 [351 – 600 kWh]	124,74	2,28	127,02
Block 4 [> 600 kWh]	137,03	2,28	139,31

In order to calculate your electricity costs, choose the block that applies to your home. For example, if your home uses 252 kWh of electricity in a month, you fall into Block 2.

Let's calculate the cost if your home used 252 kWh in April 2013.

The first 50 kWh were charged at the lower rate so:

$$50 \times 69,35 = 3\,467,5 \text{ cents}$$

The rest of the units were charged at the block 2 rate:

$$(252 - 50) = 202$$

$$\text{Therefore, } 202 \times 85,60 = 17\,291,2 \text{ cents}$$

$$\text{So, in total you would have had to pay } 3\,467,5 + 17\,291,2 = 20\,758,7 \text{ cents}$$

Remember, the tariffs are quoted in cents, not rands, so you need to do a conversion.

$$20\,758,7 \div 100 = \text{R}207,59$$

This means that your total bill would have been R207,59 for the month of April 2013.



Take note

The tiered tariff system is used to encourage people to save electricity and use it wisely as the less you use, the less you pay per unit of electricity.

What if you do not want to work out your entire bill, but just want to find out how much one particular appliance was costing you? The average per unit

price of electricity in 2013 was 71,65 c/kWh. This is the per unit price we will use for our calculations.

If we want to know how much we will pay for using a particular appliance we would use the following calculation: $\text{cost} = \text{power rating of appliance} \times \text{number of hours it was used for} \times \text{unit price of electricity}$

Let's try an example calculation for the microwave.

We want to work out the cost of using a small oven (1 500 W) for a total of 1 hour in a day. The following steps outline what you should do.

Step 1: Write down the formula

$$\text{cost} = \text{power rating} \times \text{time} \times \text{price}$$

Step 2: List all the given values in a problem

$$\text{power rating} = 1\,500\text{ W} = 1,5\text{ kW}$$

$$\text{time} = 1\text{ hour}$$

$$\text{price} = 71,65\text{ c/kWh}$$

Step 3: Substitute the given values into the formula to find the unknown

$$\text{cost} = 1,5\text{ kW} \times 1\text{ hour} \times 71,65\text{ c/kWh}$$

$$= 107,475\text{ cents}$$

$$= \text{R}1,07$$

Step 4: Write down the solution on its own line with the units.

The cost is R1,07 to run a small oven for one hour.

Let's try another example.

Have you ever notice your fridge start humming after a period of silence? Fridges are extremely energy-expensive electrical appliances. To keep the temperature at a constant cool temperature, fridges contain a thermostat that measures how cool the air is inside your fridge. When the temperature inside the fridge warms beyond a certain point, the thermostat will switch on the energy-expensive compressor and condenser. Fridges are specially insulated to keep cool air inside, and the energy demands of a fridge vary greatly depending on how often the doors are opened, and what is kept inside.

Imagine now, that you left the fridge door open by accident as you rushed to school and didn't notice until the next day! We now want to work out how much it costs to run a fridge with a power rating of 2 200 W for a day.

$$\text{cost} = \text{power rating} \times \text{time} \times \text{price}$$

$$\text{power rating} = 2\,200\text{ W} = 2,2\text{ kW}$$

$$\text{time} = 24\text{ hours}$$

$$\text{price} = 71,65\text{ c/kWh}$$

$$\text{cost} = 2,2\text{ kW} \times 24\text{ hours} \times 71,65\text{ c/kWh}$$

$$= 3\,783,12\text{ cents}$$

$$= \text{R}37,83\text{ per day}$$

The cost is R37,83 to run a fridge for one day.



Take note

Remember that we can find the power rating for an appliance on the labels.

Did you notice that both of the labels indicated power in watts (W) and not kilowatts (kW)? That means that if you use them to calculate the cost of electricity you must first convert them to kilowatts.



Did you know?

The actual cost per unit depends on whether or not you buy your electricity directly from Eskom or from your local municipality. In fact, the tariff differs from municipality to municipality as well.

ACTIVITY Calculating the cost of energy consumption

Instructions

1. Use the information in the following table to answer the questions that follow.
2. Use a cost per unit of electricity of 71,65 cents/kWh in all your calculations.

Appliance	Power rating
Microwave oven	1 360 W
Conventional oven	6 000 W
Television	105 W
Geyser	4 800 W
Incandescent light bulb	100 W
Fluorescent light tube	40 W
Vacuum cleaner	1 600 W
Washing machine	2 200 W

Questions

1. Imagine that your family has used 320 kWh of electricity this month. Calculate the cost of the 320 kWh.
2. A potato takes about 1 hour to cook in a conventional oven. In a microwave it takes approximately 12 minutes. Calculate the cost of cooking the potato in each appliance and write down which one is the cheaper option.
3. Which light bulb is cheaper to run for an hour, the incandescent light bulb or the fluorescent light bulb? Justify your answer with a calculation.
4. If you have a prepaid electricity voucher for R15, how long could you watch TV?
5. You need to vacuum your room and it takes you 30 minutes to do this. What does this cost?
6. It takes the geyser to two and a half hours to heat water from 20 °C to 65 °C. How much does it cost to heat the water?
7. What alternative appliance could a family use to heat water which would not demand such a high use of electricity?

We can see that different appliances have different power ratings and so require more electricity to run. This means that some appliances are more expensive to run than others. An incandescent light bulb, for example, is more expensive to use than a fluorescent light bulb. If you remember, an incandescent light bulb loses most of its energy as heat, instead of light.

Do you know how much electricity your family consumes? Let's find out.

ACTIVITY Home survey

Instructions

1. List, in a table similar to the one above, all the electrical appliances in your home. The 4 column headings should read: Appliance, Power rating, Number of hours and Cost.
2. Estimate the number of hours for which each appliance is used on a typical day.
3. Calculate the daily cost of each appliance, using 71,65 c/kWh as the unit price. Remember to count the number of light bulbs you have in your home and multiply that by the cost for 1 light bulb.

Questions

1. What was your total estimated electricity cost for one day?
2. If there are 30 days in an average month, what would your estimated monthly bill be?
3. Can you think of any ways that your family could reduce the amount of electricity you use?

There are many different ways in which we can try to conserve electricity in order to save money. Our choice of light bulbs can affect our electricity bills. LED light bulbs are the most energy efficient. They have lower power ratings than normal light bulbs, but most of their energy is transferred as light and very little as heat (unlike incandescent light bulbs). Light transmitted from the bulb is measured in lumens.

ACTIVITY Comparing the energy efficiency of different light bulbs

Instructions

Read the information in the table and use it to answer the questions.

	LED	Compact Fluorescent Light bulbs (CFLs)	Incandescent light bulbs
Example			
Average life span (in hours)	50 000	8 000	1 200
Watts	8	15	60
Lumens	800	800	800



Did you know?

LEDs and incandescent light bulbs also do not contain mercury, which is poisonous, while fluorescent light bulbs do.

1. Which of the three light bulbs will last the longest?
2. Which of the three light bulbs has the highest power rating?
3. How do the three light bulbs compare in terms of how much light they can provide?
4. Calculate how much it would cost to run each light bulb in a house for 5 hours a day for an entire year (365 days). Use 71,65 cents/kWh.
5. Which light bulb would you choose to use? Explain your choice.
6. Which bulb is the best for the environment? Explain your choice.

Using light bulbs which use less electricity can have a knock-on effect. If everyone is using less electricity, then there is less demand for electricity to be produced in our coal-powered power stations in South Africa. If less electricity is produced then fewer fossil fuels are burnt, and this would lead to a reduction in the production of excess greenhouse gases.

There are many important and rewarding careers in the electrical energy sector.

Let's take some time to research some of them.

ACTIVITY Career research

Instructions

1. Divide into groups of 3.
2. Use the library or the internet to research a career from the following list:
 - a) electrician
 - b) electrical engineer
 - c) IT specialist.
3. Try to find information about what someone in that career spends their day doing and what qualifications are needed to do the job.
4. Report your findings to the class.

Questions

1. Write down the career path that you found the most interesting.
2. What did you like about that particular career path?

Summary

Key concepts

- Electrical power is the rate of energy supply, measured in watts (W).
- 1 watt of power is equal to 1 joule per second.
- Different appliances use different amounts of power.
- Electrical energy is sold in units called kilowatt-hours (kWh), a measure of the energy consumption.
- 1 kWh is the energy used by a 1000 W appliance in 1 hour.
- Eskom sells electricity using tiered tariffs to discourage people from using too much electricity.
- We can calculate the cost of using a single appliance by multiplying the power rating by the number of hours and the unit cost of electricity.

Concept map

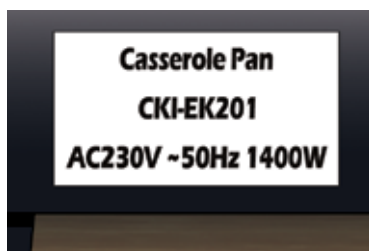
Design your own concept map to summarise this unit on the cost of electrical energy.

Cost of electrical energy

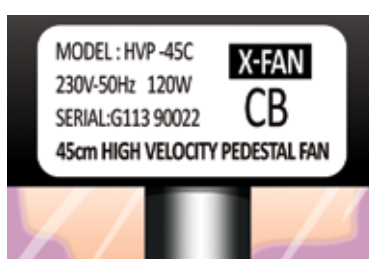
Revision

1. What is the power rating on the following two appliances? [2]

a) A frying pan



b) A fan



2. Refer to the table of power ratings for common appliances. List the appliances in sequence from those that use the least power to those that use the most. [2]

Appliance	Power rating (W)
Stove	3 600
Microwave	1 200
Washing machine	2 200
Kettle	2 200
Fridge	230
Toaster	750
Energy-saver globe	40
Incandescent light bulb	120
Vacuum cleaner	1 600

3. What is electrical power? Explain in your own words. [2]

4. Explain what is meant by 1 watt of power. [2]

5. What does it mean that a stove has a power rating of 3 600 W and a microwave has a power rating of 1 200 W? Compare these two appliances in terms of the energy supplied. [3]

6. Copy and complete the following table in your exercise books.

[8]

Joules (J)	Kilojoules (kJ)
145	
	134
1 650	
	32,12
Watts (W)	Kilowatts (kW)
1 850	
	3,79
32	
	0,485

7. An electric iron is rated at 1 500 W. If the iron is used for 3 hours every day, find the number of units of electrical energy it consumes in the month of February. [3]
8. An electric kettle is rated at 1 000 W. If it is used for 1 hour every day, find the number of units of electrical energy it consumes for the month of August. [3]
9. Your house's electricity meter reading in March was 3 456 and in April it was 4 566.
 a) How much electrical energy did your household use in that one month period? [2]
 b) If electrical energy is charged at 71,65 c/kWh, what will your bill be for that one month period? [3]
10. A 120 W electric blanket is left on for 8 hours.
 a) How many kilowatt hours of electrical energy is used by the blanket? [3]
 b) If the unit cost of electricity is 71,65 cents, what is the cost, in rands, of using the electric blanket for 8 hours? [3]
11. A 2600 W kettle in a school staffroom is used 8 times a day for five minutes each time.
 a) What is the total time that the kettle is switched on during each 5-day school week? [2]
 b) How much energy is consumed to run the kettle for this period (in kilowatt hours)? [2]
 c) If the cost of a unit is 71,65 cents per kilowatt, what is the cost of running the kettle for this period? [3]
12. If you had a prepaid electricity voucher with a value of R35, calculate the following.
 a) How long could you run a 230 W fridge for if electricity costs 71,65 cents per kWh? [5]
 b) How long could you run six 60 W incandescent light bulbs? [5]
13. Which light bulb, 15 W CFL or 8 W LED, would you choose to use? Explain your answer. [3]
14. A tumble dryer has a power rating of 4 500 W. How long did it take to dry a load of wet washing if electricity costs 71,65 cents per kWh and the cost of running the dryer is R4,84? [6]

Total [62 marks]

Glossary

acceleration the rate of change of velocity with time, as an object speeds up or slows down

alloy a mixture of different metals; the alloy will have properties from the different metals in the mixture

alternative energy a form of energy which is different to the main energy source used in the country

armature any moving part of an electrical machine in which a current is produced by a magnetic field

attraction a force which causes objects to move towards each other

battery a group of two or more electric cells connected together

circuit breaker like a fuse, a circuit breaker switches off the current in the case of an electrical fault

compression a force which attempts to flatten or deform (squash) an object

conductor a substance which allows heat, sound or electric charge to pass through it easily; a good conductor allows free passage whilst a poor conductor allows partial passage

contact force objects are in contact with each other and exert forces on each other

deformation to cause an object to change its shape

delocalised not limited to a particular place, free to move

earth leakage is a circuit breaker which will switch off all the electricity to a household or business if there is an electrical fault

earthing a circuit is earthed when there is a direct connection to the ground; this connection is usually through the earth wire in an electrical socket

electric cell a system in which chemical reactions occur to generate electricity

electric charge the physical property of matter that causes it to experience a force

when close to other electrically charged matter; there are two types of electric charges positive and negative

electric current the rate of flow of charge in an electric circuit

electrical appliance an electrical device

electrical power the rate at which energy is transferred

electrode an electric conductor used to make contact with a non-metal part of the circuit, such as a copper coin or iron nail in a lemon, or zinc or copper plates in a cell

electrolyte a special type of solution which is able to conduct electricity

electrostatic force force of attraction or repulsion between electrostatic charges

Eskom Electricity Supply Commission of South Africa

estimate a value which is not exact

excess more than is needed

field forces non-contact forces

force a push or a pull exerted on an object by an agent

free-fall when the only force acting on an object is the gravitational force

friction a force that opposes or tries to oppose the motion

fuse a safety device which switches off an appliance if the current in the circuit is too strong

generator a machine which produces an electric current by rotating a conducting coil in a magnetic field

gravitational acceleration a measure of how an object changes its speed every second; on Earth gravitational acceleration is $9,8 \text{ m/s}^2$

gravitational force force of attraction between two objects because of their masses

half cell a setup that consists of an electrode surrounded by an electrolyte; for example, a zinc half cell could consist of a zinc metal plate (the electrode) in a zinc sulphate solution (the electrolyte)

illegal forbidden by law; against the law

input energy the energy that enters a system and is altered by the system to produce an output energy

kilowatt-hour a unit of energy that is useful for measuring energy consumption

LED light emitting diode (a diode is an electrical component that allows current flow in one direction only and blocks the flow in the opposite direction)

lumens the units of measurement for light output from an LED bulb

magnet a material with a strong magnetic field around it

magnetic force a force exerted by a magnet on a ferromagnetic material

magnetic material a material which is strongly attracted to a magnet

mass a measure of the amount of matter making up an object

motor a device that can convert electrical energy into mechanical energy

national electricity grid the network of cables, pylons and transformers which transfer electricity throughout the country

net force the overall result of several forces acting on the same object at the same time

newton the unit of measurement of a force

non-contact force a force which can act over a distance without touching the object experiencing the force

normal force the reaction force of the surface to an object

nuclear fission when an atomic nucleus is split to produce two separate atomic nuclei; a large amount of energy is released during the separation

nuclear fusion when two small atomic nuclei are combined to produce one atomic nucleus; a large amount of energy is released as the nuclei are fused together

nuclear power the use of nuclear reactions to generate useful heat and electricity

ohm unit of measurement for resistance (Ω)

output energy the energy that a system produces due to an input energy

potential difference the difference in potential energy per charge between two points in an electric circuit

power consumption the amount of electrical power used by an appliance or household

power station a system for generating electricity

power surge a sudden increase in the voltage somewhere in an electric circuit which can disrupt the power supply

provision supplying something

pylon a large vertical steel tower which supports electrical power cables

radioactive the spontaneous release of a stream of particles or electromagnetic waves from an unstable nucleus

rate a ratio where one quantity is compared to time, for example km/h or m/s

repulsion a force that causes objects to move apart

reset to start something again from its start

resistance the opposition to the flow of electrical current through a material

resistor an electrical component in a circuit that opposes the flow current in the circuit

rheostat a variable resistor. The amount of resistance offered by the rheostat can be adjusted

salt bridge a device that is used to connect the two half cells in an electric cell so that their electrolytes do not mix

Sankey diagram a Sankey diagram is used to show the difference between input and output energy

series circuit a circuit which provides only one path for electric current

short circuit a short circuit is a low resistance path which causes all of the current to flow through the low resistance path and not through the rest of the circuit

speed the rate of change of distance of an object

survey information gathered from a wide range of people

tariff the amount of money charged for every unit

tension the force transmitted through a rope, string or chain. It is a contact force

tiered tariff the amount of money charged changes if more units are used; there are different levels of tariffs

transformer an electrical device to transfer energy between two parts of a circuit in the national electricity grid

transmission lines power cables which transmit electricity across the country

turbine a machine which consists of a large wheel that is made to turn by using steam

variable resistance resistance which is able to be changed

VAT Value Added Tax; this is a tax imposed by the government on all consumable goods

velocity the rate of change of the position of an object, specifying the object's speed and direction

voltage the difference in potential energy per charge between two points in an electric circuit

watt unit of measurement for power; 1 watt is 1 joule per second

weight the gravitational force of attraction exerted on an object by the Earth (or Moon or any other planet)